



Heron Link Inverter

Grid Forming Model Performance Report

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Revision History

No.	Description	Date
1	Issue for Initial Release	July 18, 2026

Executive Summary

This document was prepared to evaluate the Heron Link inverter model performance against the grid code requirements set forth in IEEE P2800.1 [1]. IEEE P2800.1 is currently under development and subject to change; this evaluation used the “D0.1” draft of the document. In addition to the IEEE P2800.1 tests, simulation tests were also performed based on the MISO grid forming (GFM) model performance test requirements [2] and selected tests from the Global Power System Transformation (G-PST) Consortium tests [3]. A generic power plant controller (PPC) model is used along with the Heron Link inverter model for modeling completeness. The model is tuned to satisfy the GFM performance test requirements evaluated herein.

The performance tests defined in this report were performed in PSCAD using electromagnetic transient (EMT) simulations. Results are presented for each test, with compliance assessed against the applicable pass/fail criteria. **The Heron Link Inverter PSCAD model and generic PPC PSCAD model presented in this report satisfied the evaluated performance test requirements for all GFM performance tests.**

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2 Test System Setup

Per IEEE P2800.1, two separate test configurations are defined to evaluate IBR compliance with grid interconnection requirements.

2.1 Test Setup 1 — Variable Impedance Source

Test Setup 1 is shown below in Figure 2-1. It employs a variable impedance $Z = R + jX$, where the magnitude $|Z|$ is defined in per-unit terms on the basis of the IBR rating as:

$$|Z| = 1 / SCR$$

The voltage source is ideal (zero impedance) with controllable magnitude, phase, and frequency. The X/R ratio is held constant at 6 for all applicable tests unless otherwise specified.

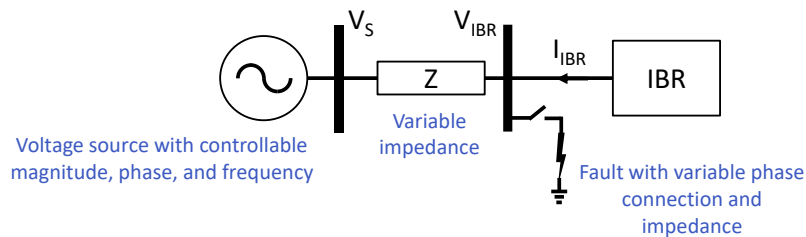


Figure 2-1: Test Setup 1: Variable Impedance Source [1]

2.2 Test Setup 2 — Islanding Load

Figure 2-2 shows Test Setup 2, in which two instances of the IBR model are connected to a constant-impedance load with a 0.95 power factor and adjustable nominal power level. The ac voltage source is disconnected to simulate loss of grid connection. Per the standard, the IBR must not receive the disconnection switch status. Unless otherwise specified, the project BESS and duplicate BESS are identical ratings.

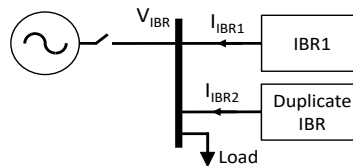


Figure 2-2: Test Setup 2: Islanding Load Configuration [1]

3 Grid Forming Model Performance Tests

The Heron Link Inverter was evaluated using Test Setup 1 or Test Setup 2, as shown in Figure 2-1 and Figure 2-2 respectively, unless otherwise specified. For each test, this section defines the setup, sequence, and success criteria, and determines the pass/fail outcome based on the simulation results.

3.1 System Strength Test (IEEE P2800.1 Clause 8.3)

This test verifies the IBR's capability to operate stably under weak-grid conditions by successively reducing the short-circuit ratio (SCR) while applying fault events. The SCR reduction with fault test evaluates GFM BESS control stability under declining system strength conditions (i.e., weak grid). The test starts with higher SCR values before introducing a fault and stepping down SCR while simultaneously clearing the fault. This sequence is repeated several times until very weak grid conditions are simulated. This test uses test system 1. The initial test setup, simulation steps, and pass/fail criteria are shown in the table below.

3.1.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Tests 1–2 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

SCR Reduction Test – Setup and Success Criteria	
Test Setup:	
The BESS project is dispatched 100% of its maximum discharge power limit per Test Setup 1 with SCR at IBR bus: 10, X/R ratio: 6 and IBR active power $P = 1.0$ pu; reactive power $Q = 0$.	
Test Sequence:	
a) Apply a three-phase bolted (zero-impedance) fault for four-line cycles. At fault clearance, step the SCR to 5, maintaining X/R constant. Allow the transient to subside. b) Repeat step a) stepping SCR from 10 to 5, 3, 1.5, and 1.2 in succession every 5 seconds. c) Repeat steps a)–b) substituting a two-phase-to-ground (2LG) fault of six-line cycles duration, with fault impedance selected such that the residual positive-sequence voltage at the IBR bus is ≤ 0.5 pu	
Success Criteria:	Pass/Fail
Plant real and reactive power output should be well controlled and plant should not trip or reduce power for any extended period of time down to the minimum SCR in the test.	PASS
The IBR shall not trip at any point during the test.	PASS
Outside fault periods, active power P shall not drop below 0.95 pu for more than 1 second.	PASS
Outside fault periods, IBR bus voltage V^{IBR} shall not drop below 0.95 pu for more than 1 second.	PASS
Any oscillations shall be damped.	PASS

3.2 Unbalance Step-Up Test (IEEE P2800.1 Clause 8.4)

This test verifies the IBR's ability to withstand voltage imbalance within the normal operating region across a range of active power levels per IEEE P2800.1 test setup in Figure 1-1.

3.2.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Tests 3–6 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria as the only pass criteria defined by IEEE P2800.1 was that the negative-sequence voltage at the IBR bus (V^{IBR}_2) shall be less than V^s_2 for all active power operating points as shown in Figure 3-1 to Figure 3-4.

Unbalance Step-Up Test – Setup and Success Criteria	
Test Setup:	
The BESS project is dispatched at 100% of its maximum discharge power limit per Test Setup 1 with SCR at IBR bus: 10, X/R ratio: 6 and IBR active power $P = 1.0$ pu; reactive power $Q = 0$.	
Test Sequence:	
a) Step the negative-sequence source voltage V^s_2 from 0 to 0.01 pu and wait 5 seconds. b) Repeat, stepping V^s_2 to 0.02 pu and 0.03 pu c) Repeat steps a)–c) three additional times, changing to 0.5 pu, –0.5 pu, and –1.0 pu	
Success Criteria:	Pass/Fail
When V^s_2 is non-zero, the negative-sequence voltage at the IBR bus (V^{IBR}_2) shall be less than V^s_2 for all active power operating points.	PASS

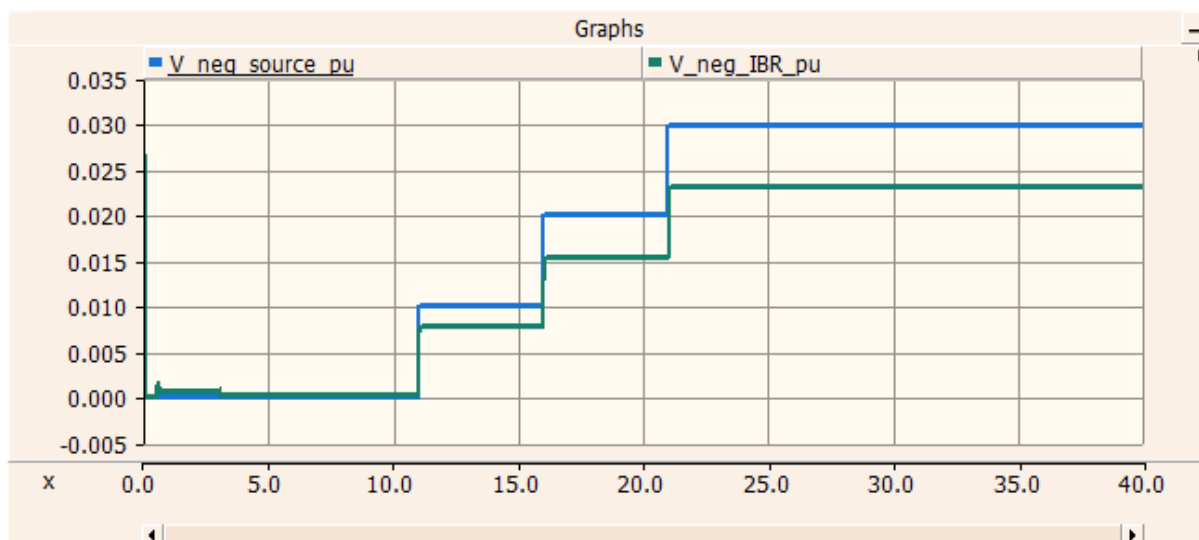


Figure 3-1: Voltage Source and IBR negative sequence voltage plotted illustrating $V^{IBR}_2 < V^s_2$ for a power dispatch of 50% in Test Setup 1.

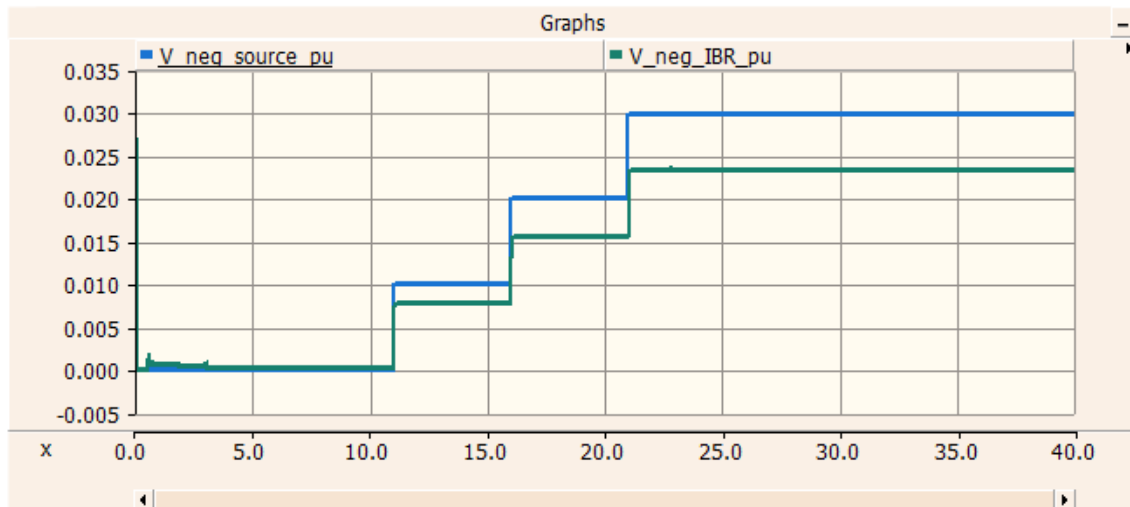


Figure 3-2 Voltage Source and IBR negative sequence voltage plotted illustrating $V^{IBR}_2 < V^s$ for a power dispatch of -50% in Test Setup 1.

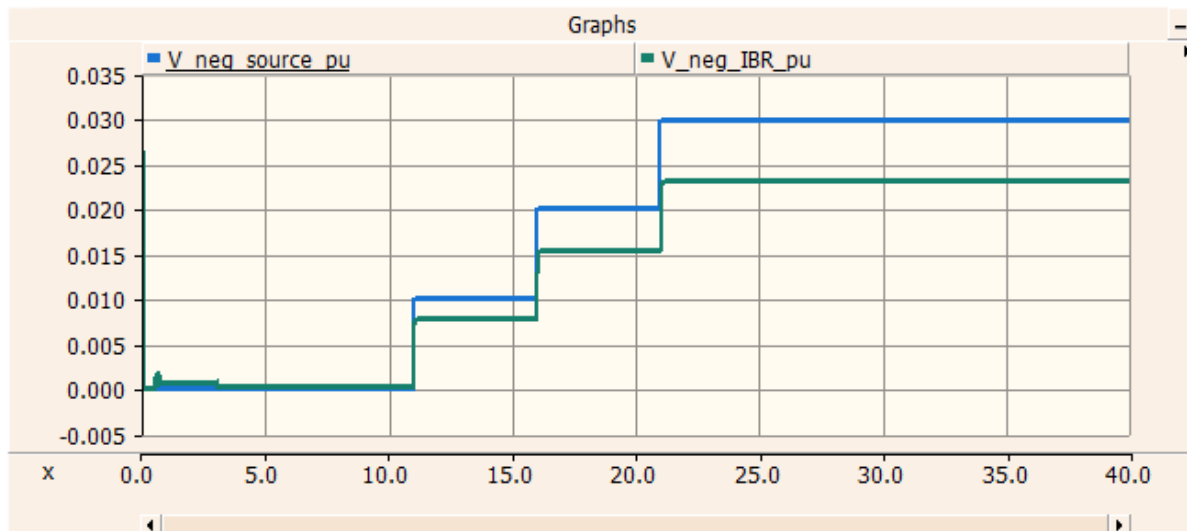


Figure 3-3 Voltage Source and IBR negative sequence voltage plotted illustrating $V^{IBR}_2 < V^s$ for a power dispatch of 100% in Test Setup 1.

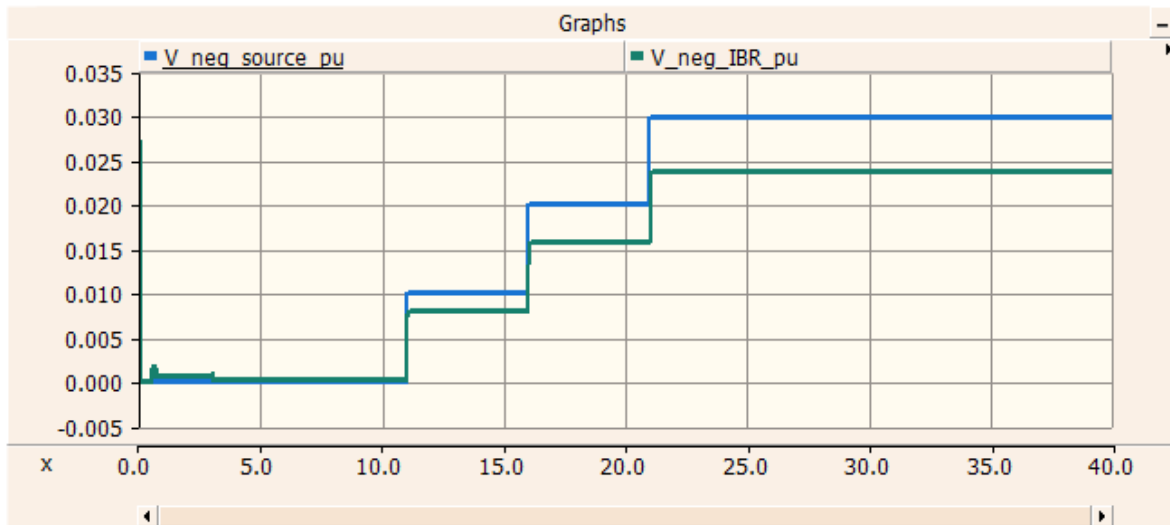


Figure 3-4 Voltage Source and IBR negative sequence voltage plotted illustrating $V^{IBR}_2 < V^s$ for a power dispatch of -100% in Test Setup 1.

3.3 Fault Response Test (IEEE P2800.1 Clause 8.5)

This test characterizes the IBR's response to selected fault types and operating conditions. It is informational only; IEEE P2800.1 does not prescribe pass/fail criteria for this test pending further industry experience and insight.

3.3.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Tests 7–14 in Appendix A: Plots of GFM Model Quality Tests.

Fault Response Test – Setup and Success Criteria	
Test Setup:	
The BESS project is dispatched at 100% of its maximum discharge power limit per Test Setup 1 with SCR at IBR bus: 10, X/R ratio: 6 and IBR active power $P = 1.0$ pu; reactive power $Q = 0$.	
Test Sequence:	
a) Allow the system to reach steady state. Apply a 3PG fault and remove it after 6 line cycles. b) Repeat for 50%, -100% and -50% power dispatches. c) Repeat (a) to (c) for 2PG faults	
Success Criteria:	Pass/Fail
This test is informational — no pass/fail determination is made per IEEE P2800.1 Clause 8.5.2, however, the model was able to ride through the faults and the results look reasonable and expected.	N/A

3.4 Response to Frequency Change Test (IEEE P2800.1 Clause 8.6)

This test verifies the IBR's capability to remain stable during frequency transients and to provide synthetic inertia response quantified by the equivalent inertia constant H.

3.4.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 15 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

Response to Frequency Change Test – Setup and Success Criteria	
Initial Dispatch	
The BESS project is dispatched at 0% of its maximum discharge power limit per Test Setup 1 SCR at IBR bus: 3, X/R ratio: 6, IBR active power P = 0; reactive power Q = 0	
Test Sequence:	
a) Run until the system is stable at the given power flow conditions, without oscillations. b) Ramp frequency down at 1 Hz/s for 1 second. Hold for 2 seconds. Ramp frequency up at 1 Hz/s for 1 second (returning to nominal). Hold for 6 seconds. c) Step frequency down 0.3 Hz below nominal. Hold for 5 seconds. Step back to nominal. Hold for 5 seconds. d) Step frequency up 0.3 Hz above nominal. Hold for 5 seconds. Step back to nominal. Hold for 5 seconds.	
Success Criteria	
	Pass/Fail
Active power P and reactive power Q shall be well controlled throughout.	PASS
Voltage and frequency at the IBR bus shall not deviate from steady-state values by more than 5% for more than 100 ms.	PASS
When frequency is not ramping, V^{IBR} shall settle within the normal operating region and P shall settle according to the IBR's droop and deadband settings.	PASS
All oscillations shall be damped.	PASS
The calculated inertia constant H shall be greater than 2.5 seconds as shown in Figure 3-5, where: $H = f_n \times \Delta E = f_n \times (E_1 - E_0)$ where f_n is the nominal system frequency (60 Hz), E_1 is the integral of active power during the first 500 ms of the ramp event, and E_0 is the integral during the 500 ms preceding the ramp, both per-unitized on the IBR rated active power.	PASS

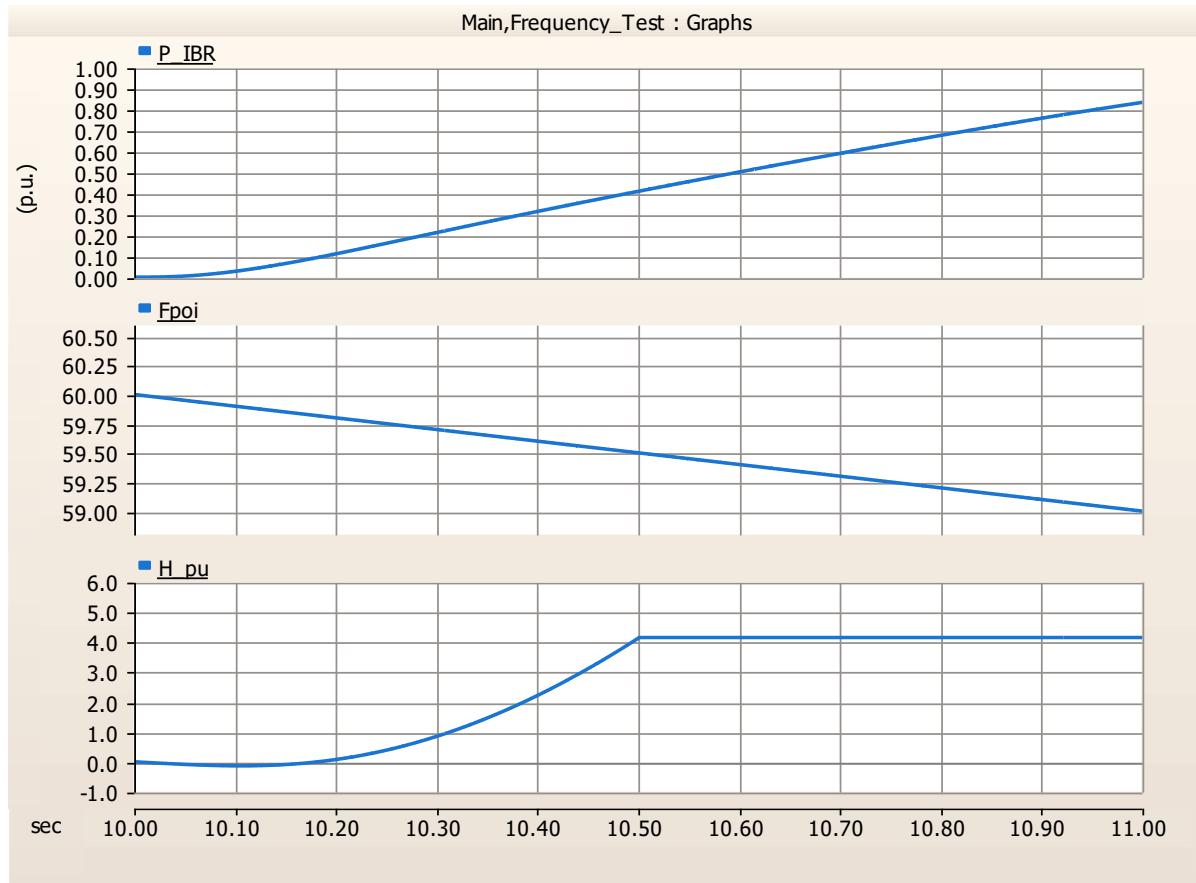


Figure 3-5 Response to ROCOF test and Inertia Calculation

3.5 Phase Angle Jump Test (IEEE P2800.1 Clause 8.7)

This test verifies the IBR's stability and active power response during sudden changes in positive-sequence voltage phase angle (all three phases moving together).

3.5.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 16 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

Phase Angle Jump Test – Setup and Success Criteria	
Initial Dispatch	
The BESS project is dispatched at 100% of its maximum discharge power limit per Test Setup 1 with SCR at IBR bus: 3, X/R ratio: 6 and IBR active power $P = 1.0$ pu; reactive power $Q = 0$.	
Test Sequence:	
a) Run until the system is stable at the given power flow conditions, without oscillations. b) Step the source voltage phase angle down 10°. Allow the transient to subside. c) Step the source voltage phase angle up 10°. Allow the transient to subside. d) Step the source voltage phase angle down 25°. Allow the transient to subside. e) Step the source voltage phase angle up 25°. Allow the transient to subside.	
Success Criteria	
	Pass/Fail
For the ($\pm 10^\circ$) phase jumps: The positive-sequence active power shall change by at least 0.2 pu in the direction opposing the phase jump, unless limited by IBR current or power limits.	PASS
At least 90% of the required 0.2 pu change shall occur within the first line cycle following the phase jump.	PASS
The change in active power shall be sustained for at least 3 line cycles.	PASS
For the ($\pm 25^\circ$) phase jumps: Any oscillations shall be damped.	PASS

3.6 Small Voltage Step Test (IEEE P2800.1 Clause 8.8)

This test verifies the IBR's reactive power response to sudden small changes in voltage magnitude with an infinite source voltage.

3.6.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 17 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

Small Voltage Step Test – Setup and Success Criteria	
Initial Dispatch	
The project BESS is dispatched at 100% of its maximum discharge power limit per Test Setup 1	
SCR is infinite since the impedance is ideal, i.e., $Z_{sc} = 0$ (ideal voltage source directly connected)	
Test Sequence:	
a) Step the source voltage magnitude down by 0.03 p.u for 10 seconds. b) Step up by 0.03 pu for 10 seconds. c) Step up by an additional 0.03 pu for 10 seconds. d) Step down by 0.03 p.u for 10 seconds.	
Success Criteria	
	Pass/Fail
Run until the system is stable at the given power flow conditions, without oscillations.	PASS
Following each voltage step, the positive-sequence reactive power shall change within 6 line cycles by at least 0.03 pu in the direction opposing the voltage change.	PASS
At least 90% of the IBR's initial reactive power change (ie: the change that occurs in the first six cycles) shall occur within the first cycle.	PASS
All oscillations shall be damped.	PASS

3.7 Loss of Grid Connection BESS Discharging (IEEE P2800.1 Clause 8.9)

The Loss of Grid Connection Limit Test evaluates the stability of the GFM in response to loss of grid connection during discharging conditions. This test uses test system 2 and the initial test setup, simulation steps, and pass/fail criteria are shown in the table below

3.7.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 18 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

Loss of Grid Connection Test (BESS Discharging) – Setup and Success Criteria	
Initial Dispatch	
The project BESS is dispatched at 10% of its maximum discharge power limit.	
The duplicate BESS is dispatched at 30% of its maximum discharge power limit.	
The load is set to 130% of project BESS active power.	
Test Sequence:	
Initiate simulation and run until system is stable at the given power flow conditions, without oscillations	
Trip the voltage source by opening the grid breaker	
Success Criteria	
Pre-Trip:	Pass/Fail
Both BESS plants active power outputs match dispatched levels.	PASS
Voltage source active power output matches the rest of the load.	PASS
Frequency is 1 pu.	PASS
Voltage at bus 1 is within 5% of nominal (i.e., 0.95 to 1.05 pu).	PASS
Phase voltage and current waveforms are not distorted.	PASS
Oscillations are not present in RMS quantities.	PASS
Reactive power output from all devices should be within limits.	PASS
Post-Trip:	Pass/Fail
Plant output is well controlled with no significant frequency/voltage oscillations.	PASS
Voltage settles to a stable operating point. Voltage shall not deviate beyond [0.8, 1.10] pu for more than 0.1 s at any point during the test.	PASS
Final voltage is expected based on droop and deadband settings.	PASS
Voltage and frequency shall not oscillate excessively and shall settle within 5 seconds inside their continuous operating regions.	PASS
Final frequency is expected based on droop and deadband settings.	PASS
All oscillations shall be damped within 10 seconds.	PASS
Distortion observed in phase quantities dissipates over time.	PASS
Active power immediately moves to meet load requirement and settle according to its frequency droop setting. Instantaneous power measurements (or filtered with time constant ≤ 1 ms) shall be used for criterion evaluation.	PASS

The combined active and reactive power of both IBRs shall increase by 90% of the load deficit (at $V = 0.8$ pu) within 1 cycle or less. Note: Active power load deficit at 0.8 pu: $0.8^2 \times 0.9 \times (P_3 - P_1 - P_2)$ where P_3 is the load initialize power, P_2 is the duplicate BESS initial power and P_1 is the project BESS initial power.

PASS

3.8 Loss of Grid Connection BESS Charging Test (IEEE P2800.1 Clause 8.9)

The Loss of Grid Connection Limit Test evaluates the stability of the GFM in response to loss of grid connection during charging conditions. This test uses test system 2 and the initial test setup, simulation steps, and pass/fail criteria are shown in the table below.

3.8.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 19 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

Loss of Grid Connection BESS Charging Test (Charging) – Setup and Success Criteria	
Initial Dispatch	
The project BESS is dispatched at -40% of its maximum discharge power limit.	
The duplicate BESS is dispatched at -60% of its maximum discharge power limit.	
The load is set to 70% of project BESS active power.	
Test Sequence:	
Initiate simulation and run until system is stable at the given power flow conditions, without oscillations.	
Trip the voltage source by opening the grid breaker.	
Success Criteria	
Pre-Trip:	Pass/Fail
Both BESS plants active power outputs match dispatched levels.	PASS
Voltage source active power output matches the rest of the load.	PASS
Frequency is 1 pu.	PASS
Voltage at bus 1 is within 5% of nominal (i.e., 0.95 to 1.05 pu).	PASS
Phase voltage and current waveforms are not distorted.	PASS
Oscillations are not present in RMS quantities.	PASS
Reactive power output from all devices should be within limits.	PASS
Post-Trip:	Pass/Fail
Plant output is well controlled with no significant frequency/voltage oscillations.	PASS
Voltage settles to a stable operating point. Voltage shall not deviate beyond [0.8, 1.10] pu for more than 0.1 s at any point during the test.	PASS
Final voltage is expected based on droop and deadband settings.	PASS
Voltage and frequency shall not oscillate excessively and shall settle within 5 seconds inside their continuous operating regions.	PASS
Final frequency is expected based on droop and deadband settings.	PASS
All oscillations shall be damped within 10 seconds.	PASS
Distortion observed in phase quantities dissipates over time.	PASS
Active power immediately moves to meet load requirement and settle according to its frequency droop setting. Instantaneous power measurements (or filtered with time constant ≤ 1 ms) shall be used for criterion evaluation.	PASS

The combined active and reactive power of both IBRs shall increase by 90% of the load deficit (at $V = 0.8$ pu) within 1 cycle or less. Note: Active power load deficit at 0.8 pu: $0.8^2 \times 0.9 \times (P_3 - P_1 - P_2)$ where P_3 is the load initialize power, P_2 is the duplicate BESS initial power and P_1 is the project BESS initial power.

PASS

3.9 Loss of Grid Connection Limit Test (IEEE P2800.1 Clause 8.9)

The Loss of Grid Connection Limit Test evaluates the stability of the GFM in response to loss of grid connection to test the limits of the BESS. This test uses test system 2 and the initial test setup, simulation steps, and pass/fail criteria are shown in the table below.

3.9.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 20 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

Loss of Grid Connection Limit Test – Setup and Success Criteria	
Initial Dispatch	
The project BESS is dispatched at 0% of its maximum discharge power limit.	
The duplicate BESS is dispatched at 100% of its maximum discharge power limit.	
The load is set to 165% of project BESS active power.	
Test Sequence:	
Initiate simulation and run until system is stable at the given power flow conditions, without oscillations.	
Trip the voltage source by opening the grid breaker.	
Success Criteria	
Pre-Trip:	Pass/Fail
Both BESS plants active power outputs match dispatched levels.	PASS
Voltage source active power output matches the rest of the load.	PASS
Frequency is 1 pu.	PASS
Voltage at bus 1 is within 5% of nominal (i.e., 0.95 to 1.05 pu).	PASS
Phase voltage and current waveforms are not distorted.	PASS
Oscillations are not present in RMS quantities.	PASS
Reactive power output from all devices should be within limits.	PASS
Post-Trip:	Pass/Fail
Plant output is well controlled with no significant frequency/voltage oscillations.	PASS
Voltage settles to a stable operating point. Voltage shall not deviate beyond [0.8, 1.10] pu for more than 0.1 s at any point during the test.	PASS
Final voltage is expected based on droop and deadband settings.	PASS
Voltage and frequency shall not oscillate excessively and shall settle within 5 seconds inside their continuous operating regions.	PASS
Final frequency is expected based on droop and deadband settings.	PASS
All oscillations shall be damped within 10 seconds.	PASS
Distortion observed in phase quantities dissipates over time.	PASS
Active power immediately moves to meet load requirement and settle according to its frequency droop setting. Instantaneous power measurements (or filtered with time constant ≤ 1 ms) shall be used for criterion evaluation.	PASS

The combined active and reactive power of both IBRs shall increase by 90% of the load deficit (at $V = 0.8$ pu) within 1 cycle or less. Note: Active power load deficit at 0.8 pu: $0.8^2 \times 0.9 \times (P_3 - P_1 - P_2)$ where P_3 is the load initialize power, P_2 is the duplicate BESS initial power and P_1 is the project BESS initial power.

PASS

3.10 Rate of Change of Frequency Test

The ROCOF test evaluates the stability of the GFM in response to changing frequency as per [2]. The test procedure includes both frequency increases and decreases. The initial test setup, simulation steps, and pass/fail criteria are shown in the table below.

3.10.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 21 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

ROCOF Test – Setup and Success Criteria	
Test Setup:	
The short circuit ratio (SCR) at the GFM BESS connection point shall be set to 10.	
The system equivalent X/R shall be set to 6.	
The BESS is dispatched to 50% of continuous rating.	
Test Sequence:	
Ramp frequency from 60 to 61 Hz at 4 Hz/s. Remain at 61 Hz for 5 seconds.	
Ramp frequency from 61 to 60 Hz at 4 Hz/s. Remain at 60 Hz for 5 seconds.	
Ramp frequency from 60 to 59 Hz at 4 Hz/s. Remain at 59 Hz for 5 seconds.	
Ramp frequency from 59 to 60 Hz at 4 Hz/s.	
Success Criteria:	Pass/Fail
Plant output is well controlled with no significant frequency/voltage oscillations.	PASS
Voltage settles to stable operating point when frequency is not ramping.	PASS
Active power should settle based on frequency droop and deadband settings when frequency is not ramping.	PASS
Oscillations should be adequately damped.	PASS

3.11 Consecutive Voltage Disturbance Ride Through Test

The test evaluates the stability of the GFM in response to multiple voltage disturbance ride through events to verify that the inverter can ride through.

3.11.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 22 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

Consecutive Voltage Disturbance Test – Setup and Success Criteria	
Test Setup:	
The short circuit ratio (SCR) at the GFM BESS connection point shall be set to 10.	
The system equivalent X/R shall be set to 6.	
The BESS is dispatched to 50% of continuous rating.	
Test Sequence:	
The voltage at the infinite bus goes from 1 pu to 0.02 for 320 ms and returns to 1 pu for 1 second.	
The voltage at the infinite bus goes from 1 pu to 0.25 for 880 ms and returns to 1 pu for 1 second.	
The voltage at the infinite bus goes from 1 pu to 0.50 for 1.80 sec and returns to 1 pu for 1 second.	
The voltage at the infinite bus goes from 1 pu to 0.70 for 3.00 sec and returns to 1 pu for 1 second.	
Pass/Fail Criteria:	
The BESS rides through the faults in a stable manner and does not trip.	PASS
Any oscillation should be settled.	PASS
Any distortion observed in phase quantities should dissipate over time.	PASS
Active and reactive power are recovered to the pre-fault values after the fault clearance.	PASS

3.12 Load Step Changes during Islanded and Grid Connected Conditions

This test evaluates the impact of load step changes on the GFM model during islanded and grid connected conditions as defined in [3].

3.12.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 23 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

Load Step Changes – Setup and Success Criteria	
Test Setup:	
The short circuit ratio (SCR) at the GFM BESS connection point shall be set to 20.	
The system equivalent X/R shall be set to 6.	
The BESS is dispatched to 50% of continuous rating.	
Test Sequence:	
(a) Run until the system is stable at the given power flow conditions, without oscillations.	
(b) The load is increased by 25% of the initial load at $t = 3$ s.	
(c) The load is increased by 100% of the initial load at $t = 11$ s.	
(d) All the additional loads are tripped at $t = 19$ s.	
(e) The voltage source is tripped at $t = 20$ s.	
(f) The load is increased by 25% of the initial load at $t = 25$ s.	
(g) The load is increased by 50% of the initial load at $t = 28$ s.	
(h) All additional loads are tripped at $t = 32$ s.	
Pass/Fail Criteria:	
Each BESS's active power output matches dispatched levels.	PASS
SG active power output matches the rest of the load.	PASS
Frequency should be 1 pu.	PASS
Voltage at Bus 1 should be within 5% of nominal.	PASS
Phase voltage and current waveform should not be distorted.	PASS
Oscillations should not be present in the RMS quantities.	PASS
Reactive power output from all devices should be within limits.	PASS
After Load profile Application	
Immediately following the load increase, BESS output should be well controlled. System frequency and voltage should not oscillate excessively or deviate from steady-state levels for any significant amount of time.	PASS
Voltage settles to a stable and acceptable operating point following load changes.	PASS
The final voltage is as expected based on the droop and deadband settings.	PASS
Frequency should finally settle at the nominal value after additional loads are disconnected.	PASS
Any oscillation should be settled.	PASS
Any distortion observed in phase quantities should dissipate over time.	PASS
Active power from BESS should move immediately to meet the load requirement and settle according to its frequency droop setting.	PASS
Reactive power from each BESS should move immediately and settle according to its voltage droop setting.	PASS

3.13 Fault Ride Through

The test evaluates the stability of the GFM in response to multiple voltage disturbance ride through events to verify that the inverter can ride through after loss of synchronism as defined in [3].

3.13.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 24 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

Fault Ride Through – Setup and Success Criteria	
Test Setup:	
The BESS project is dispatched 100% of its maximum discharge power limit per Test Setup 1 with SCR at IBR bus: 10, X/R ratio: 6 and IBR active power $P = 1.0$ pu; reactive power $Q = 0$.	
Test Sequence:	
(a) Run until the system is stable at the given power flow conditions, without oscillations. (b) A bolted single-line-to-ground fault is applied at $t = 2$ s for six cycles. (c) 5 seconds later, a bolted line-to-line fault is applied for six cycles. (d) 5 seconds later, a bolted three-phase-to-ground fault is applied for six cycles.	
Pass/Fail Criteria:	
The BESS rides through the faults in a stable manner and does not trip.	PASS
Any oscillation should be settled.	PASS
Any distortion observed in phase quantities should dissipate over time.	PASS
Active and reactive power are recovered to the pre-fault values after the fault clearance	PASS

3.14 Fault Ride Through with Loss of Synchronism

The test evaluates the stability of the GFM in response to multiple voltage disturbance ride through events to verify that the inverter can ride through after loss of synchronism as defined in [3].

3.14.1 Test Procedure and Results

The following test setup, sequence, and success criteria were applied. Based on the waveform responses shown in Test 25 in Appendix A: Plots of GFM Model Quality Tests, the Heron Link Inverter satisfied the applicable criteria.

Fault Ride Through with loss of Synchronism – Setup and Success Criteria	
Test Setup:	
Test Setup 2 is modified with the duplicate BESS rated for 1.1 of the MVA of the BESS project. Both the duplicate and BESS project are dispatched to 90% MW output. The voltage source is set with an SCR of 20, X/R ratio: 6 with a load set to 10% larger than the combined MW output of the BESS project and duplicate BESS project.	
Test Sequence:	
(a) Run until the system is stable at the given power flow conditions, without oscillations. (b) The voltage source is tripped at $t = 2$ s. (c) A bolted single-line-to-ground fault is applied at $t = 2$ s for six cycles. (d) 5 seconds later, a bolted line-to-line fault is applied for six cycles. (e) 5 seconds later, a bolted three-phase-to-ground fault is applied for six cycles.	
Pass/Fail Criteria:	Pass/Fail?
The BESS rides through the faults in a stable manner and does not trip.	PASS
Any oscillation should be settled.	PASS
Any distortion observed in phase quantities should dissipate over time.	PASS
Following the trip and the single-line-to-ground fault, BESS output should be well controlled. System frequency and voltage should not oscillate excessively or deviate from steady-state levels for any significant amount of time.	PASS
The final voltage after the trip is as expected based on the droop and deadband settings.	PASS
Frequency settles to a stable operating point.	PASS
The final frequency after the trip is as expected based on the droop and deadband settings.	PASS
Active power from each BESS should move immediately to meet the load requirement and settle according to its frequency droop setting.	PASS
Reactive power from each BESS should move immediately and settle according to its voltage droop setting.	PASS

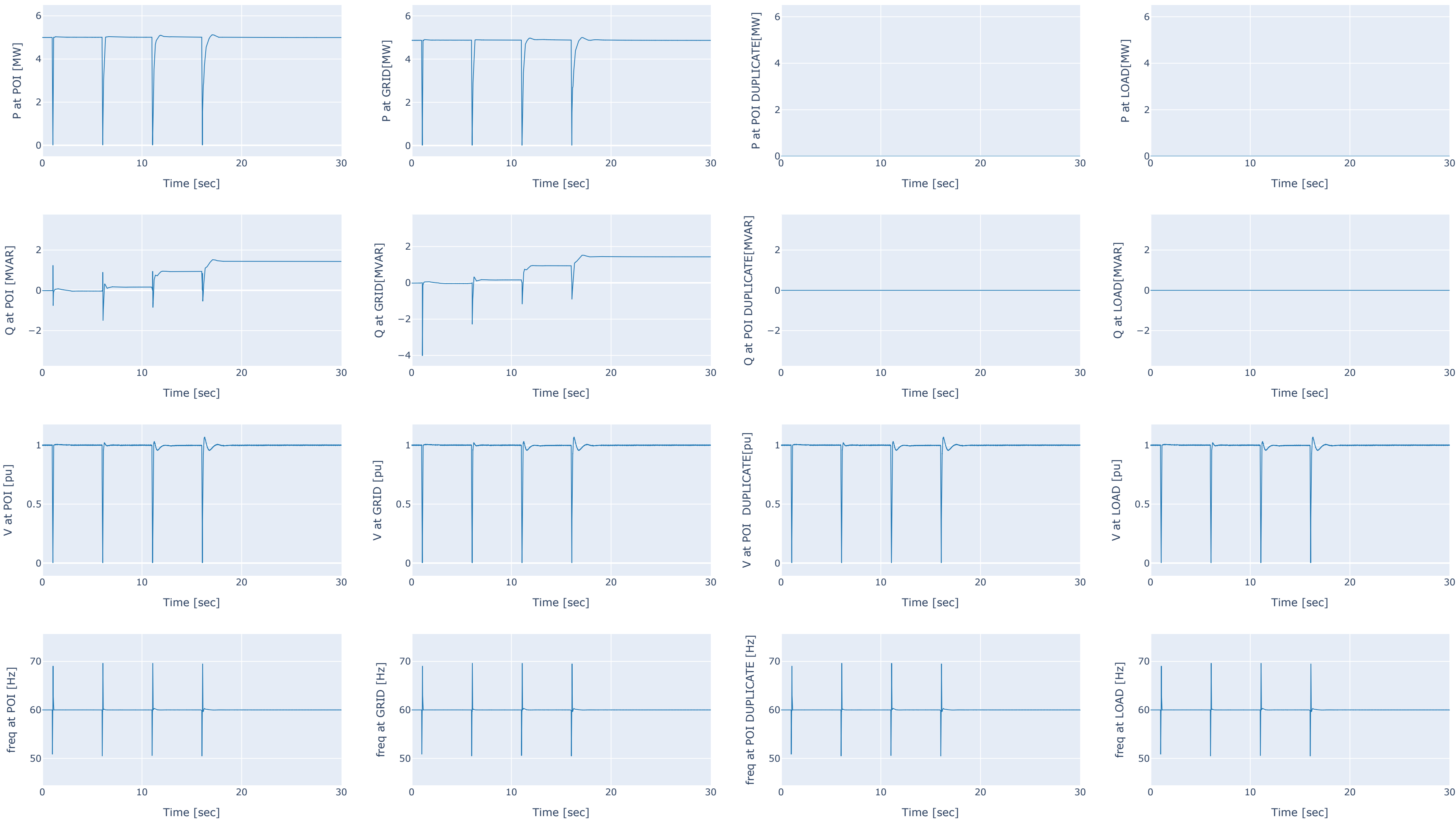
References

- [1] "IEEE 2800.1 Draft," 2026.
- [2] "MISO, Business Practices Manual, Generator Interconnection, Manual No. 015, 2025."
- [3] G. P. Consortium, "A Review of the Generic Grid-Forming Model Used by the System Operator in Chile," December 2024.

Appendix A: Plots of GFM Model Quality Tests

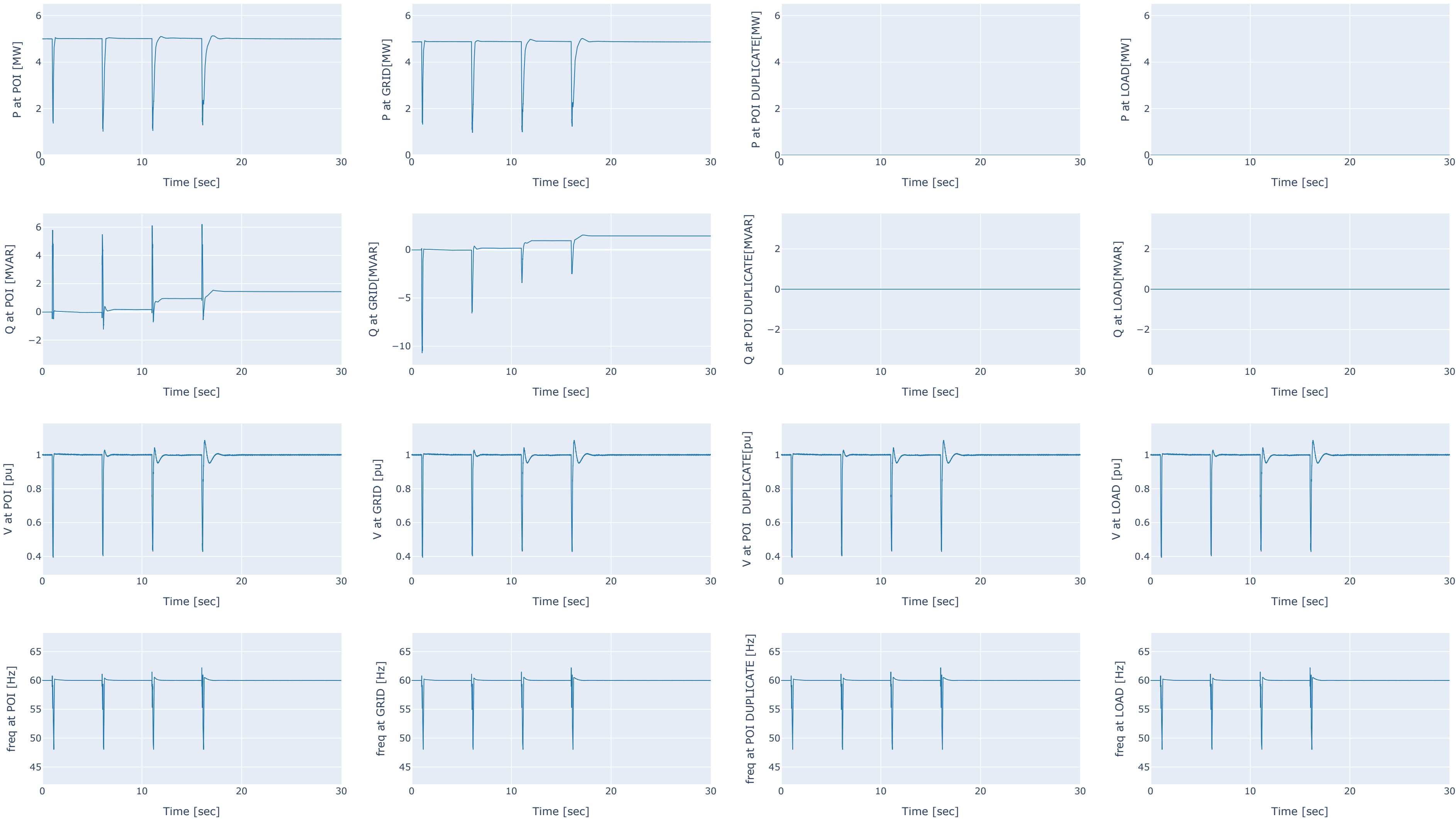
System Strength Test (IEEE 2800.1 Clause 8.3) 3PG four cycles

— PSCAD (Jul1_BB)



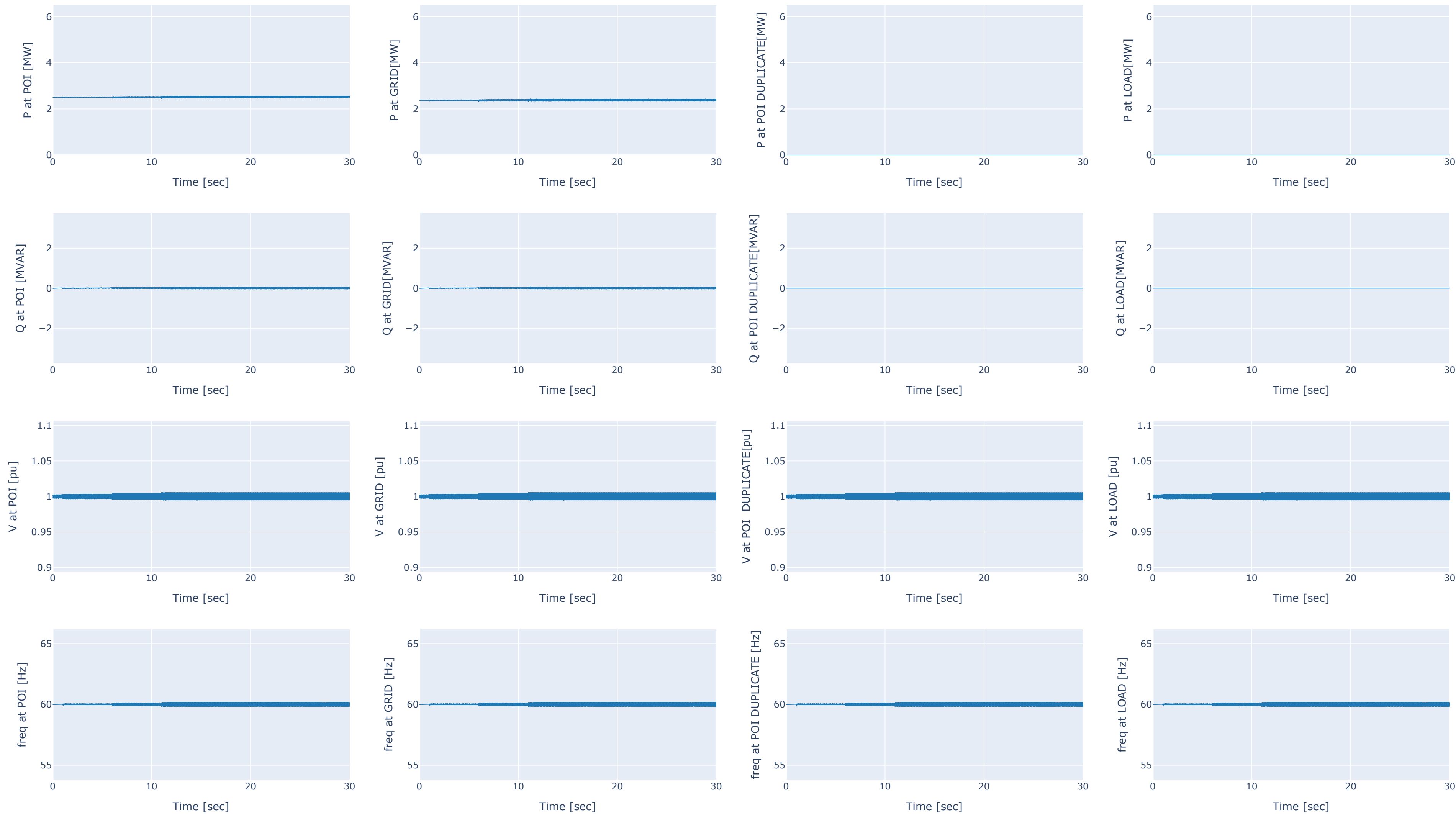
System Strength Test (IEEE 2800.1 Clause 8.3) 2PG six cycles

PSCAD (Jul1_BB)



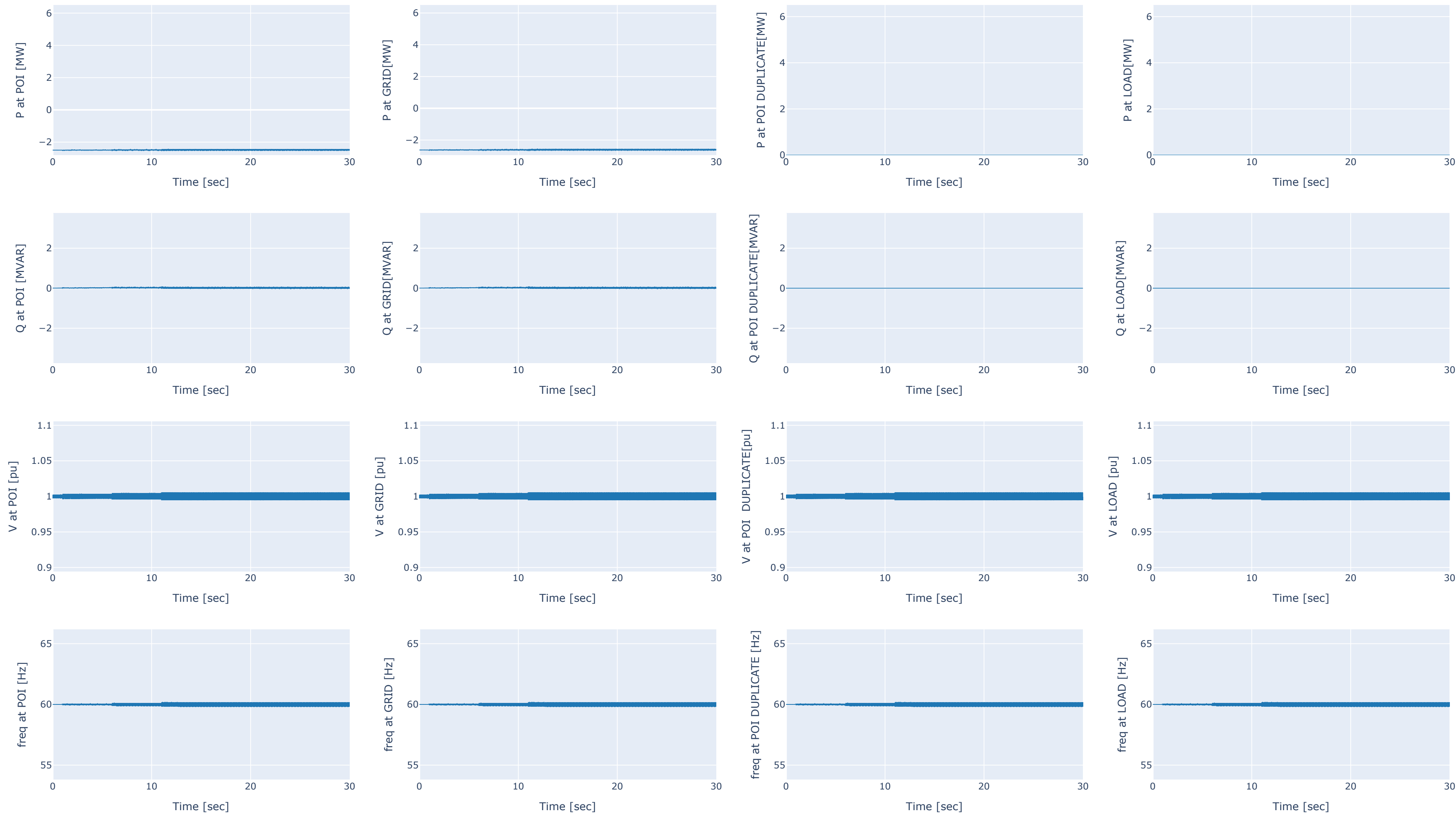
Unbalance Step-Up Test (IEEE 2800.1 Clause 8.4) 50% Active Power Dispatch

PSCAD (Jul1_BB)



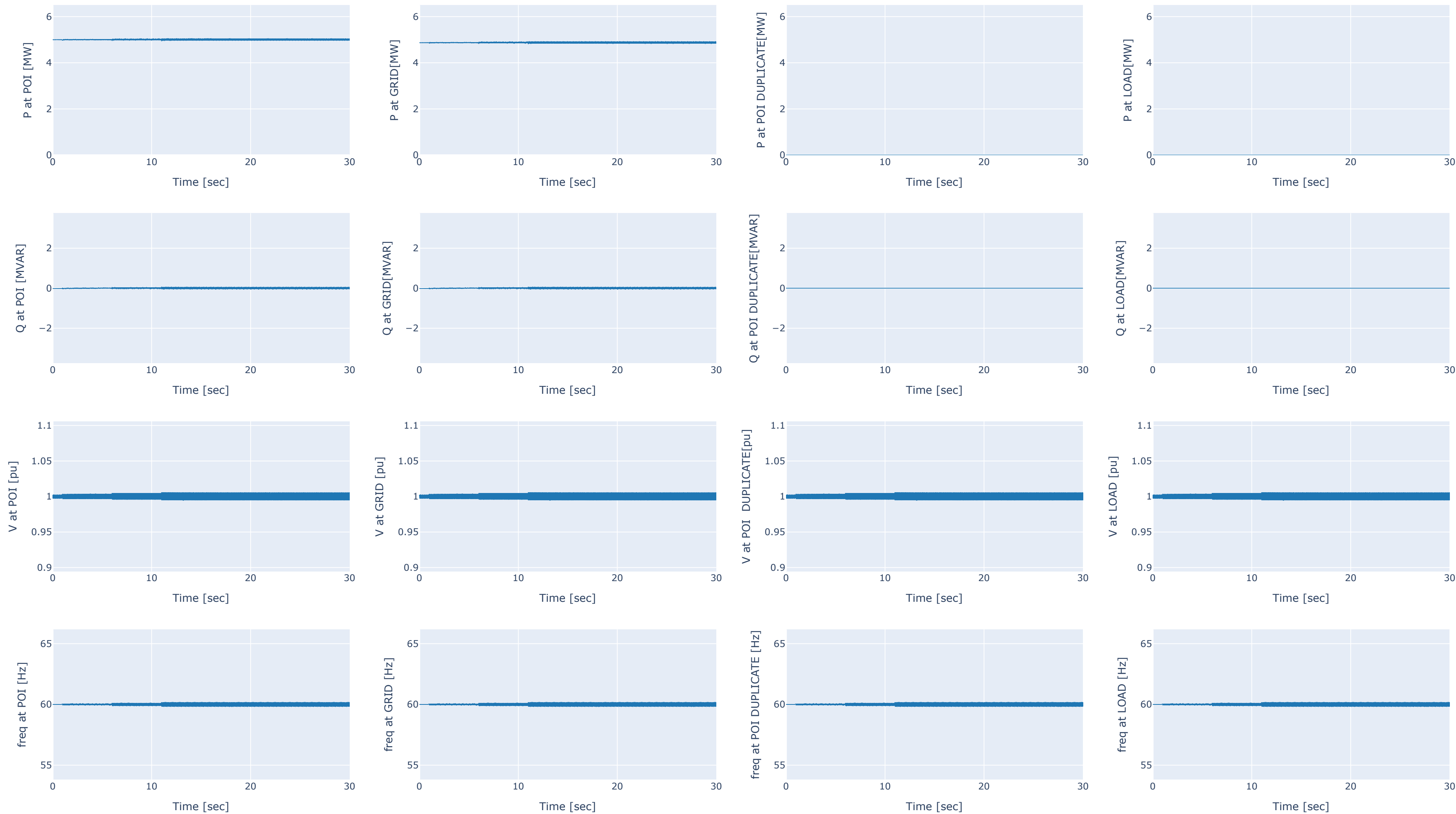
Unbalance Step-Up Test (IEEE 2800.1 Clause 8.4) -50% Active Power Dispatch

PSCAD (Jul1_BB)



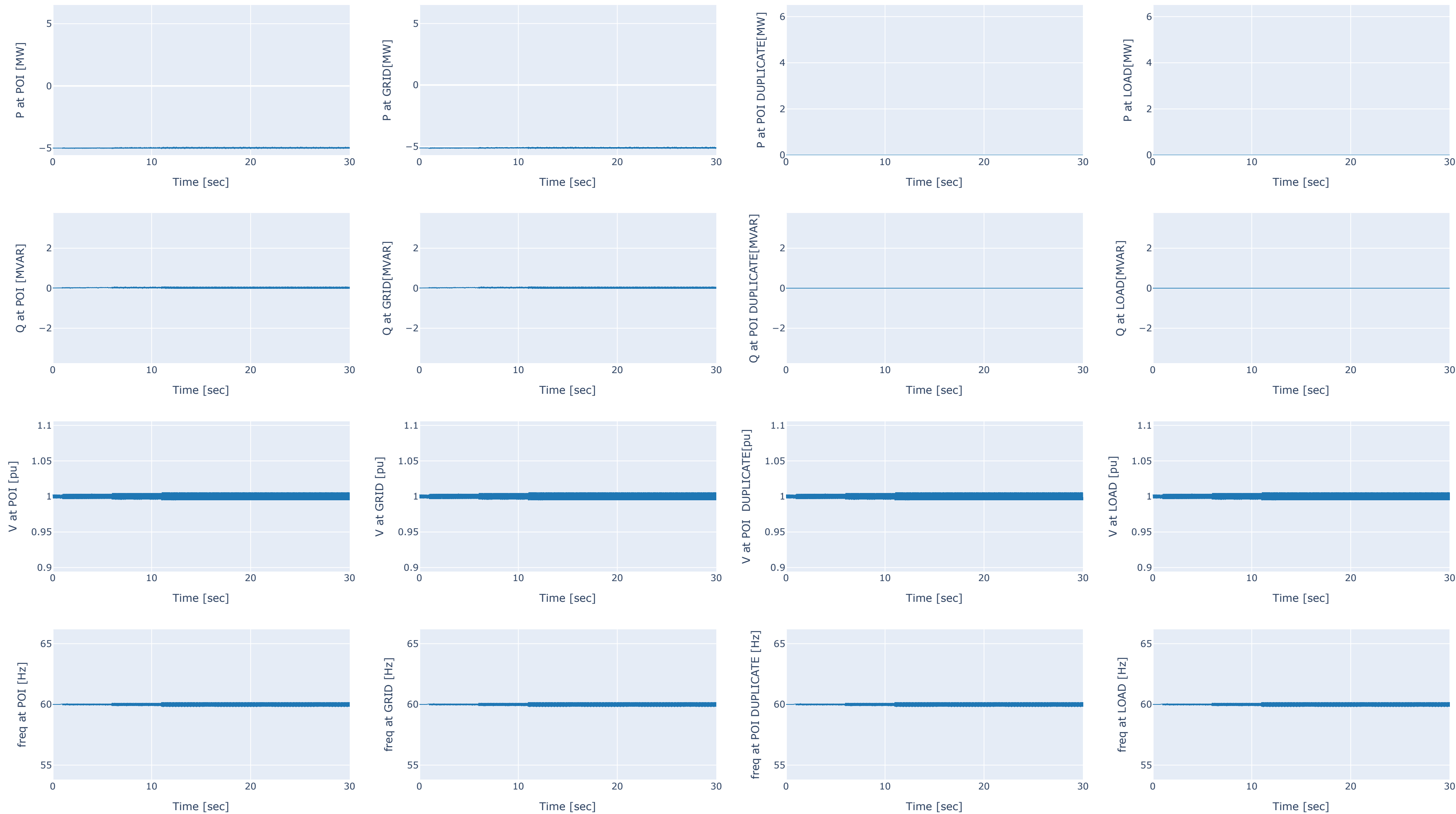
Unbalance Step-Up Test (IEEE 2800.1 Clause 8.4) 100% Active Power Dispatch

— PSCAD (Jul1_BB)



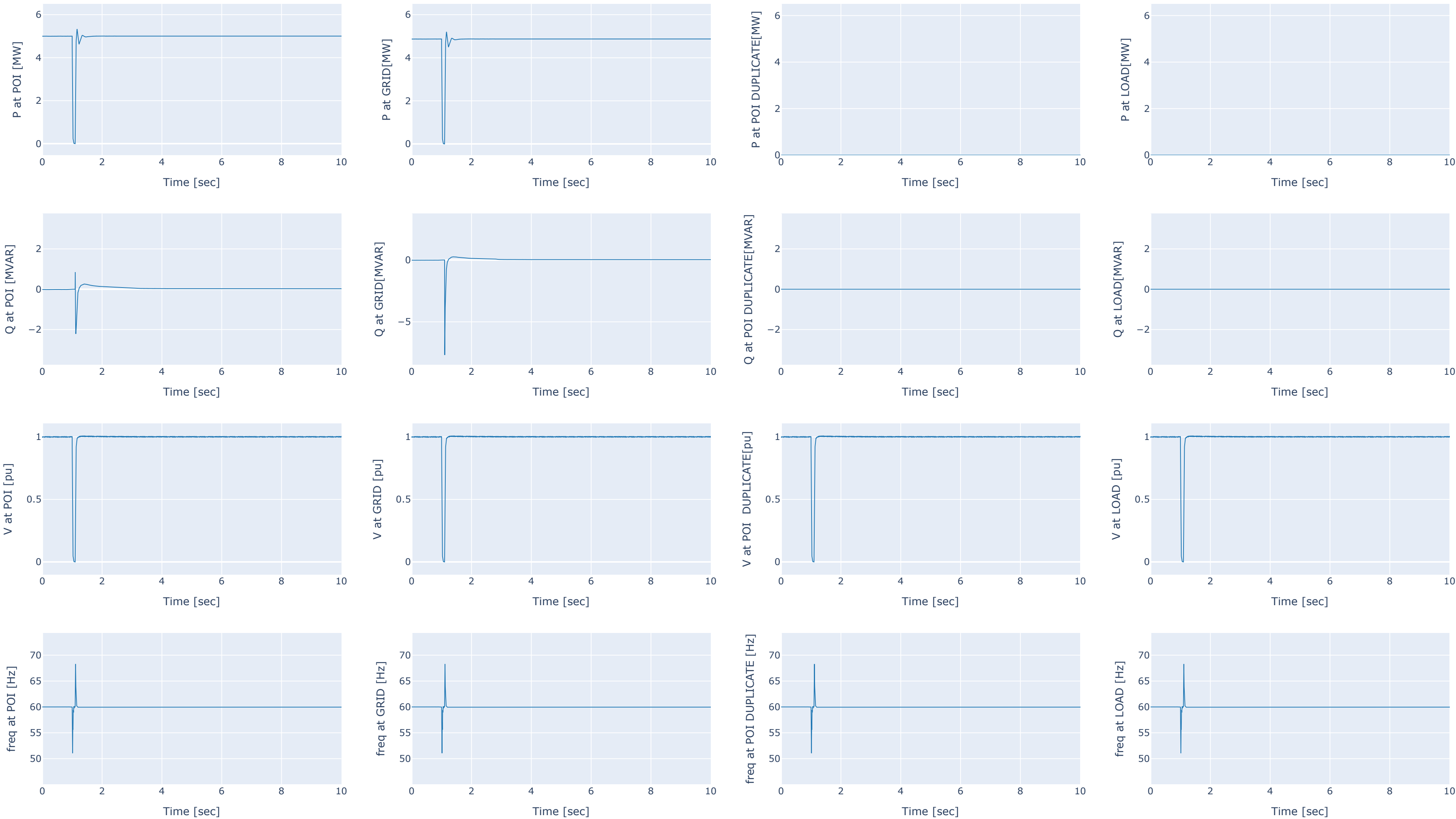
Unbalance Step-Up Test (IEEE 2800.1 Clause 8.4) -100% Active Power Dispatch

— PSCAD (Jul1_BB)



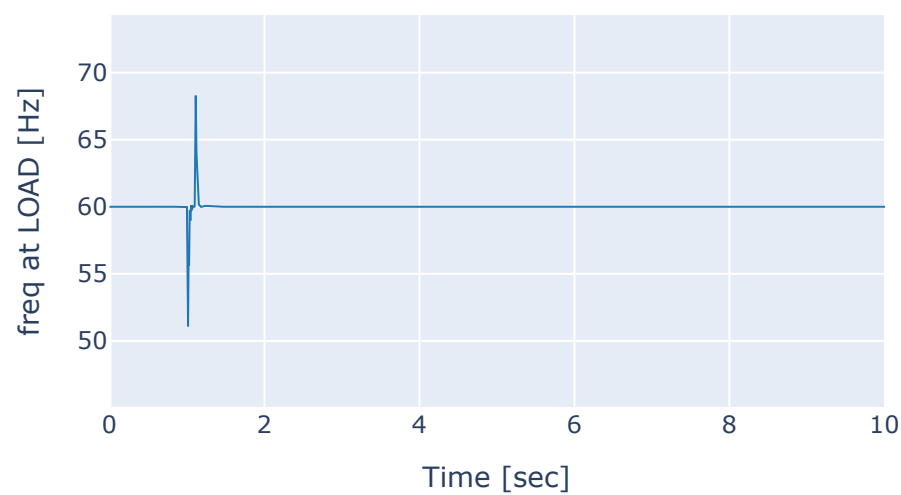
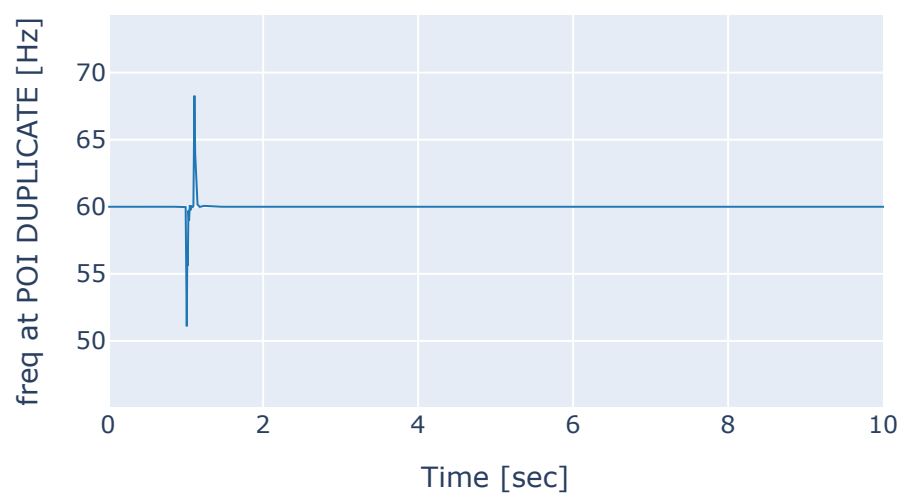
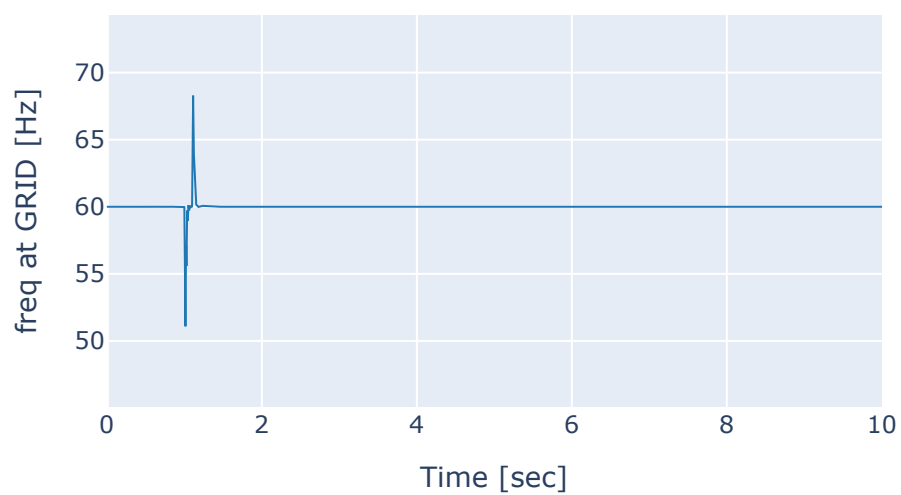
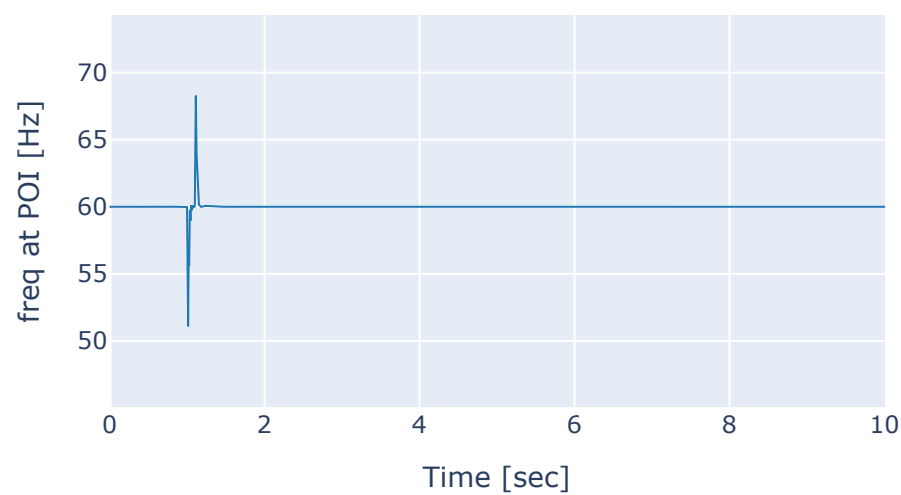
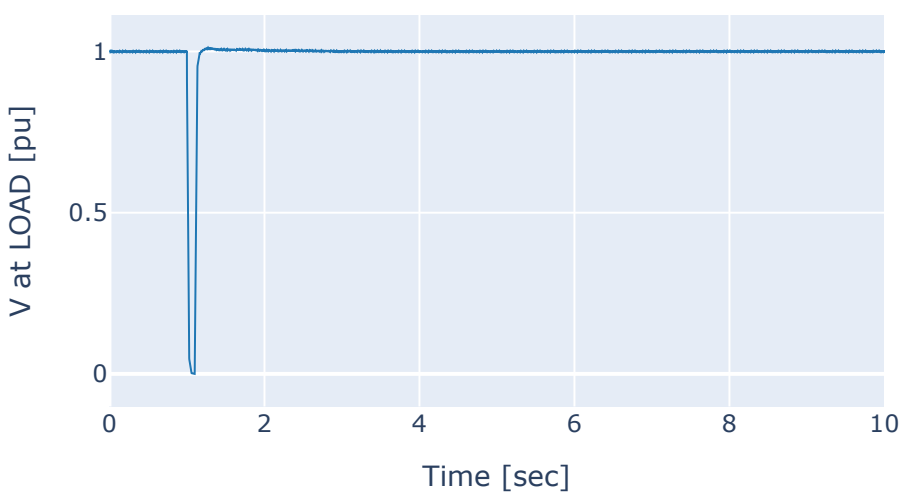
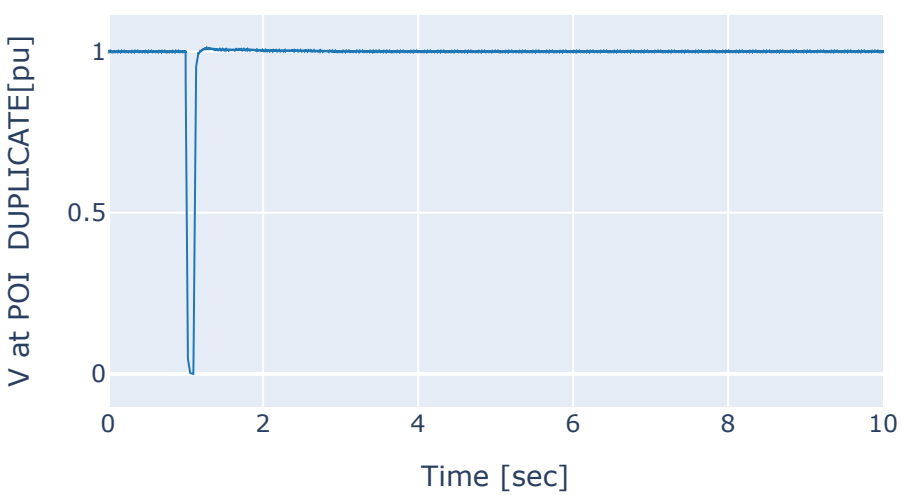
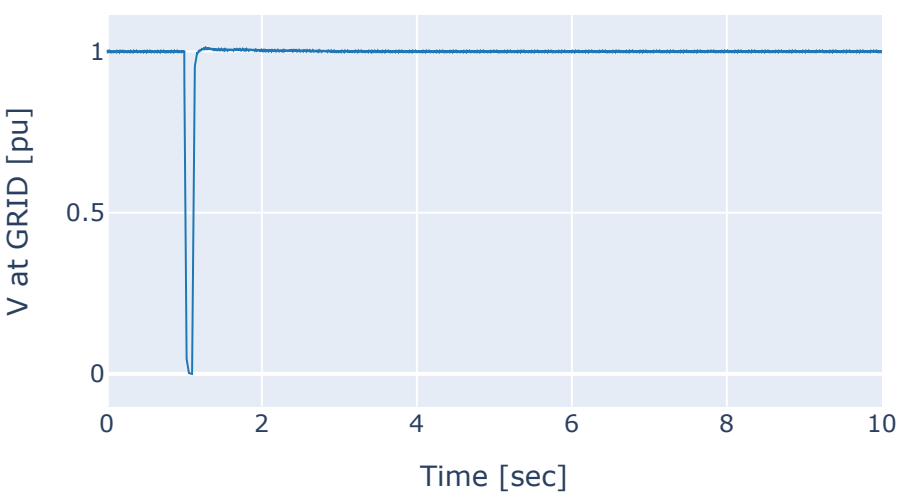
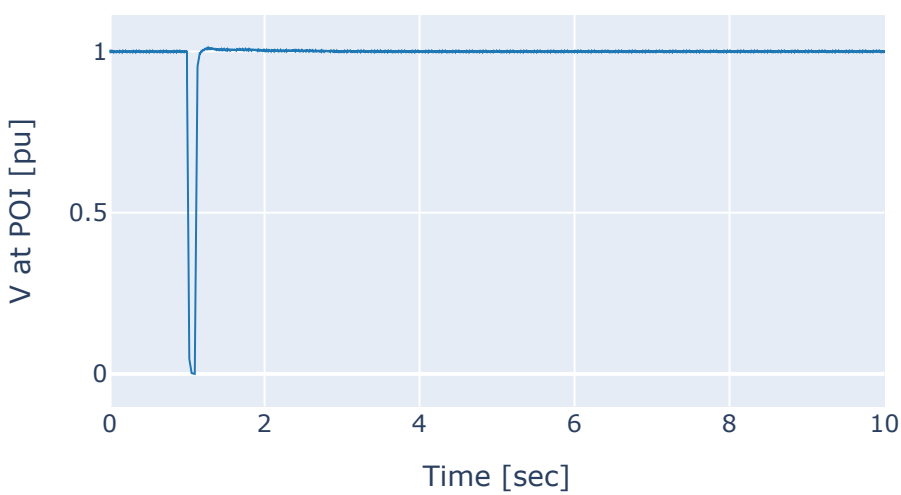
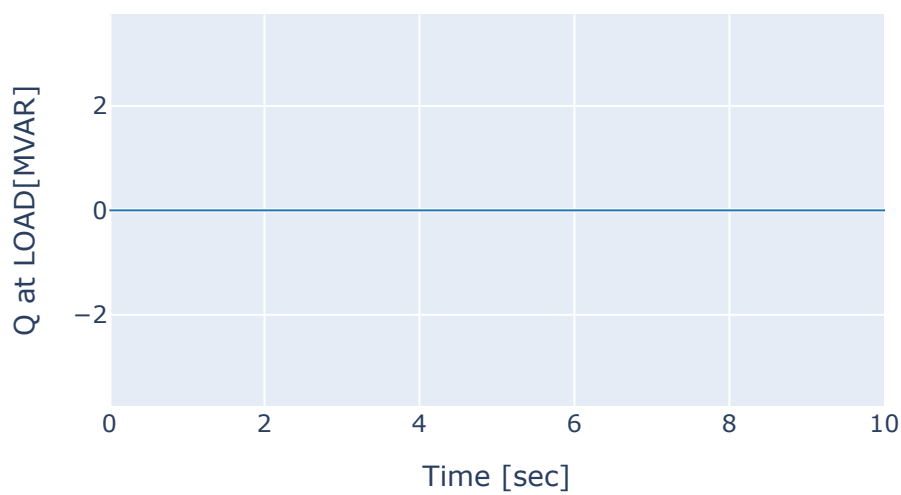
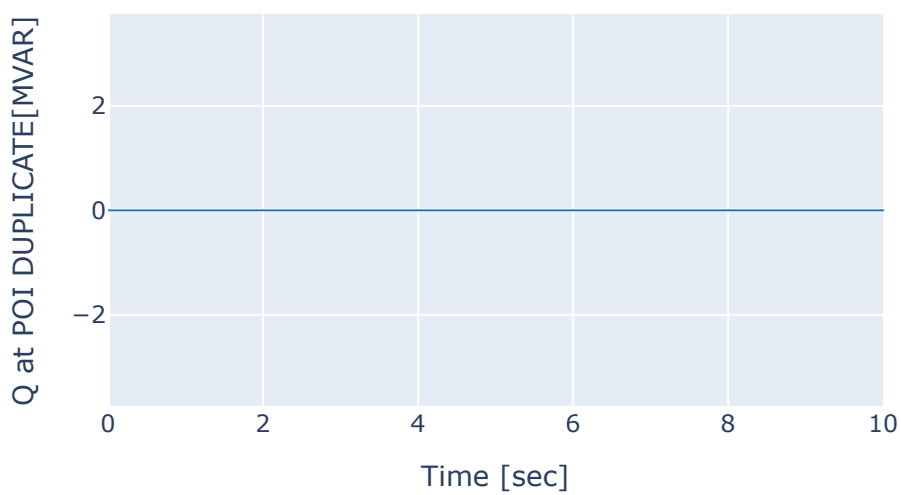
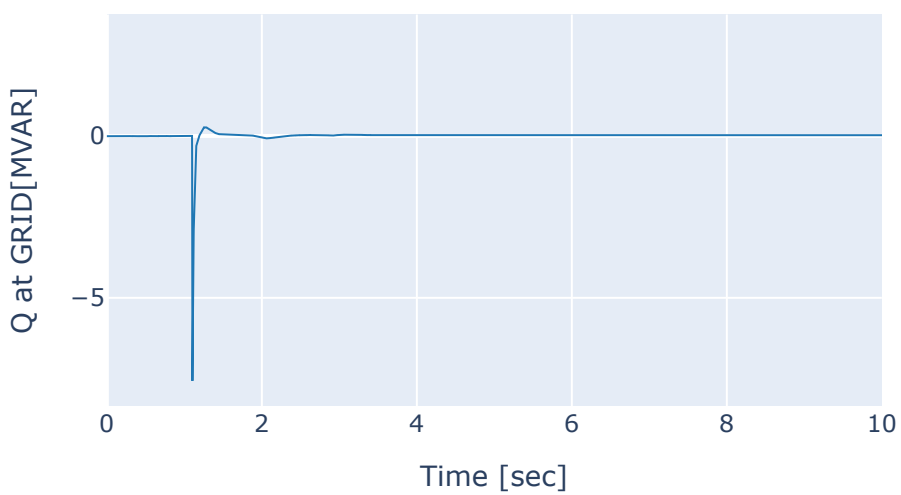
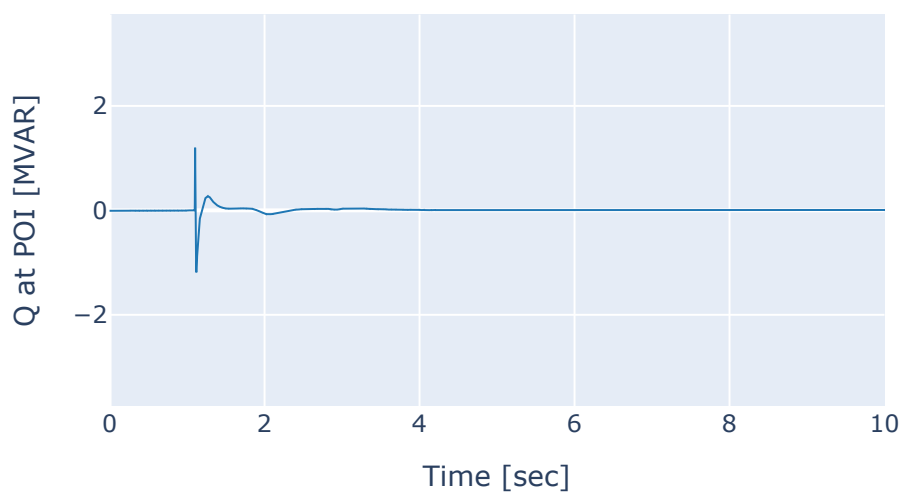
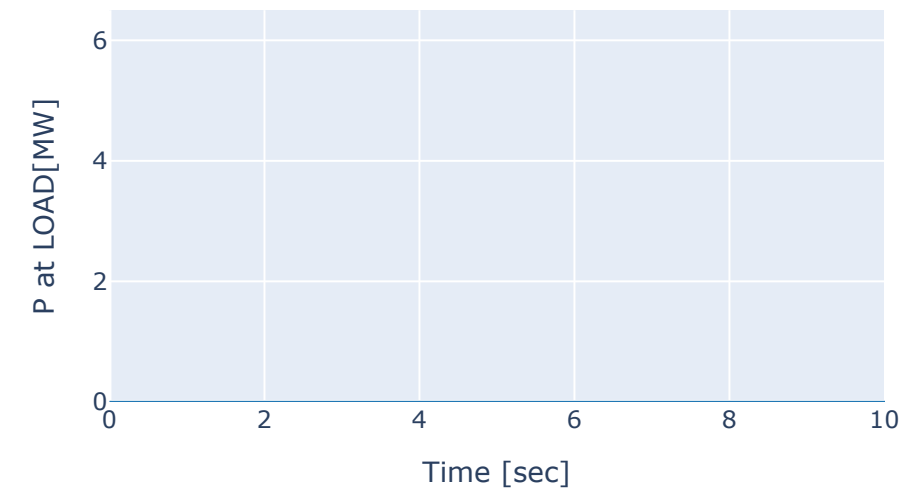
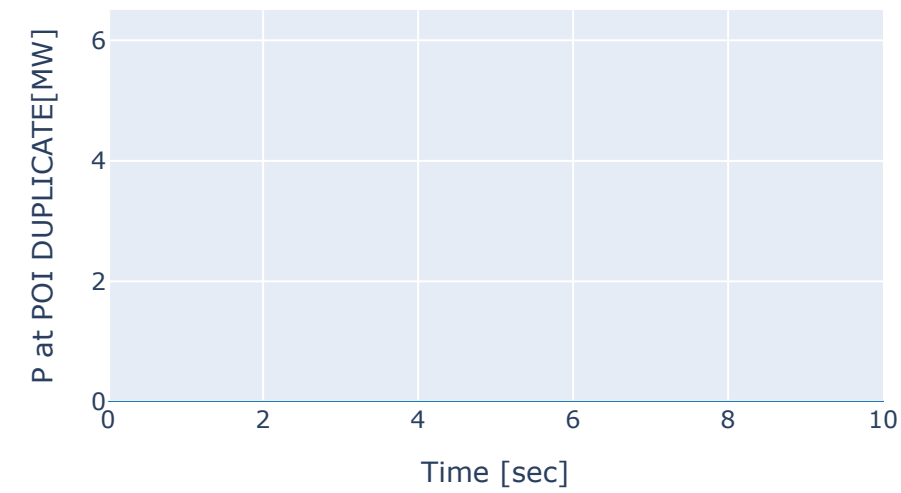
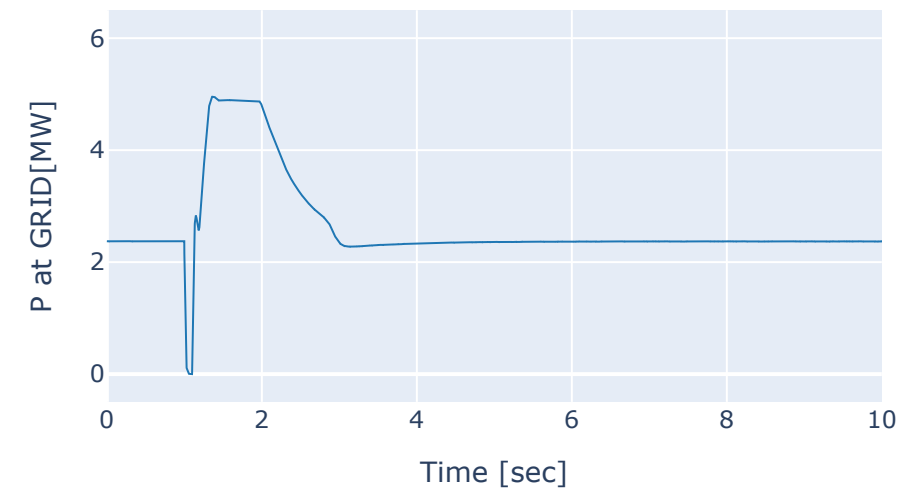
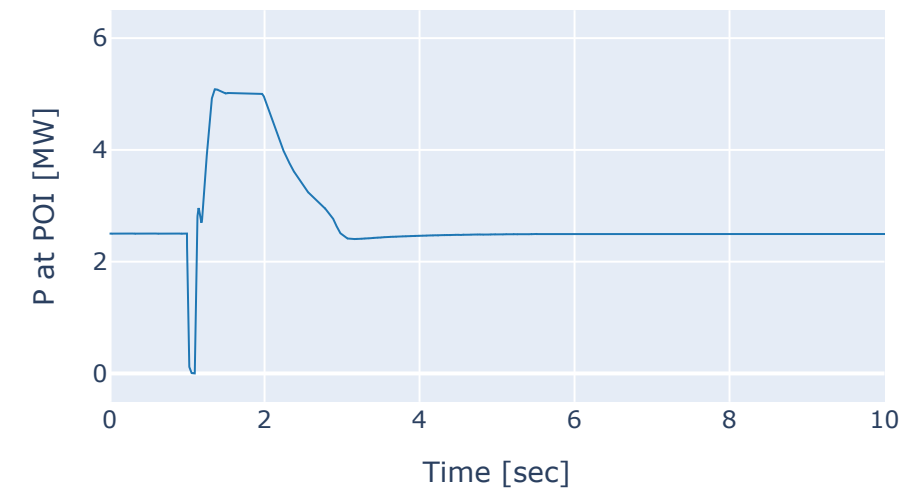
Fault Response Test (IEEE 2800.1 Clause 8.5) 3PG six cycles 100% Active Power Dispatch

— PSCAD (Jul1_BB)



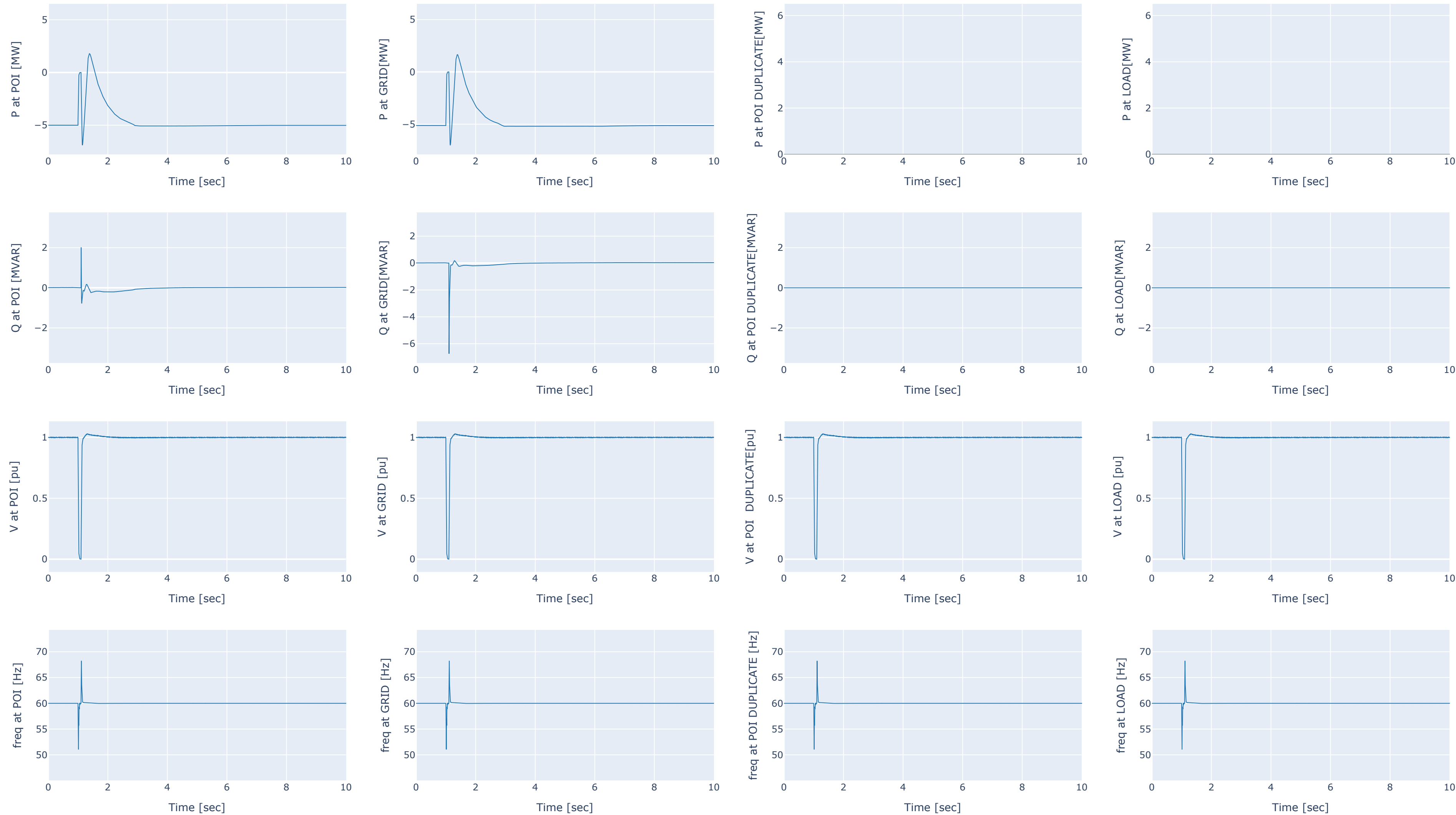
Fault Response Test (IEEE 2800.1 Clause 8.5) 3PG six cycles 50% Active Power Dispatch

— PSCAD (Jul1_BB)



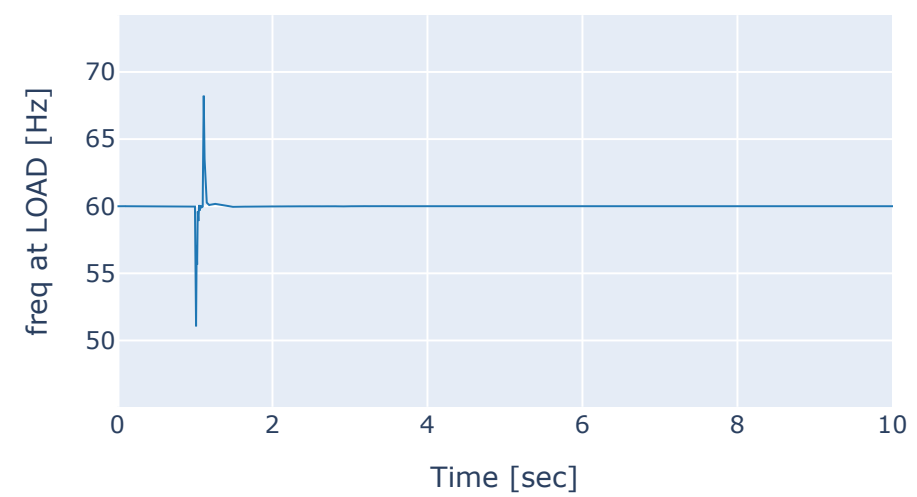
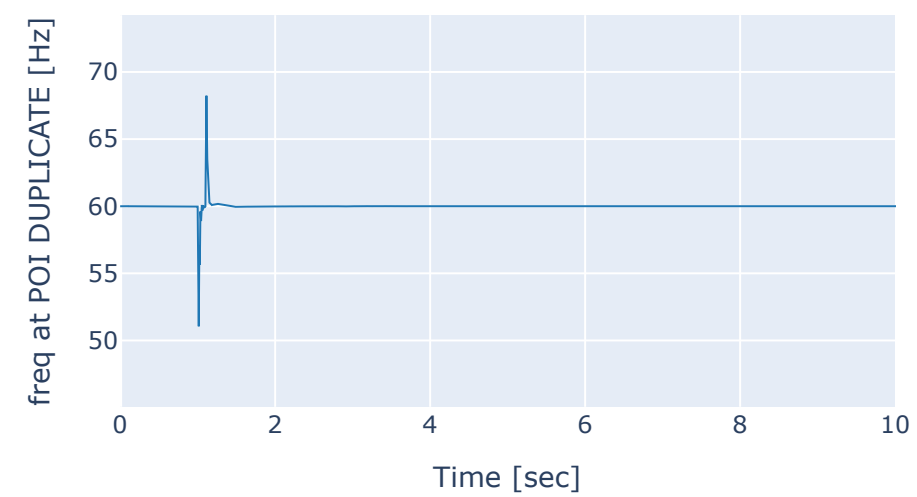
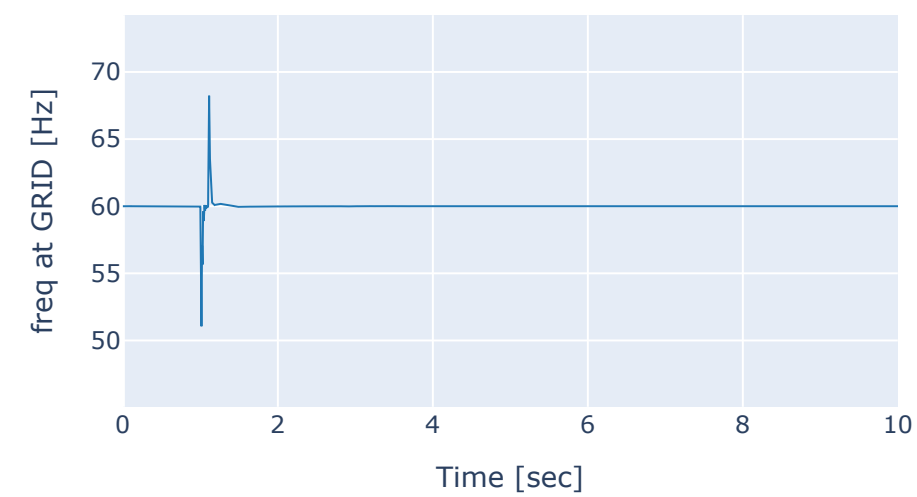
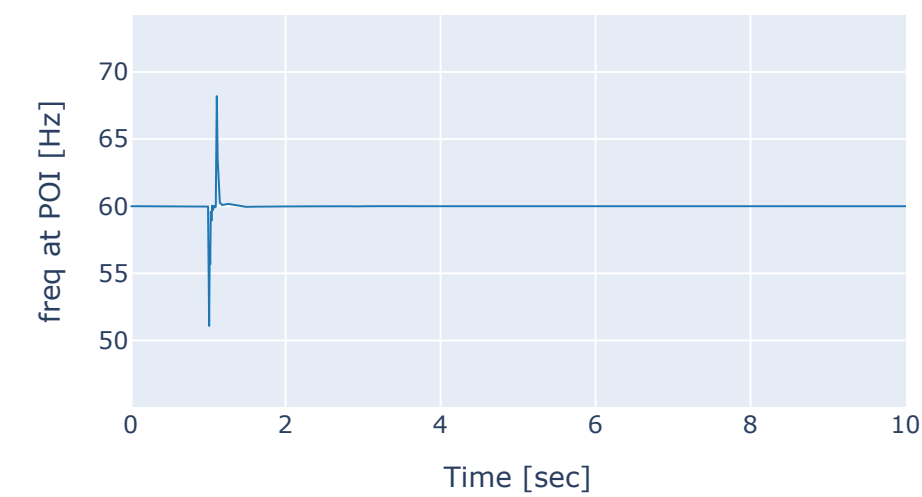
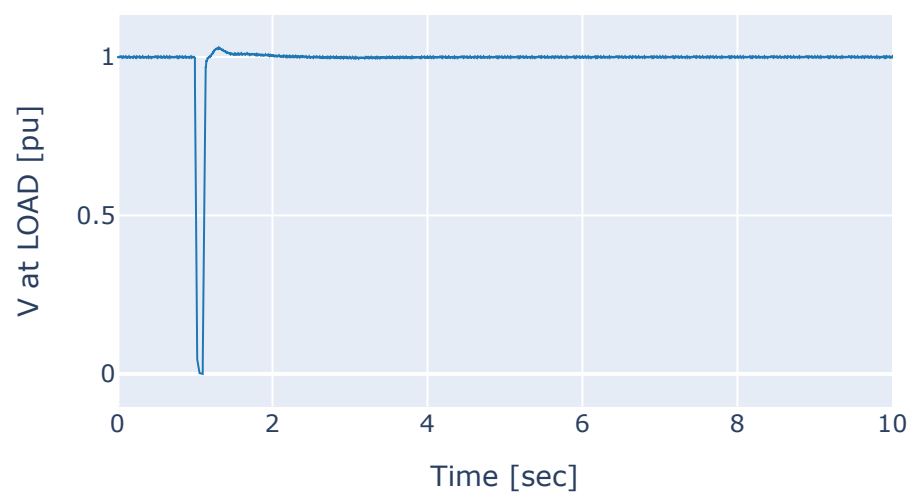
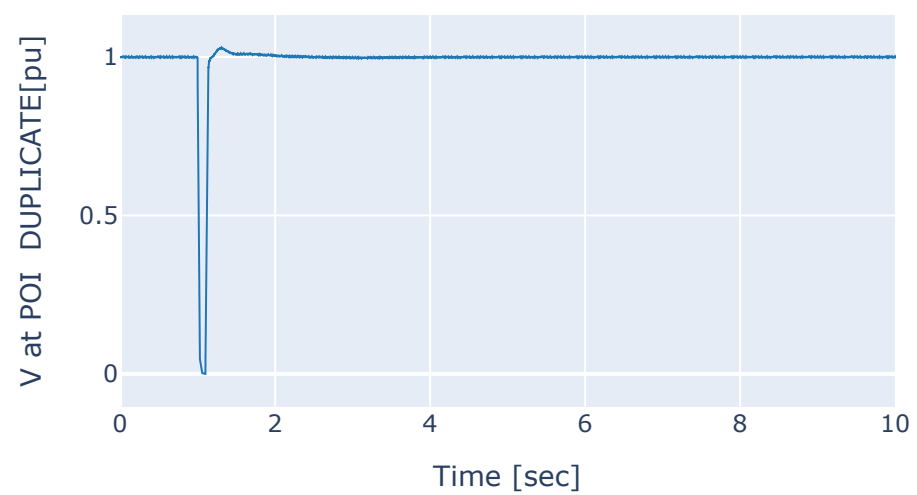
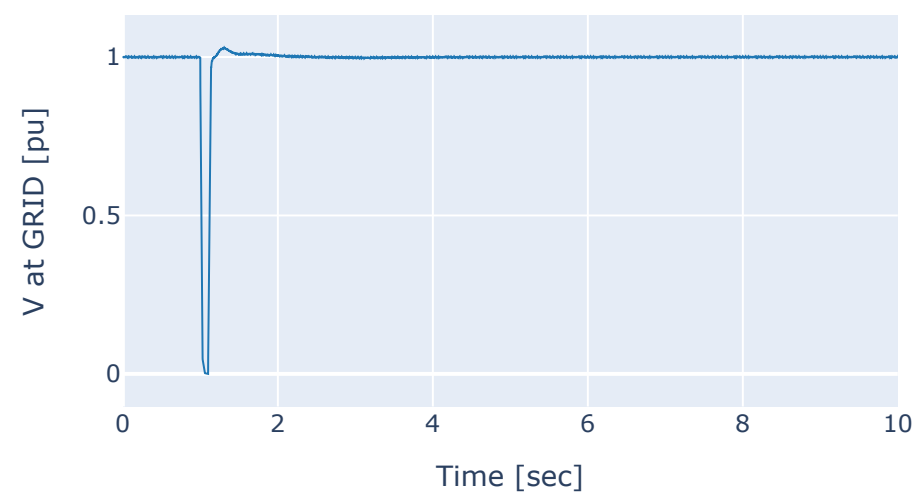
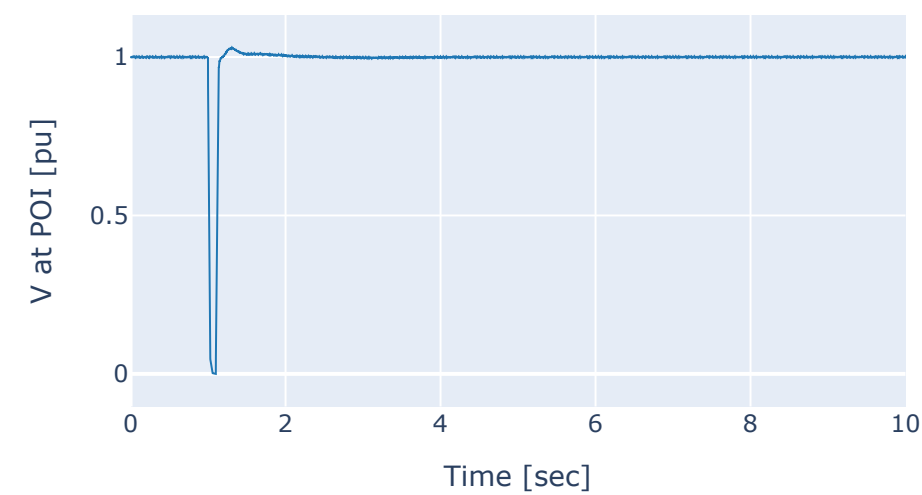
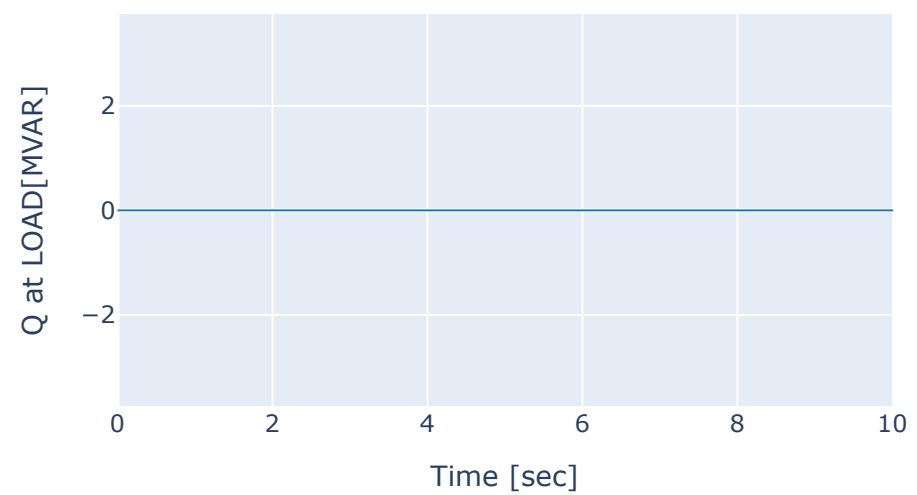
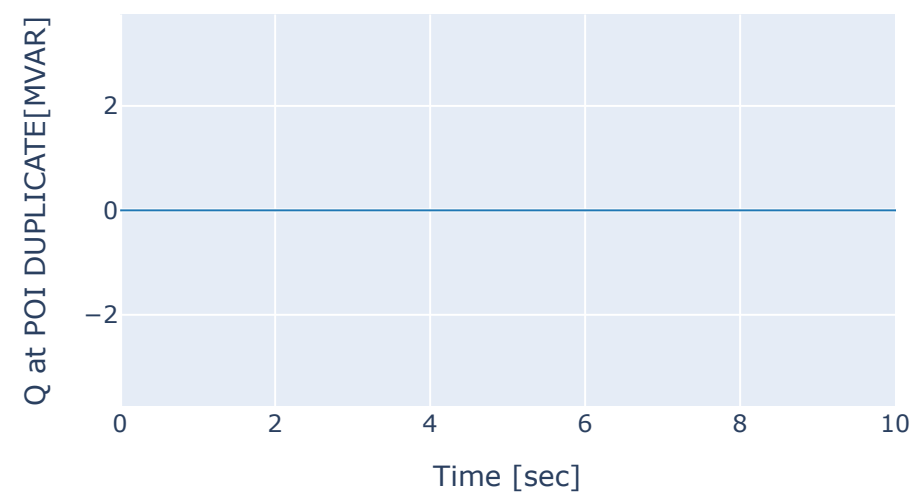
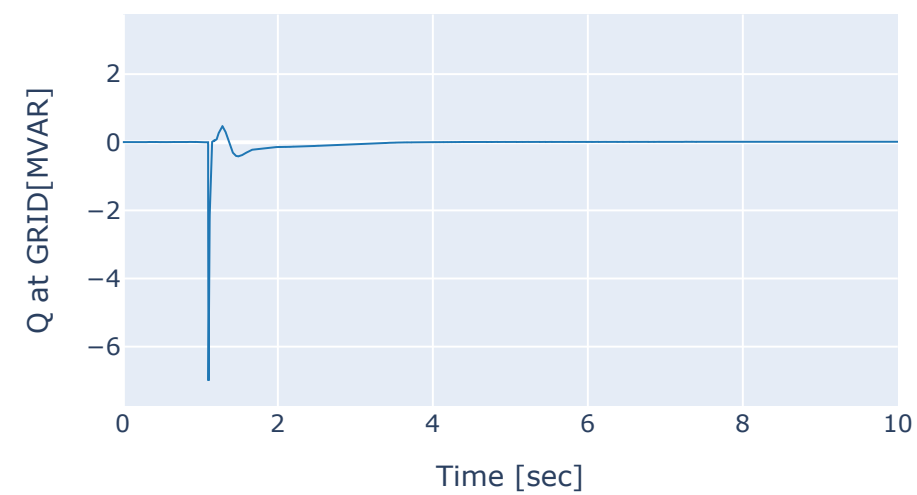
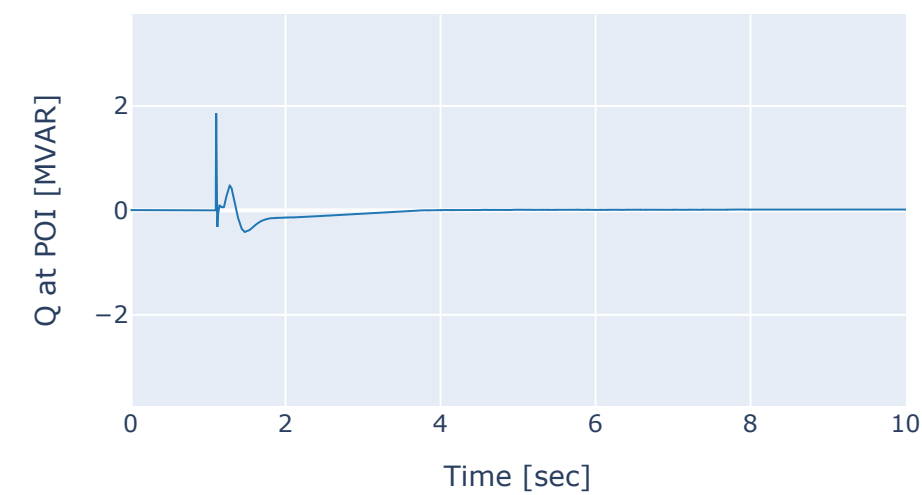
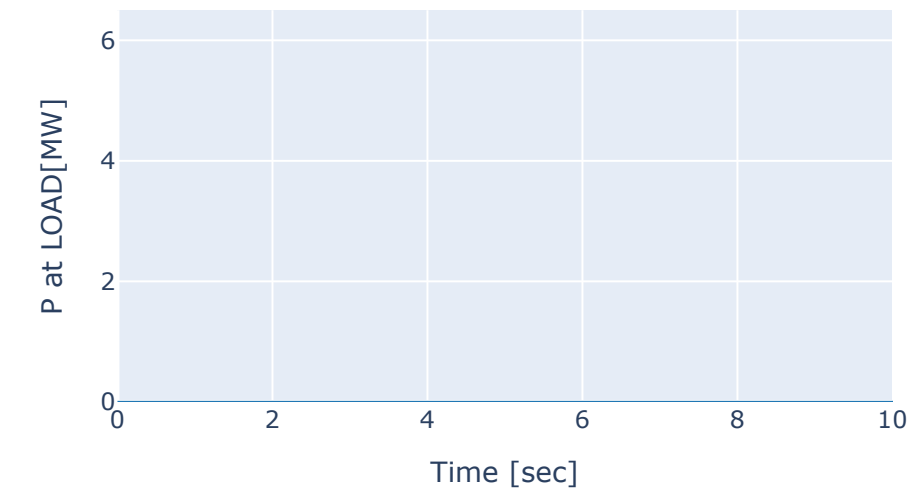
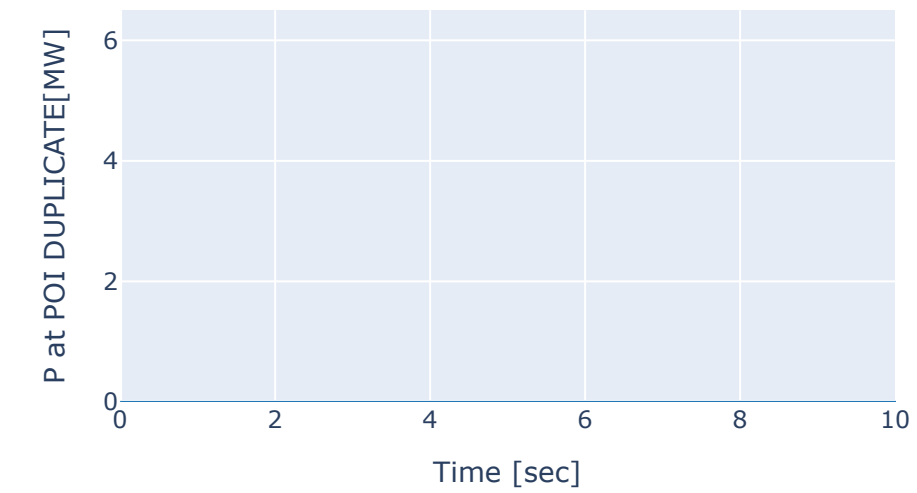
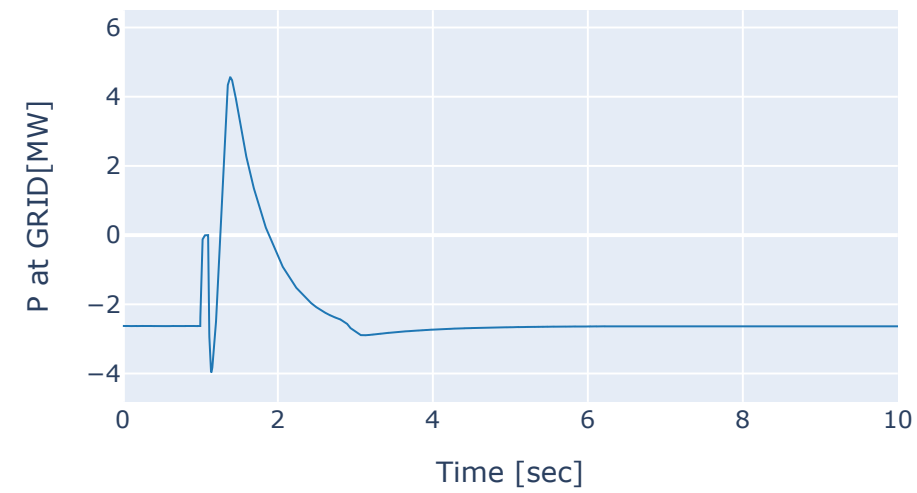
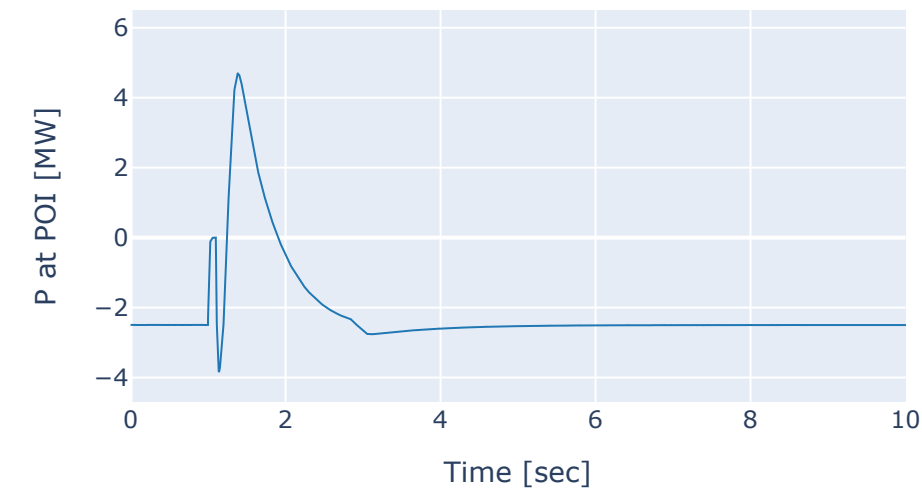
Fault Response Test (IEEE 2800.1 Clause 8.5) 3PG six cycles -100% Active Power Dispatch

PSCAD (Jul1_BB)



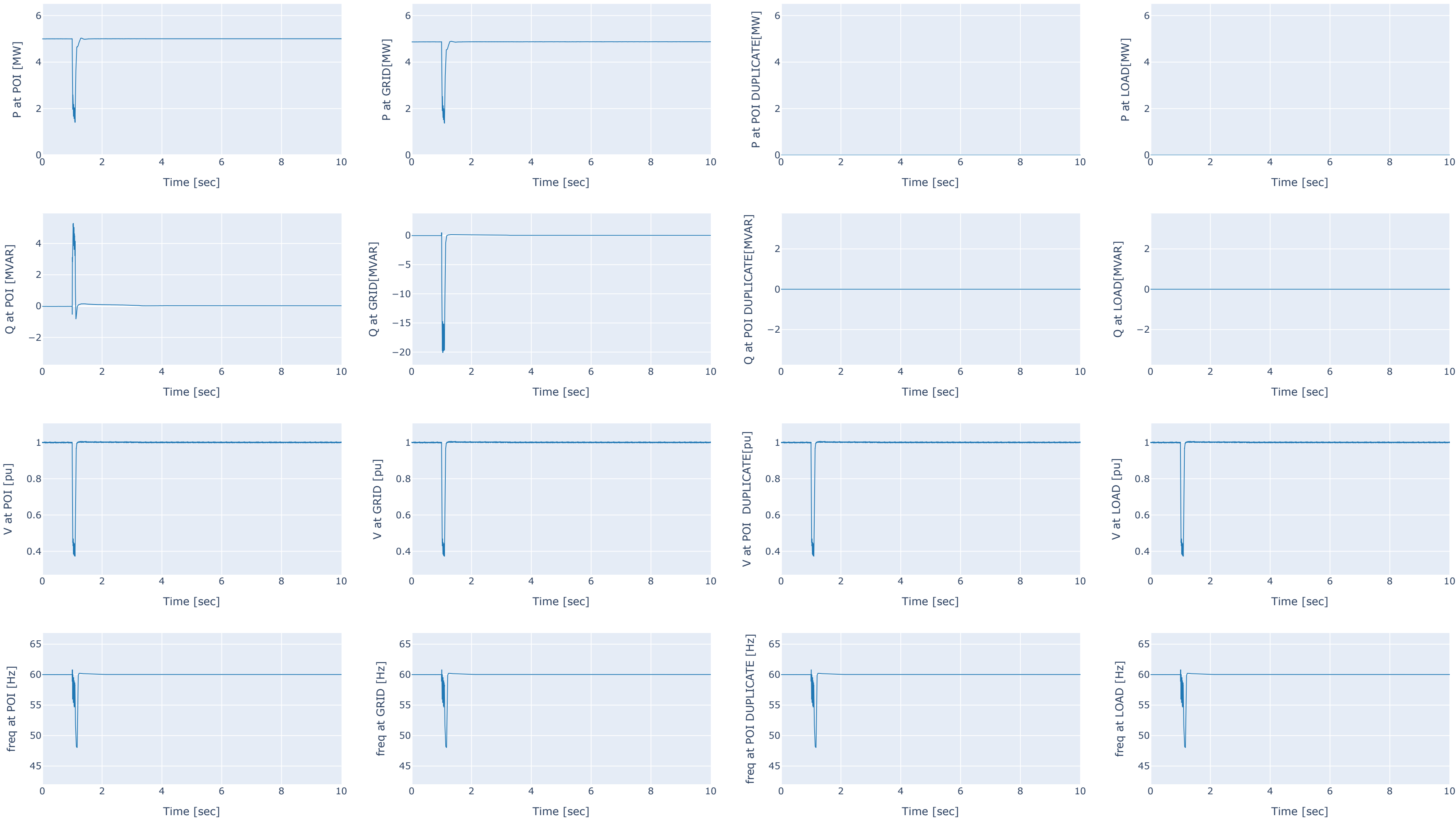
Fault Response Test (IEEE 2800.1 Clause 8.5) 3PG six cycles -50% Active Power Dispatch

PSCAD (Jul1_BB)



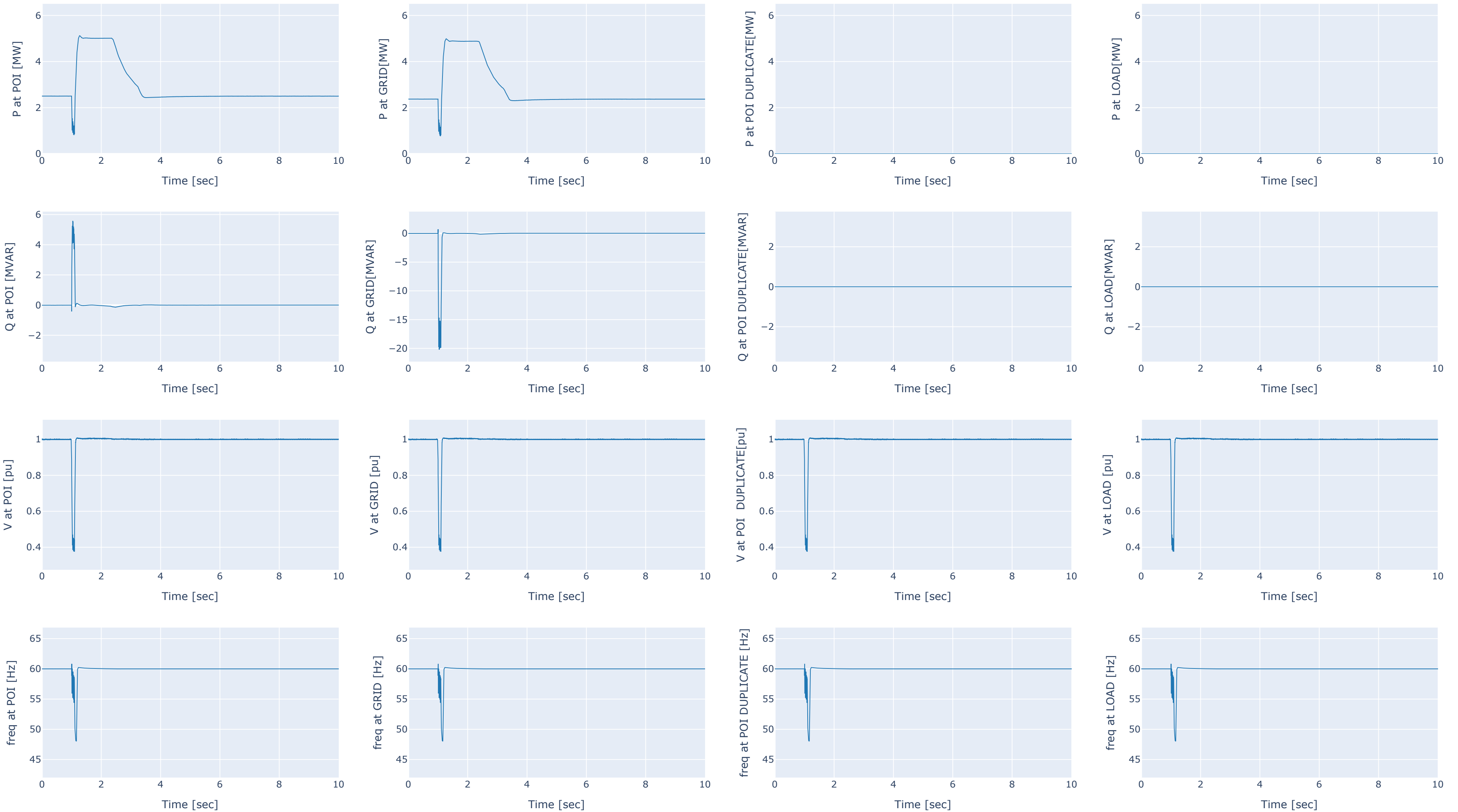
Fault Response Test (IEEE 2800.1 Clause 8.5) 2PG six cycles 100% Active Power Dispatch

PSCAD (Jul1_BB)



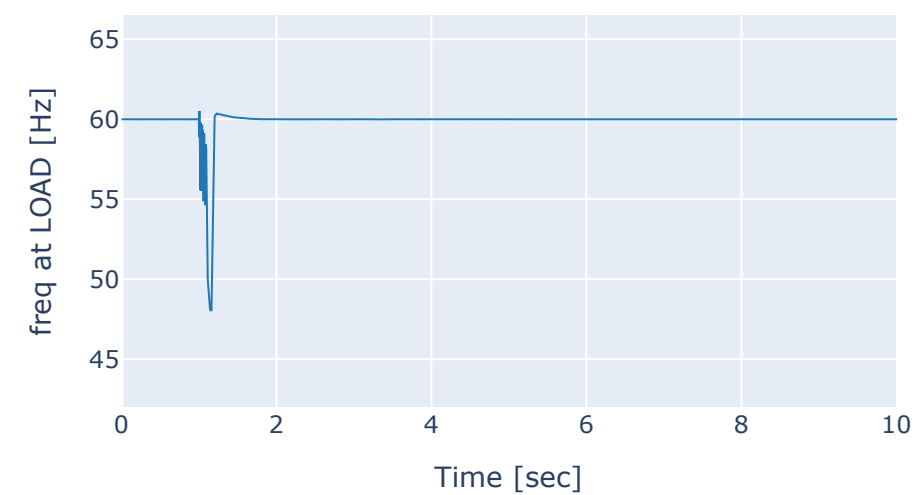
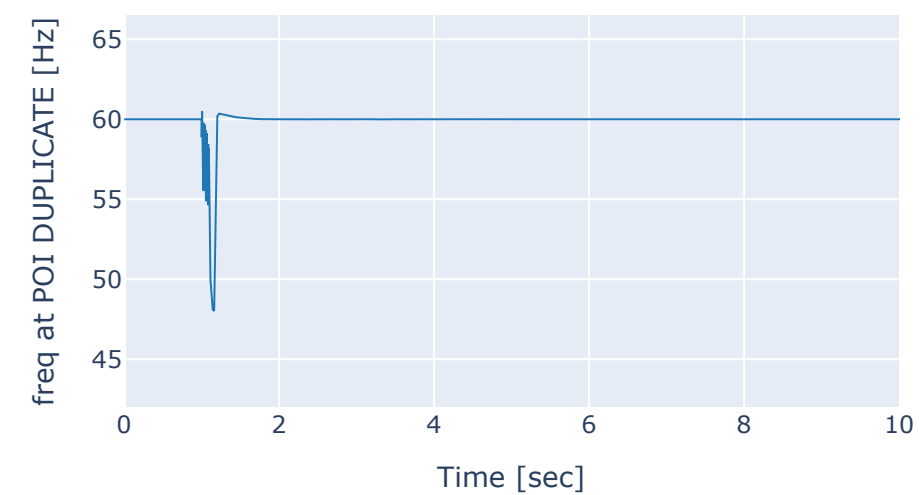
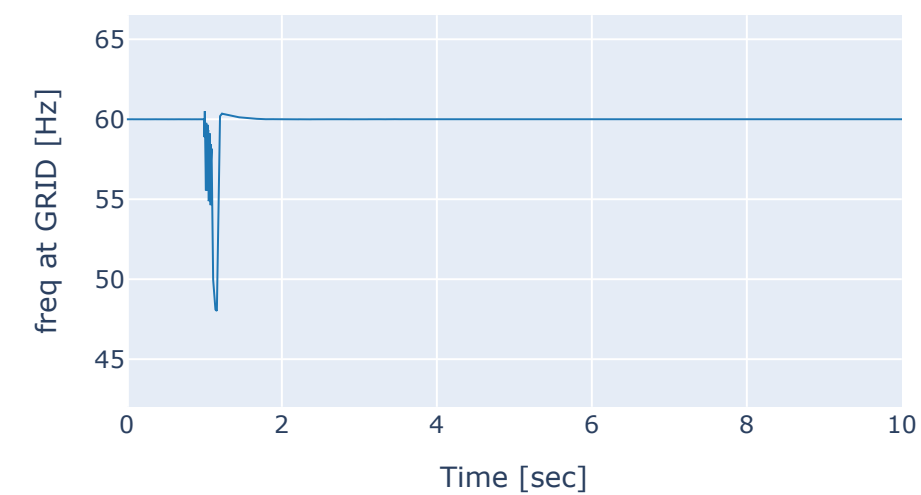
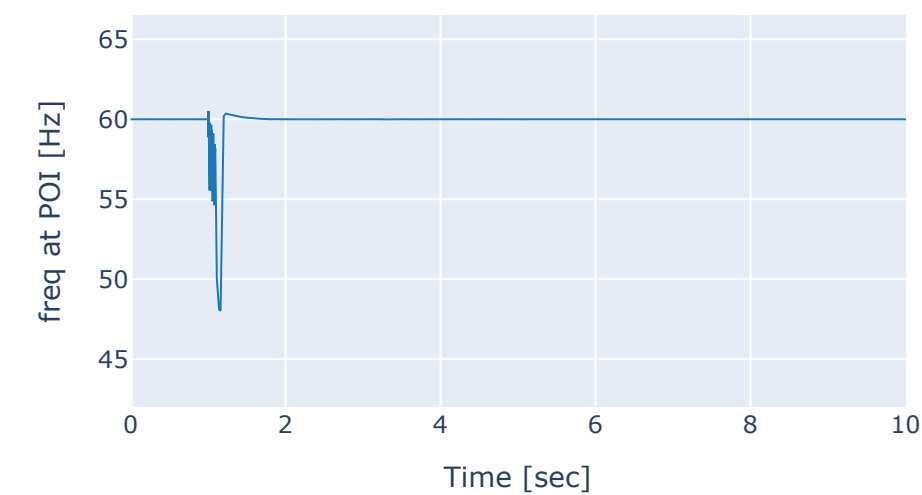
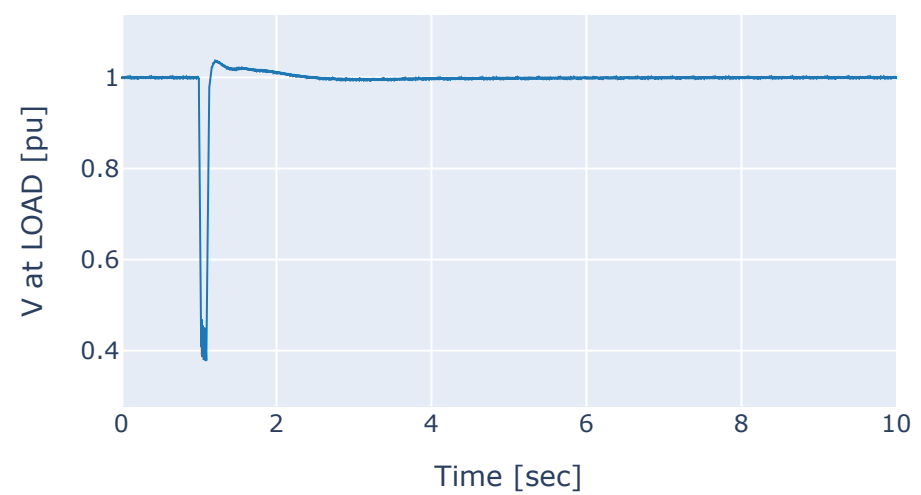
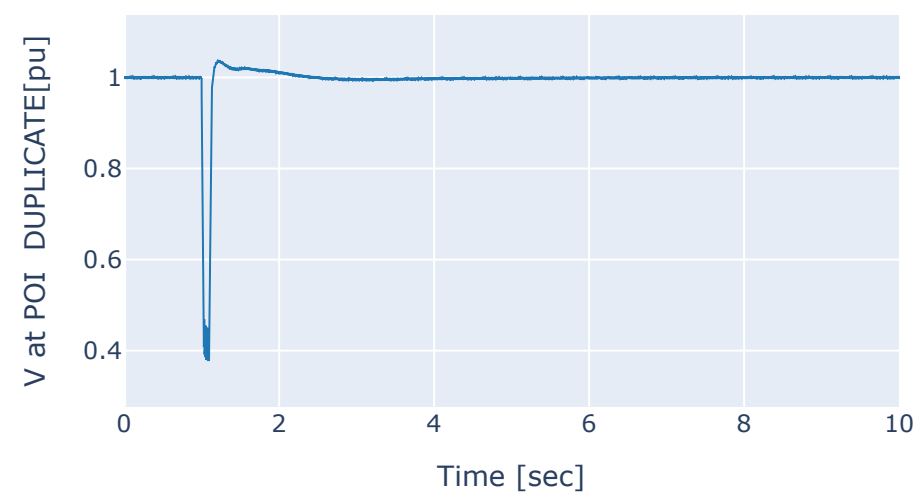
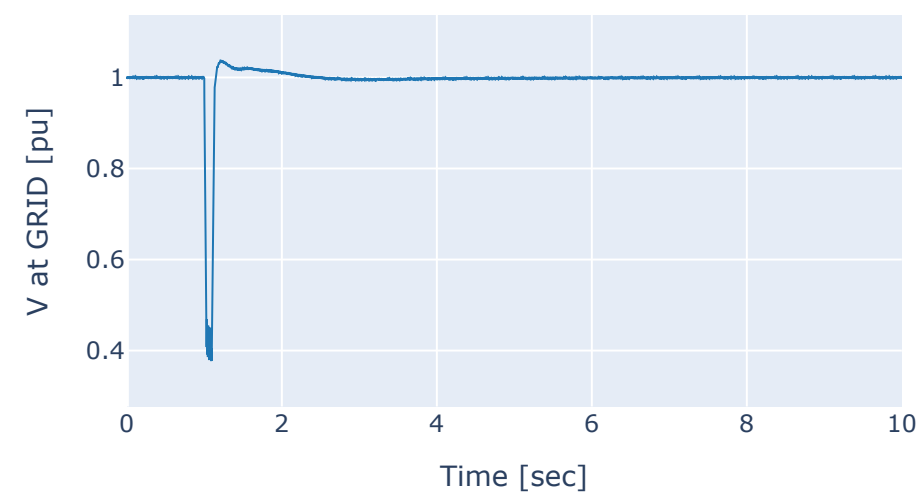
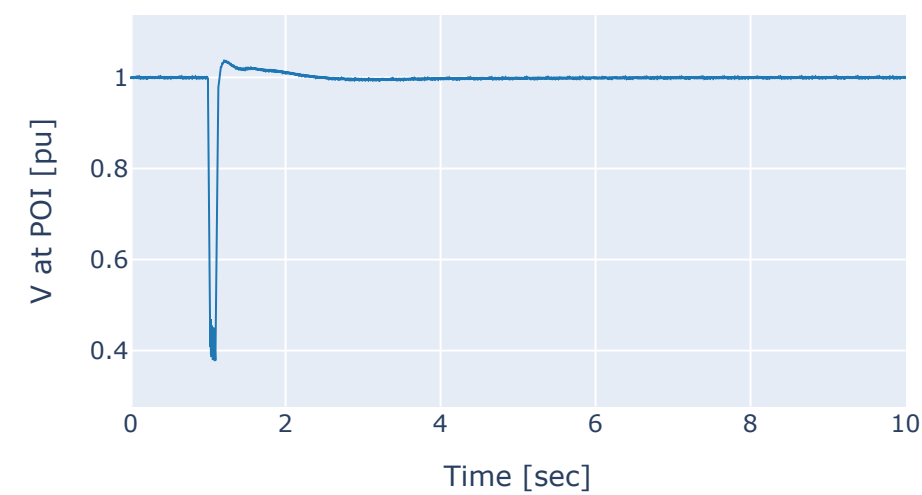
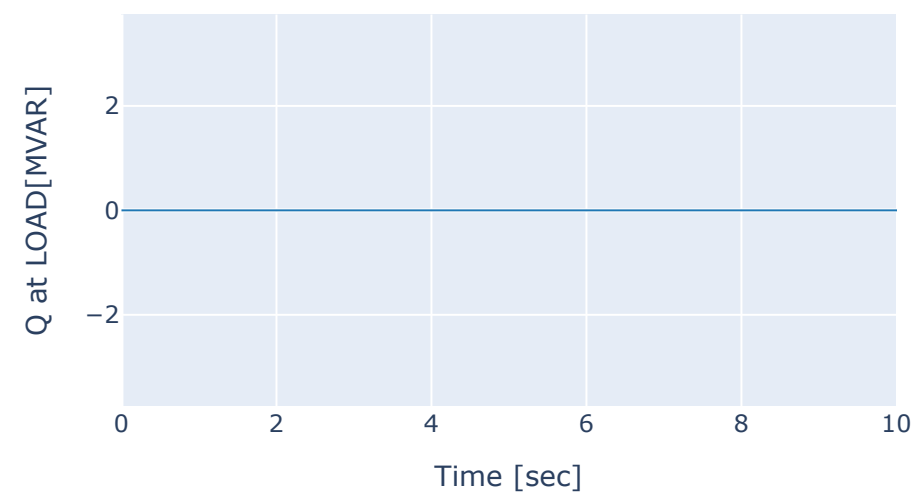
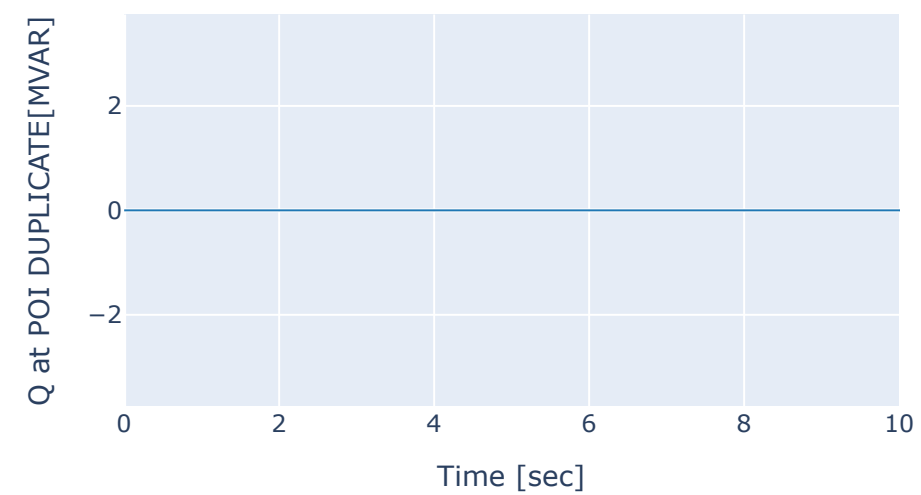
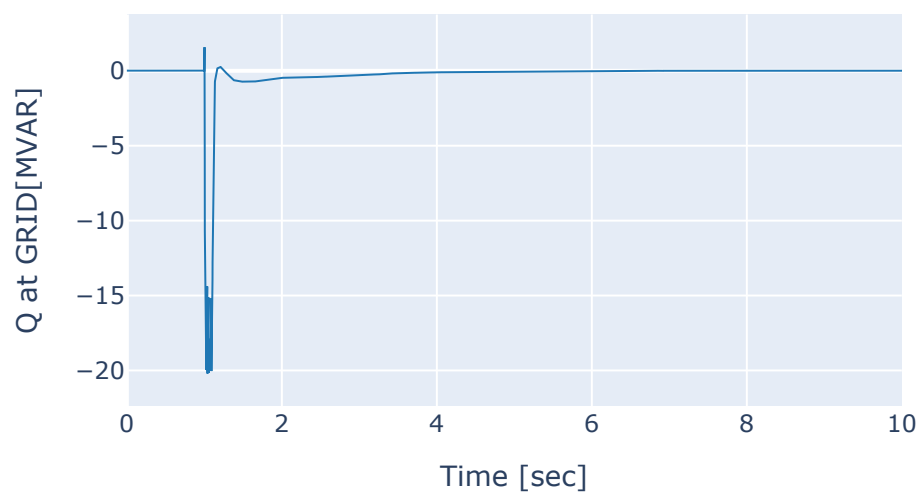
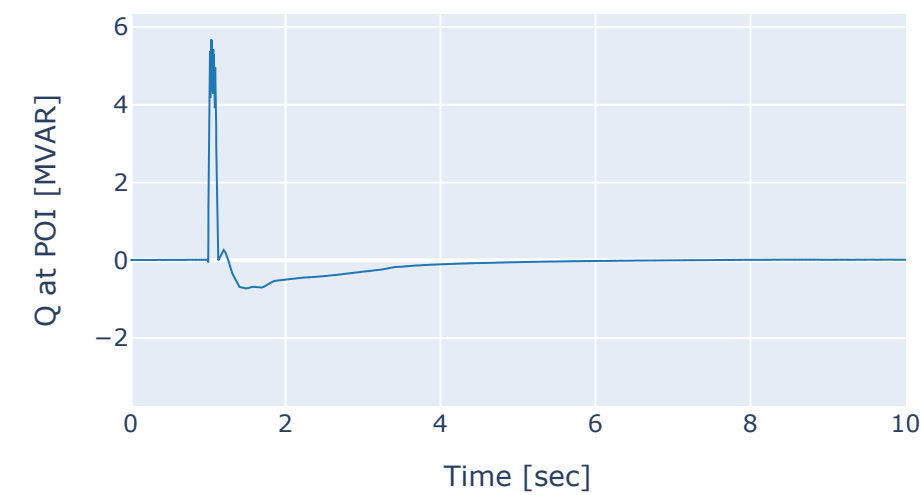
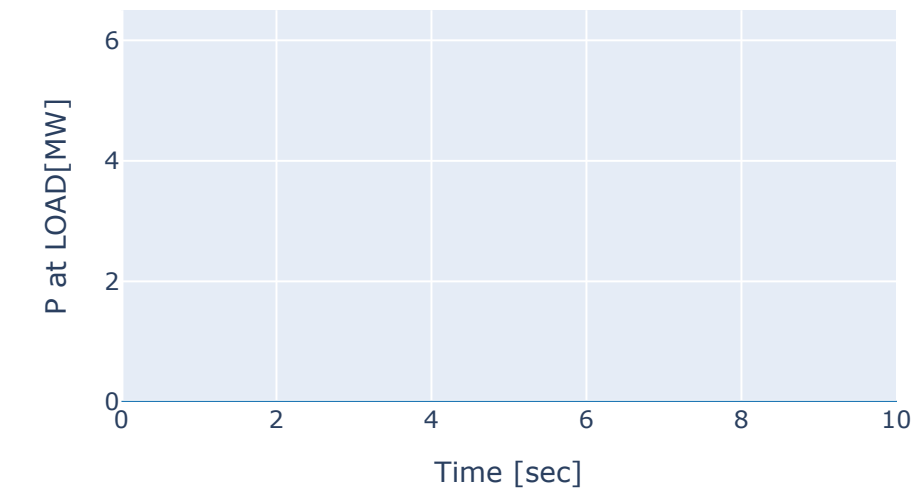
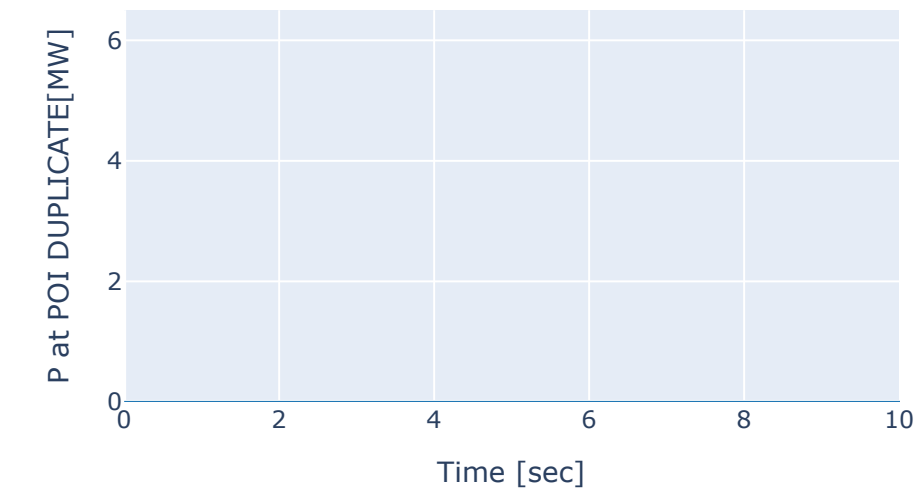
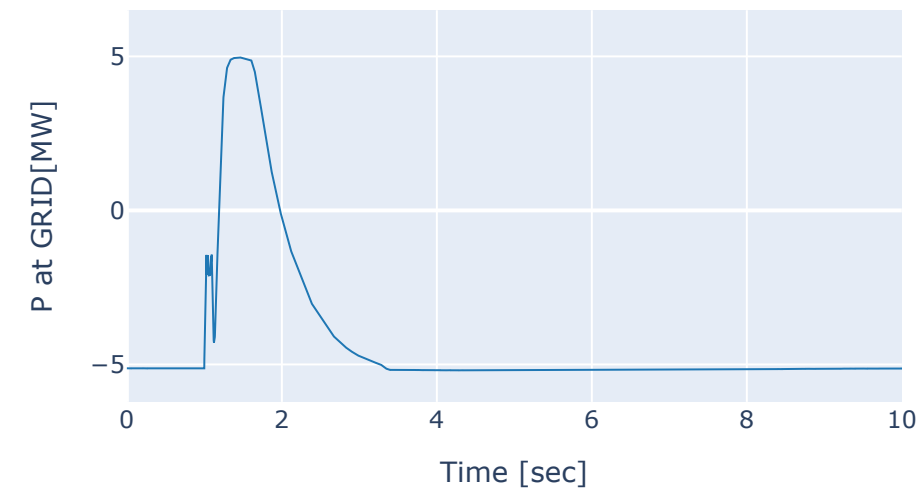
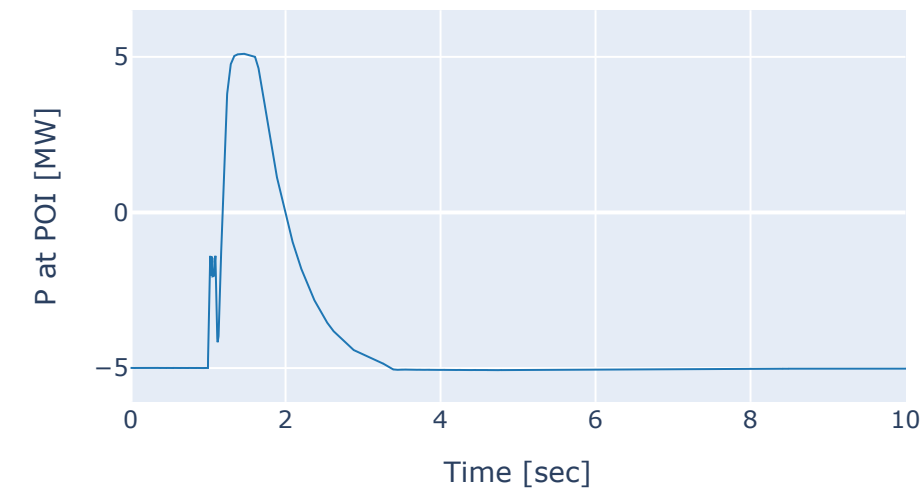
Fault Response Test (IEEE 2800.1 Clause 8.5) 2PG six cycles 50% Active Power Dispatch

PSCAD (Jul1_BB)



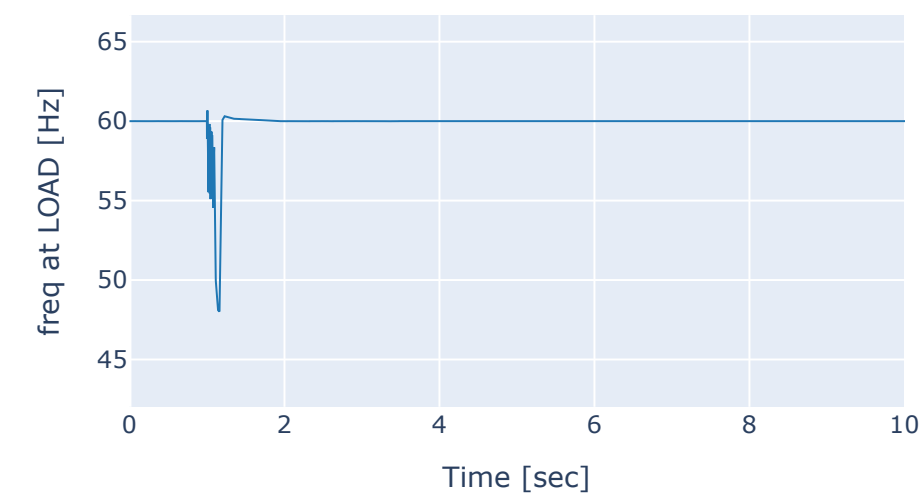
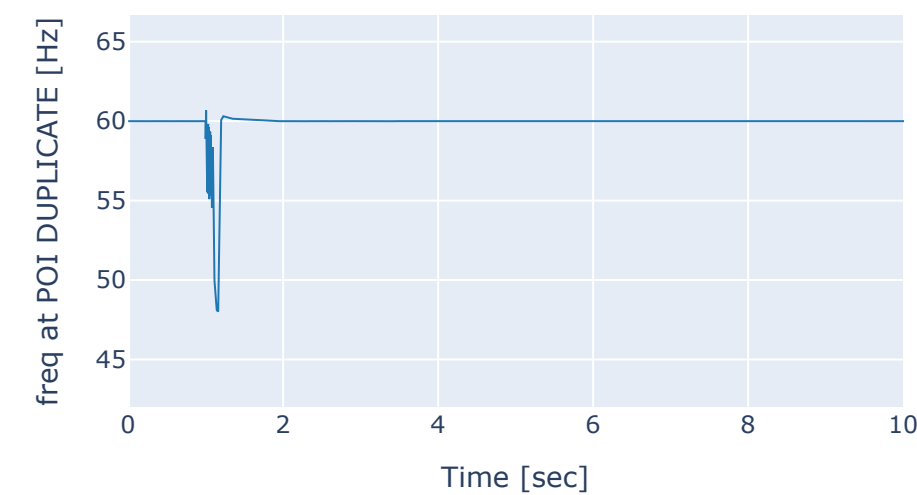
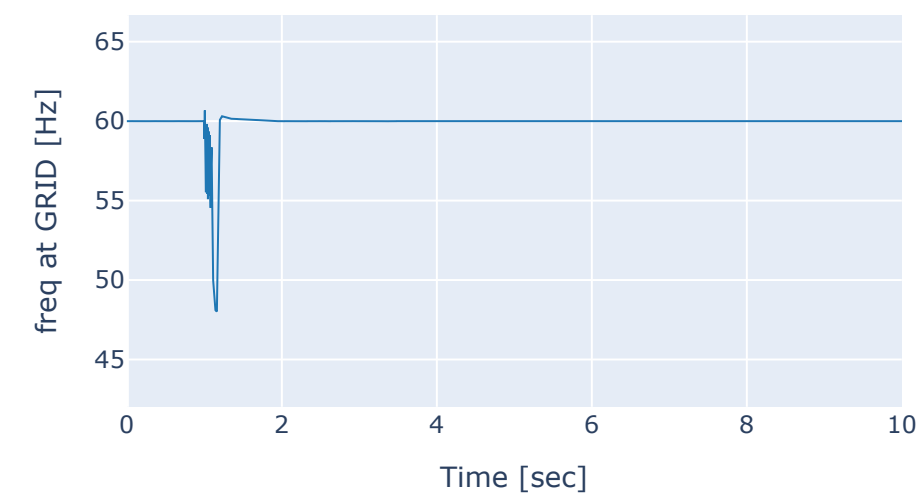
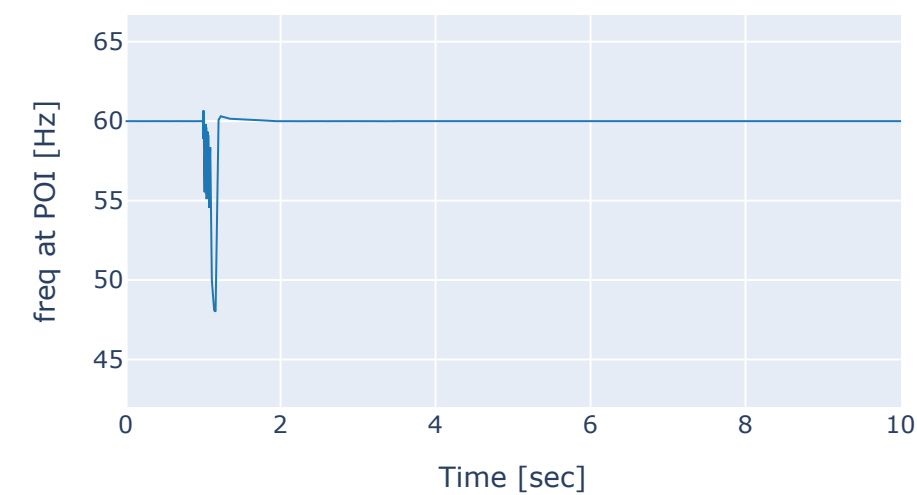
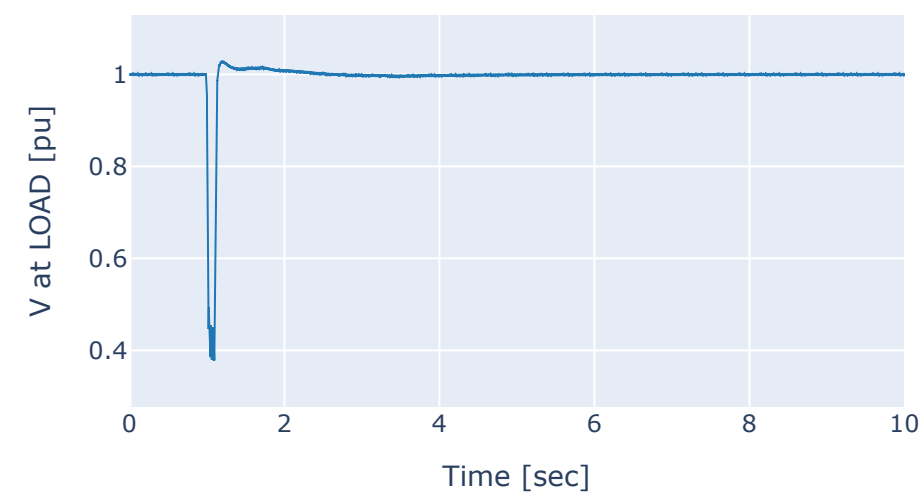
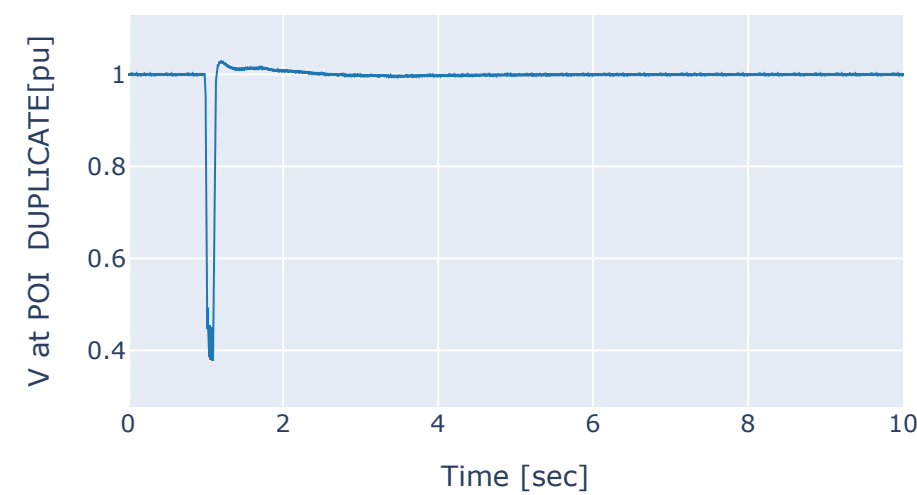
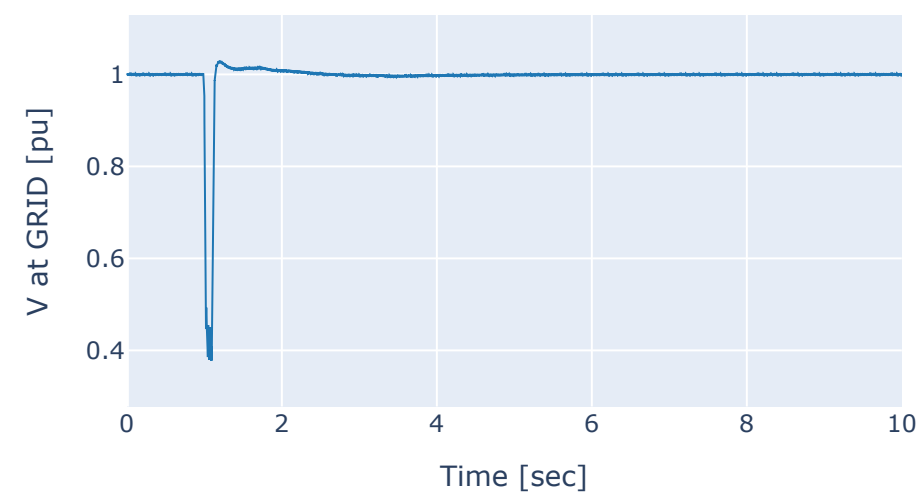
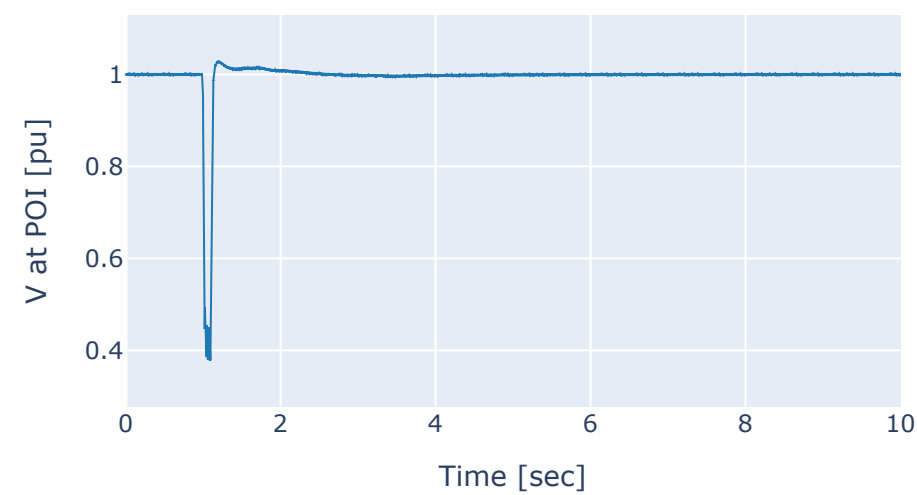
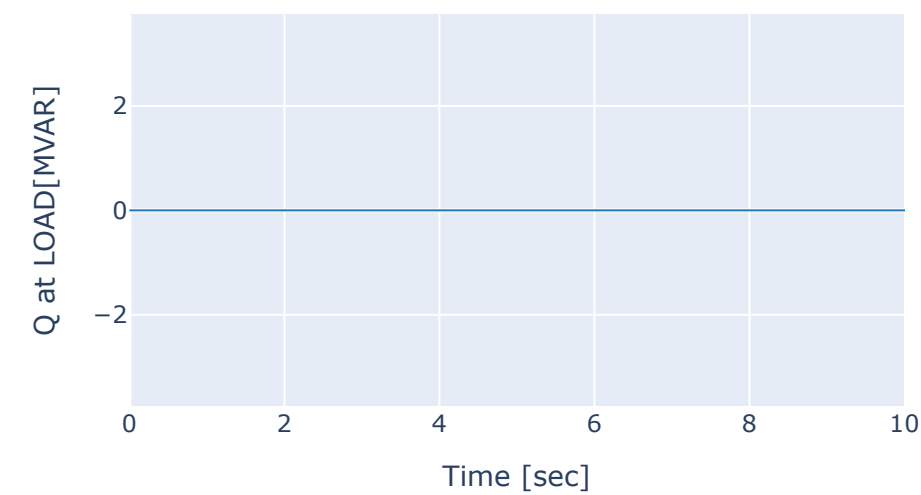
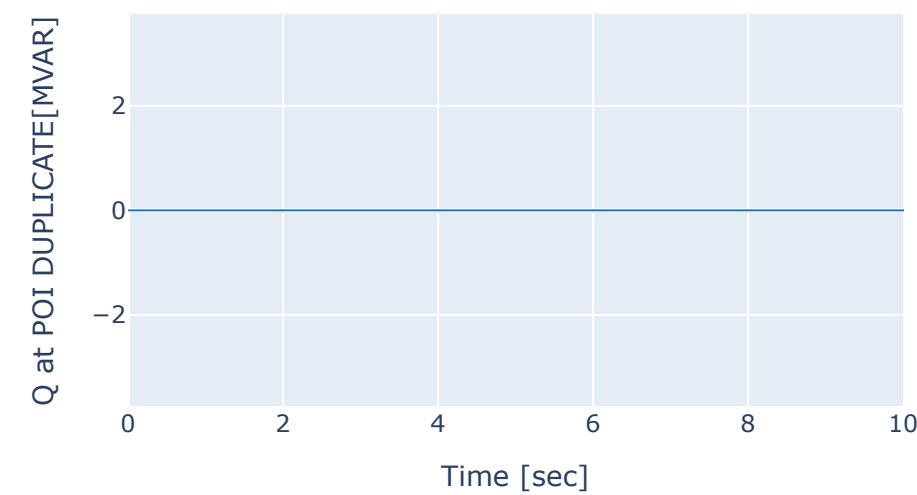
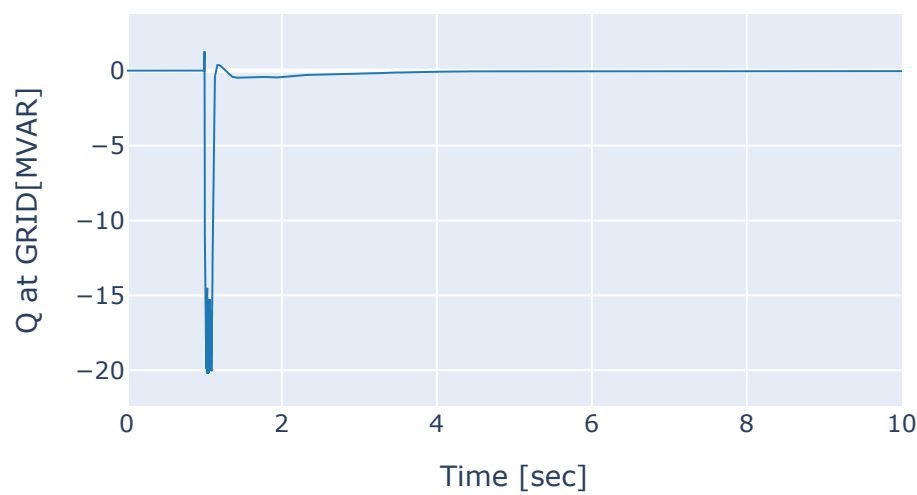
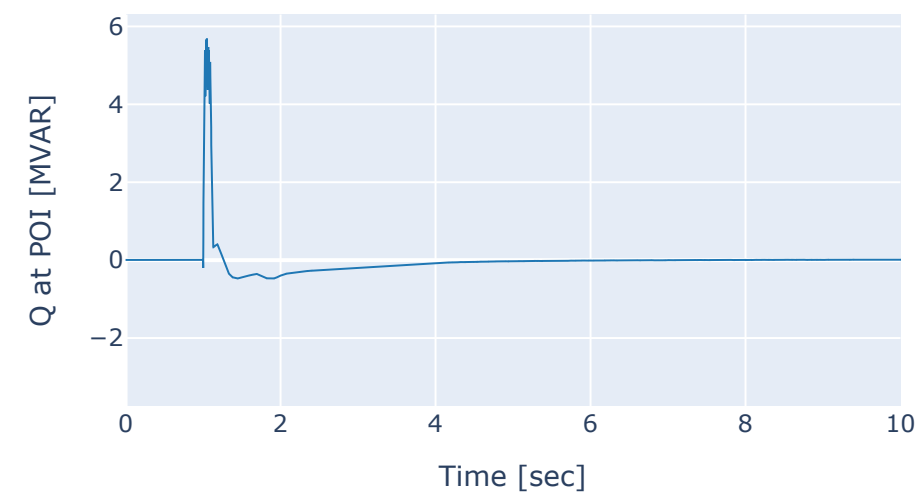
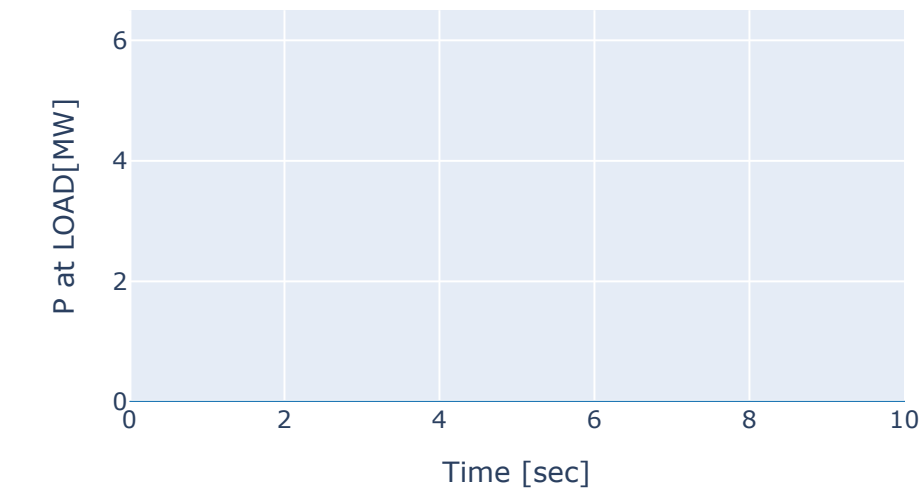
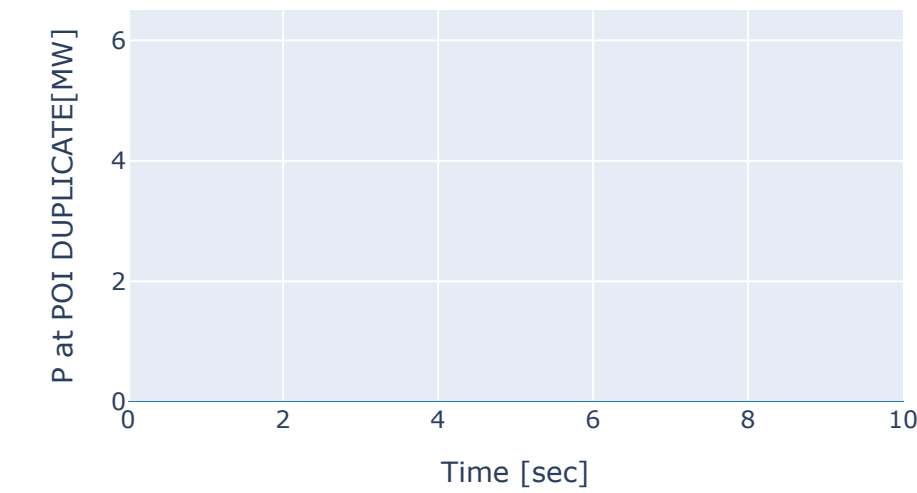
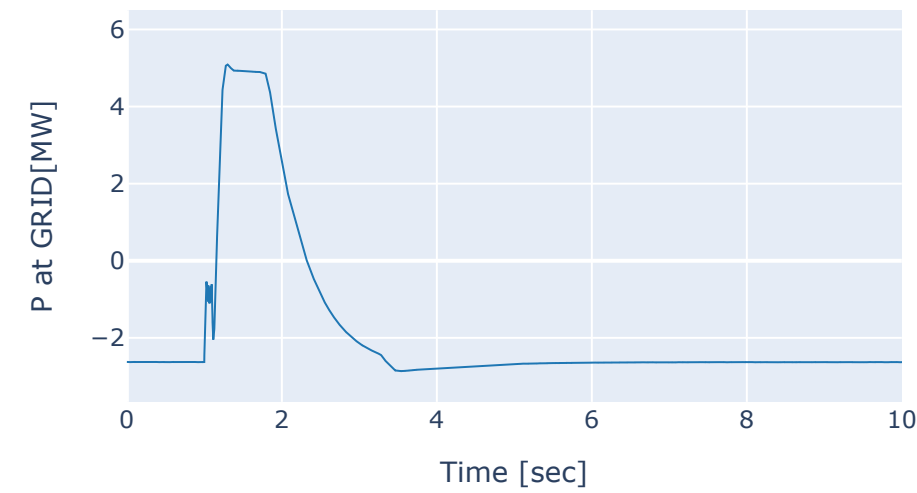
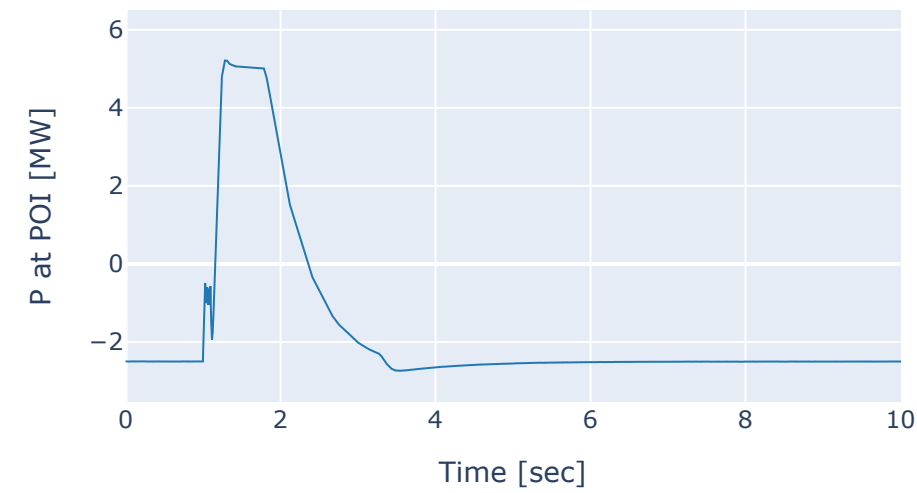
Fault Response Test (IEEE 2800.1 Clause 8.5) 2PG six cycles -100% Active Power Dispatch

— PSCAD (Jul1_BB)



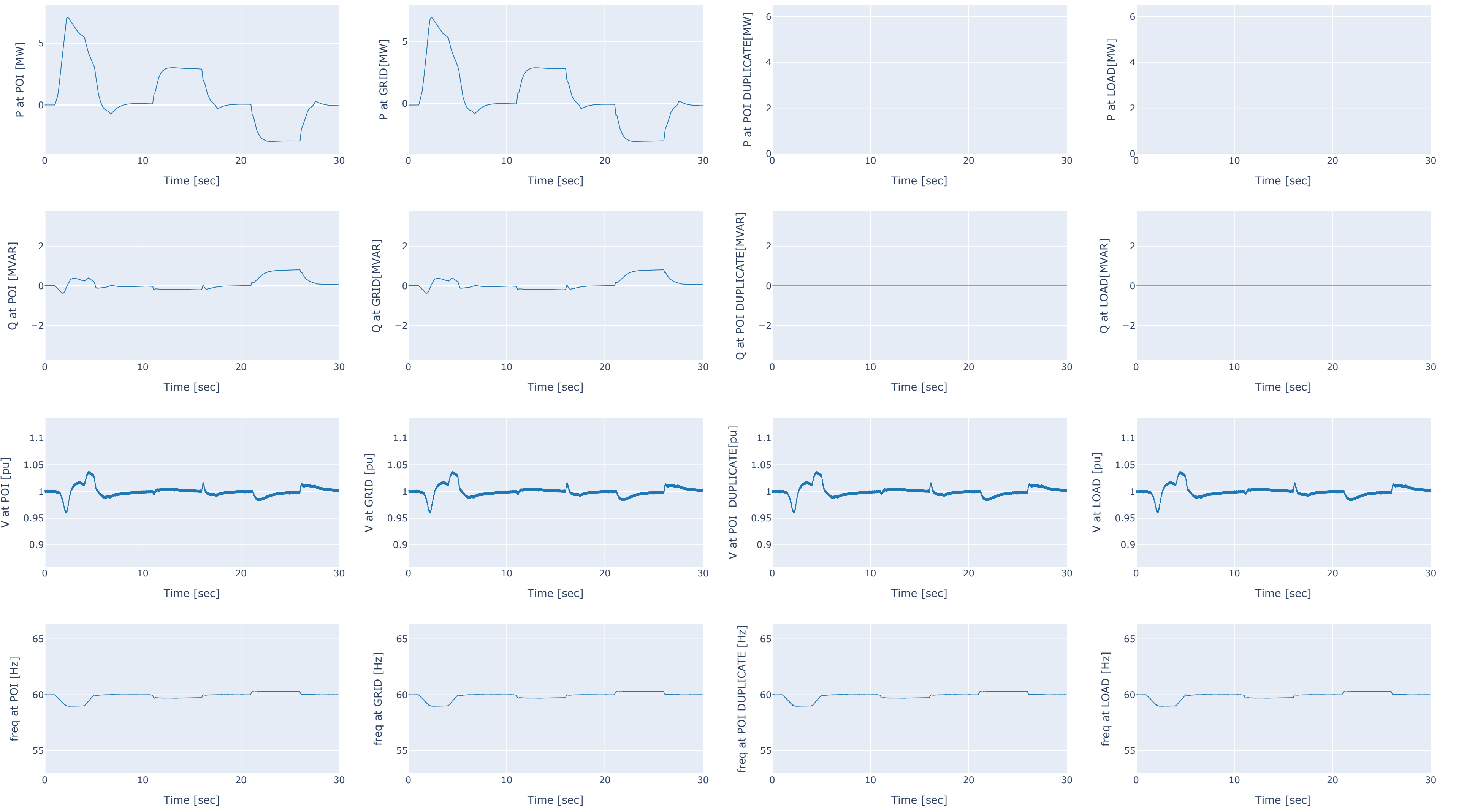
Fault Response Test (IEEE 2800.1 Clause 8.5) 2PG six cycles -50% Active Power Dispatch

— PSCAD (Jul1_BB)



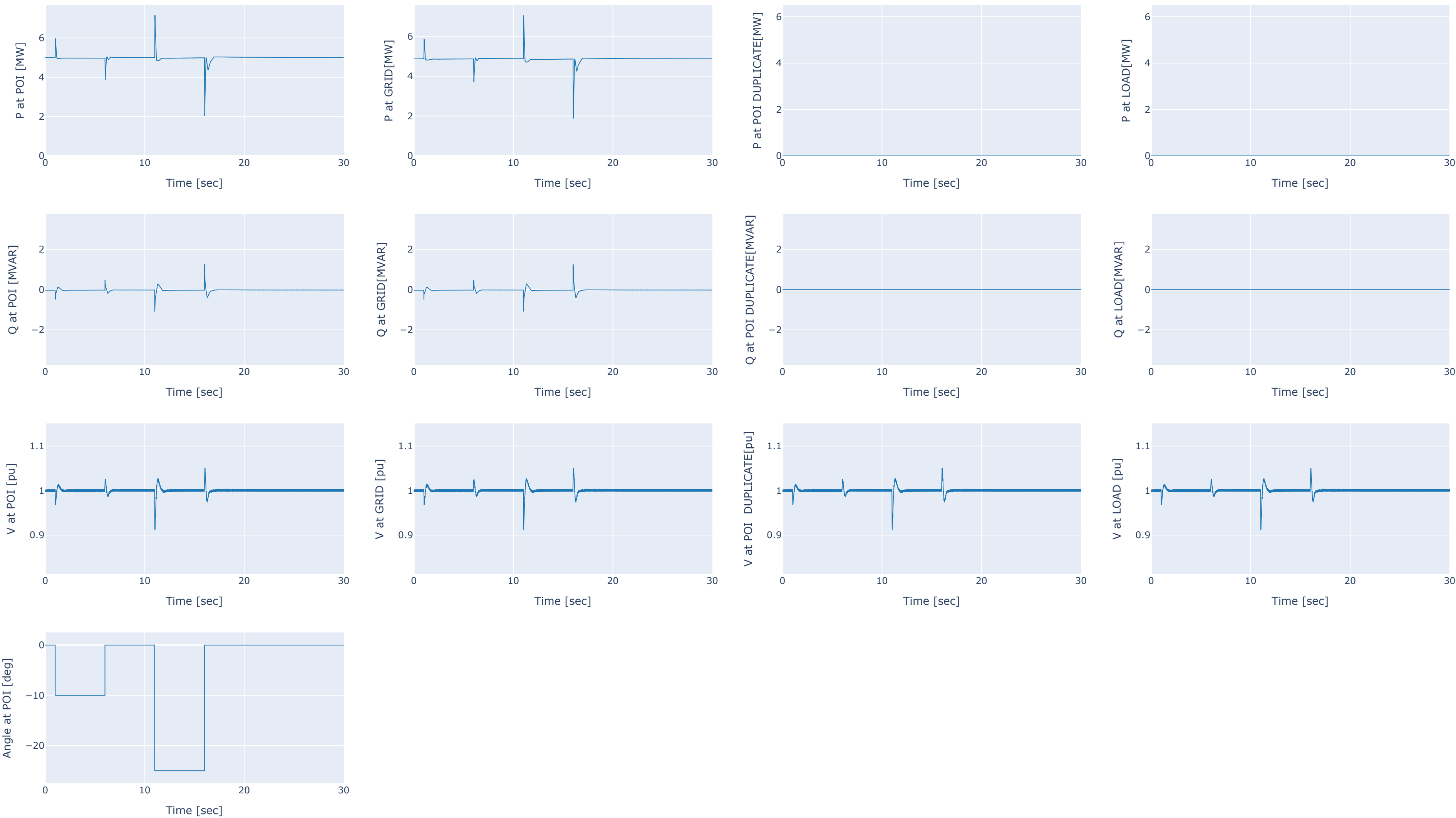
Response to Frequency Change Test (IEEE 2800.1 Clause 8.6)

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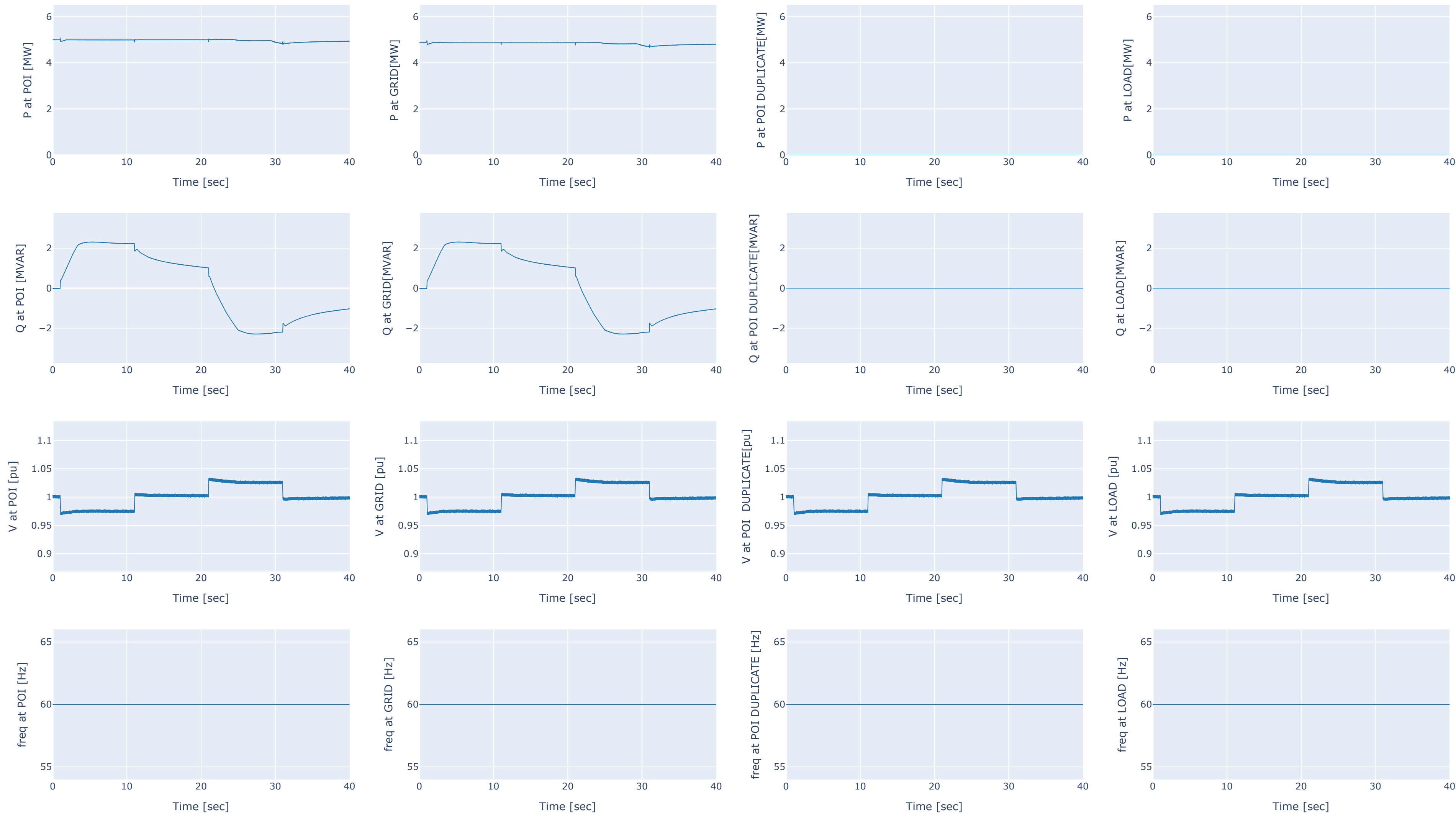
Phase Angle Jump Test (IEEE 2800.1 Clause 8.7)

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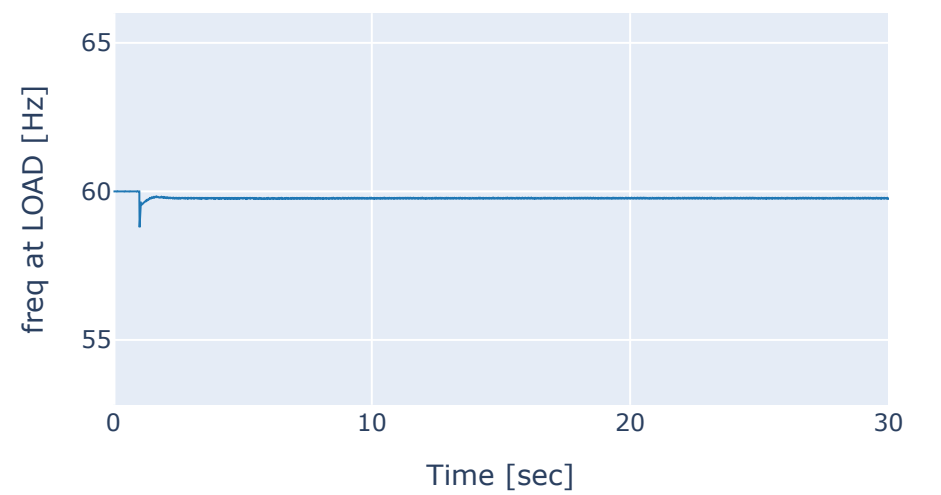
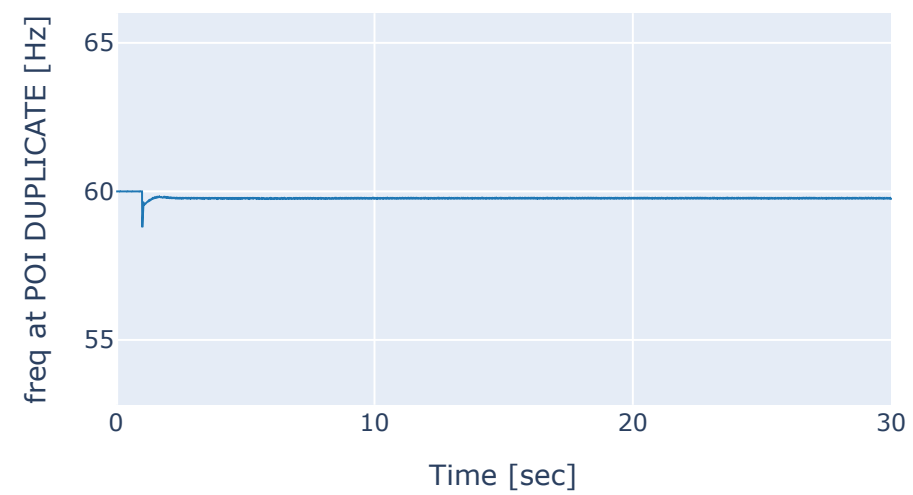
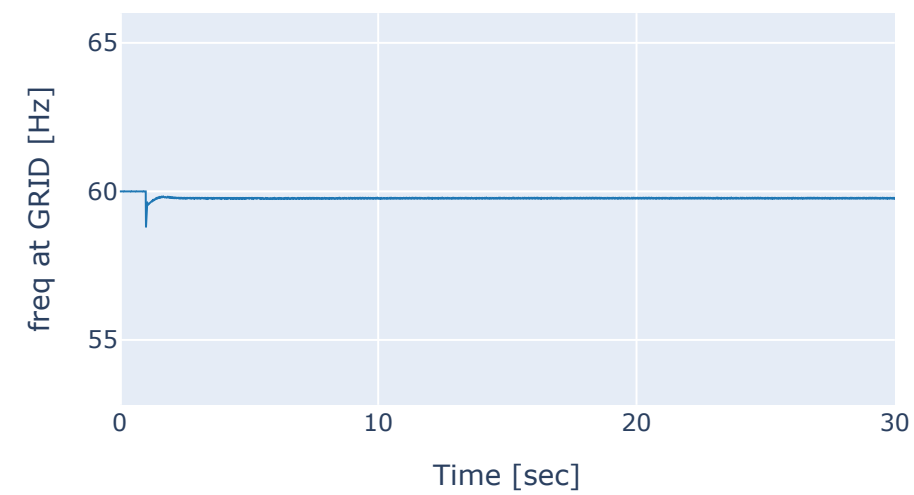
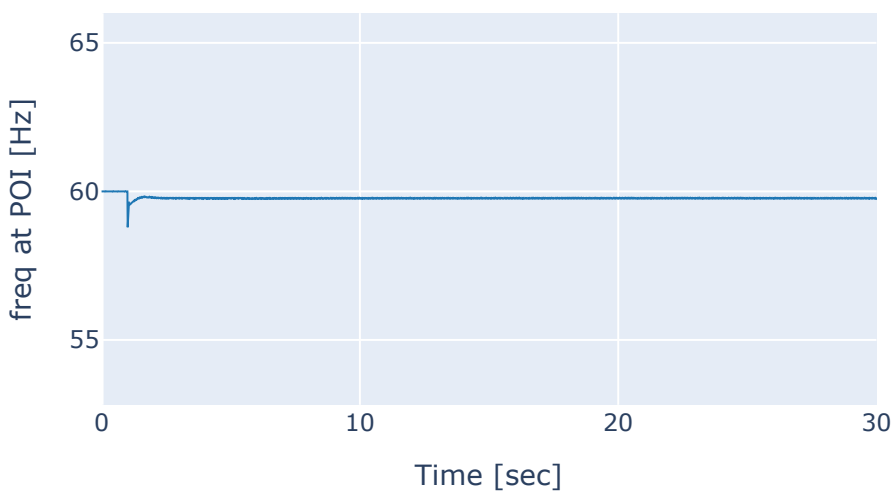
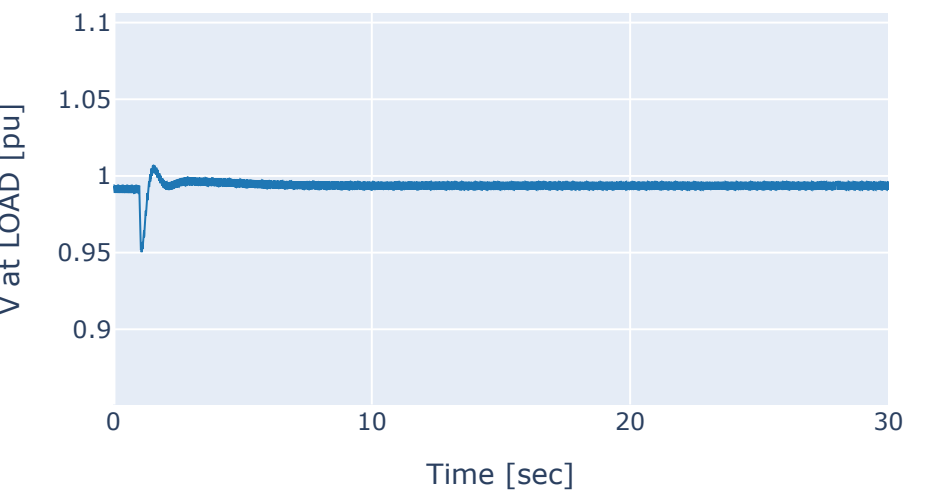
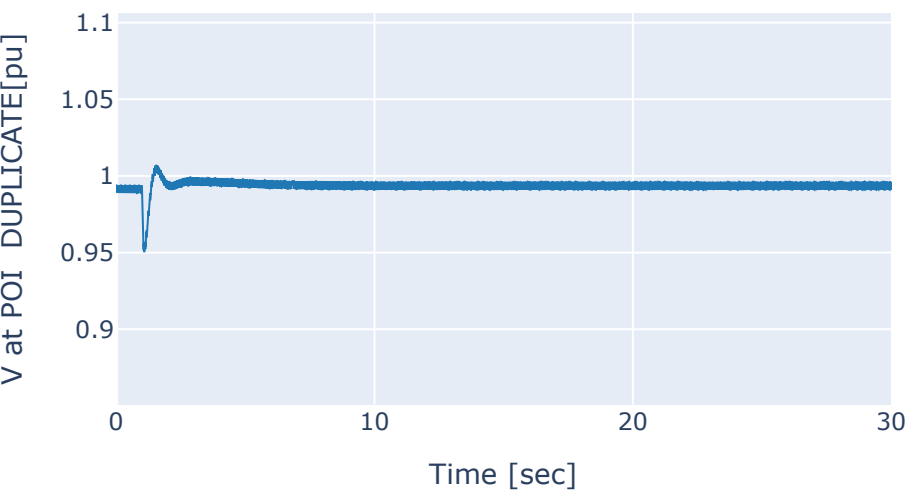
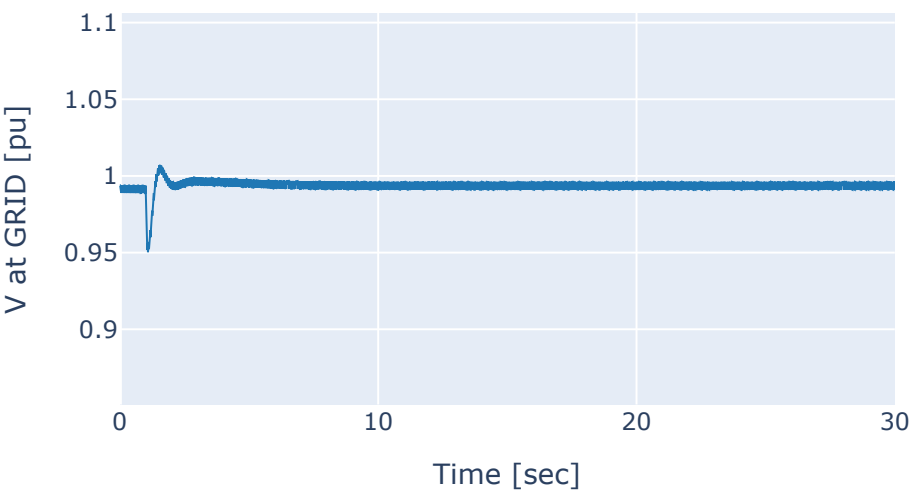
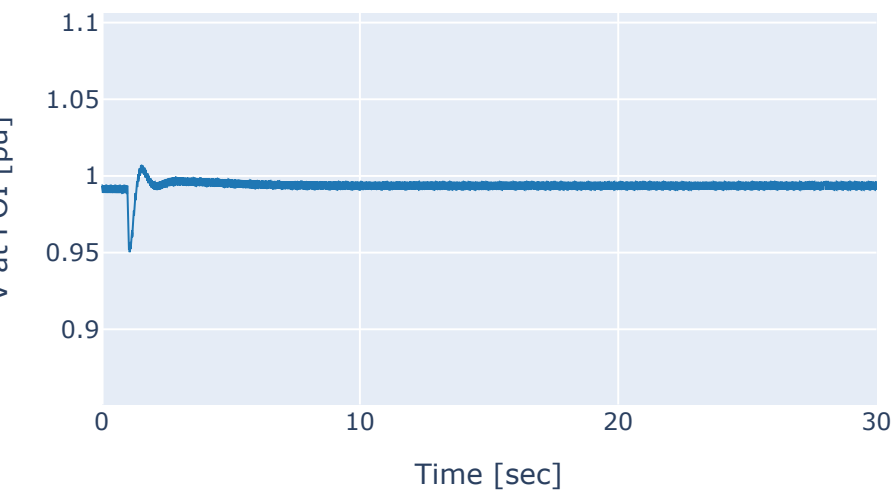
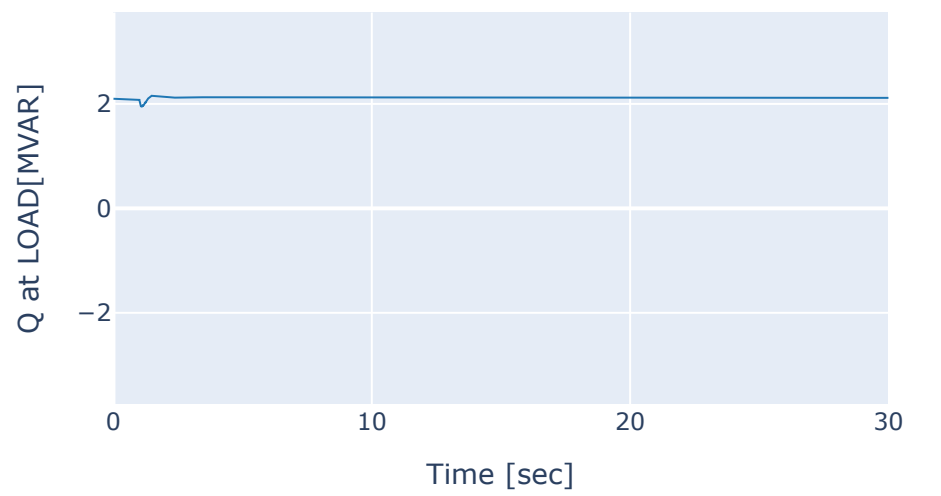
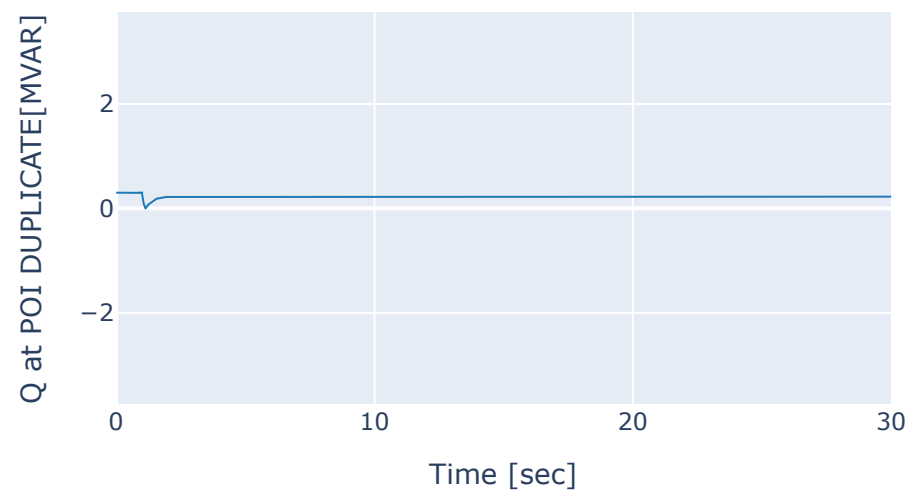
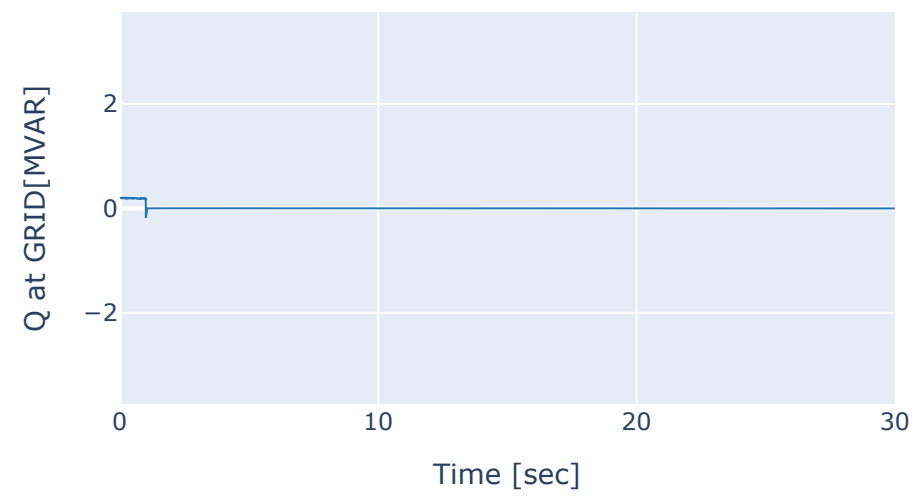
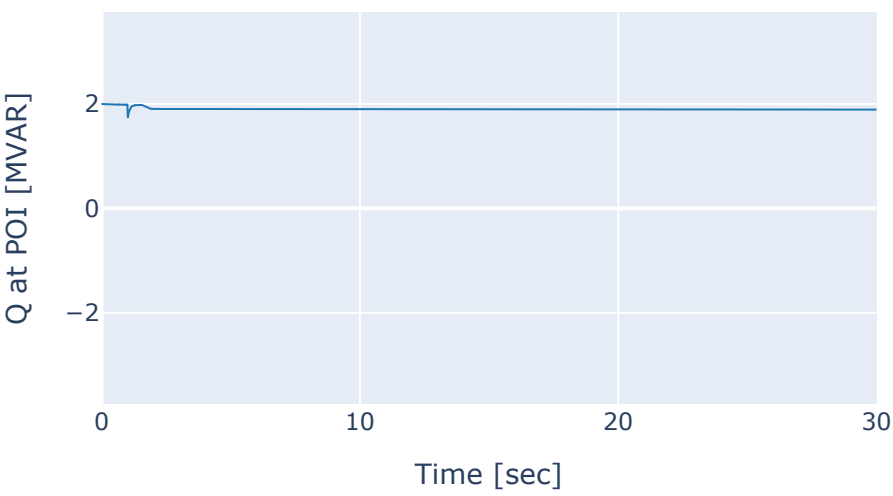
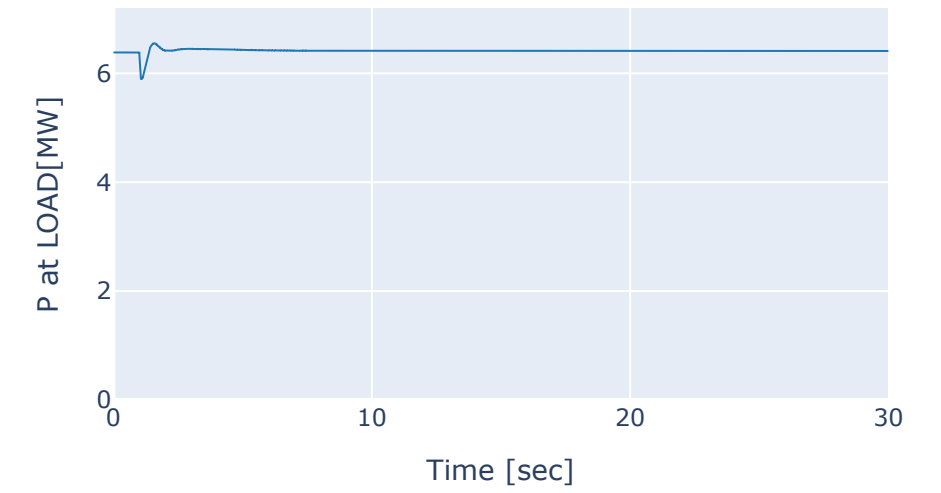
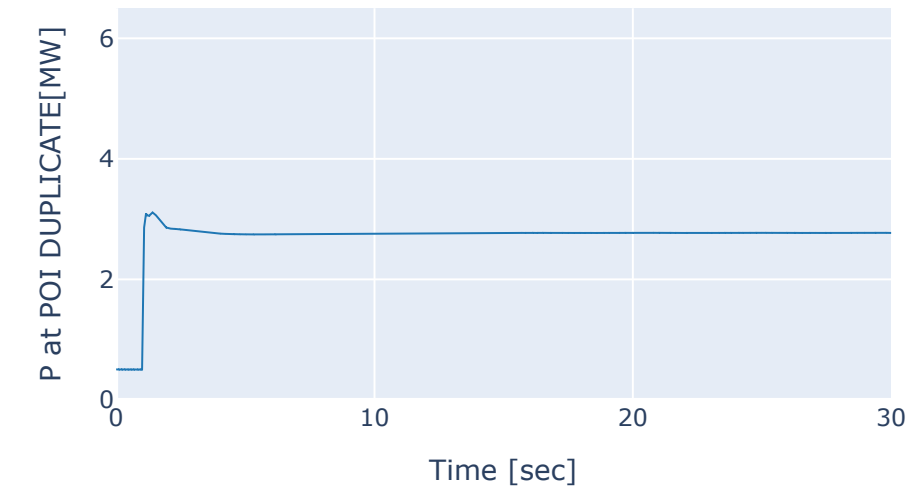
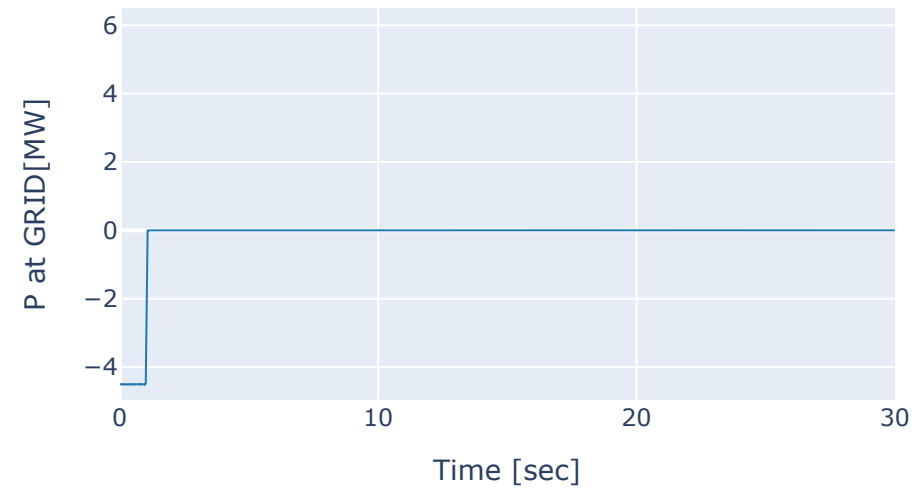
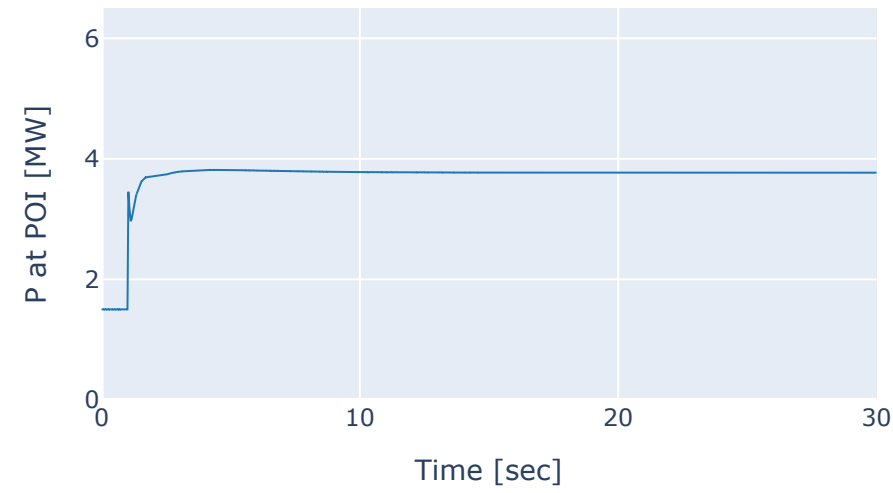
Small Voltage Step Test (IEEE 2800.1 Clause 8.8)

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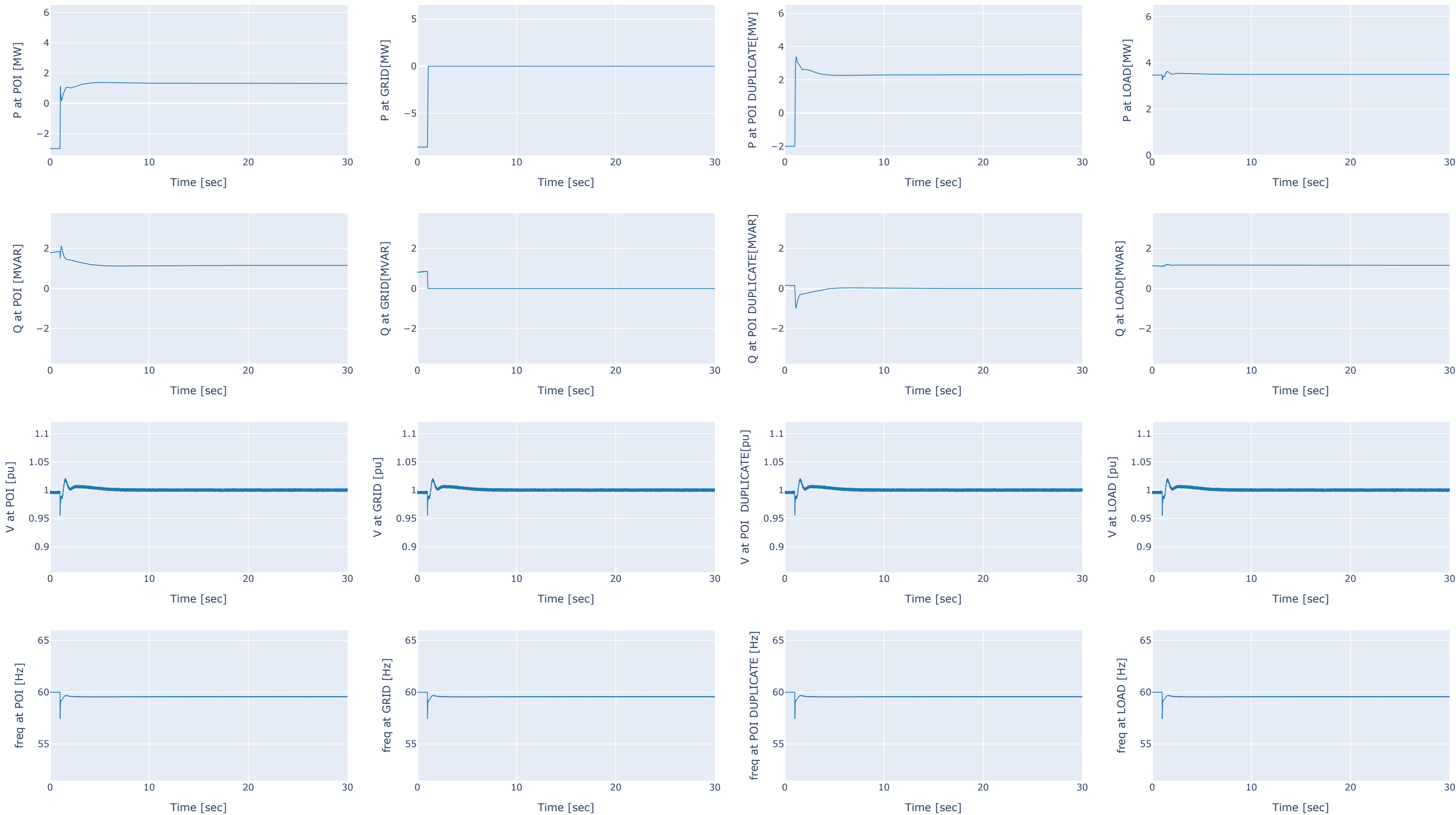
LLSM Bess Discharging

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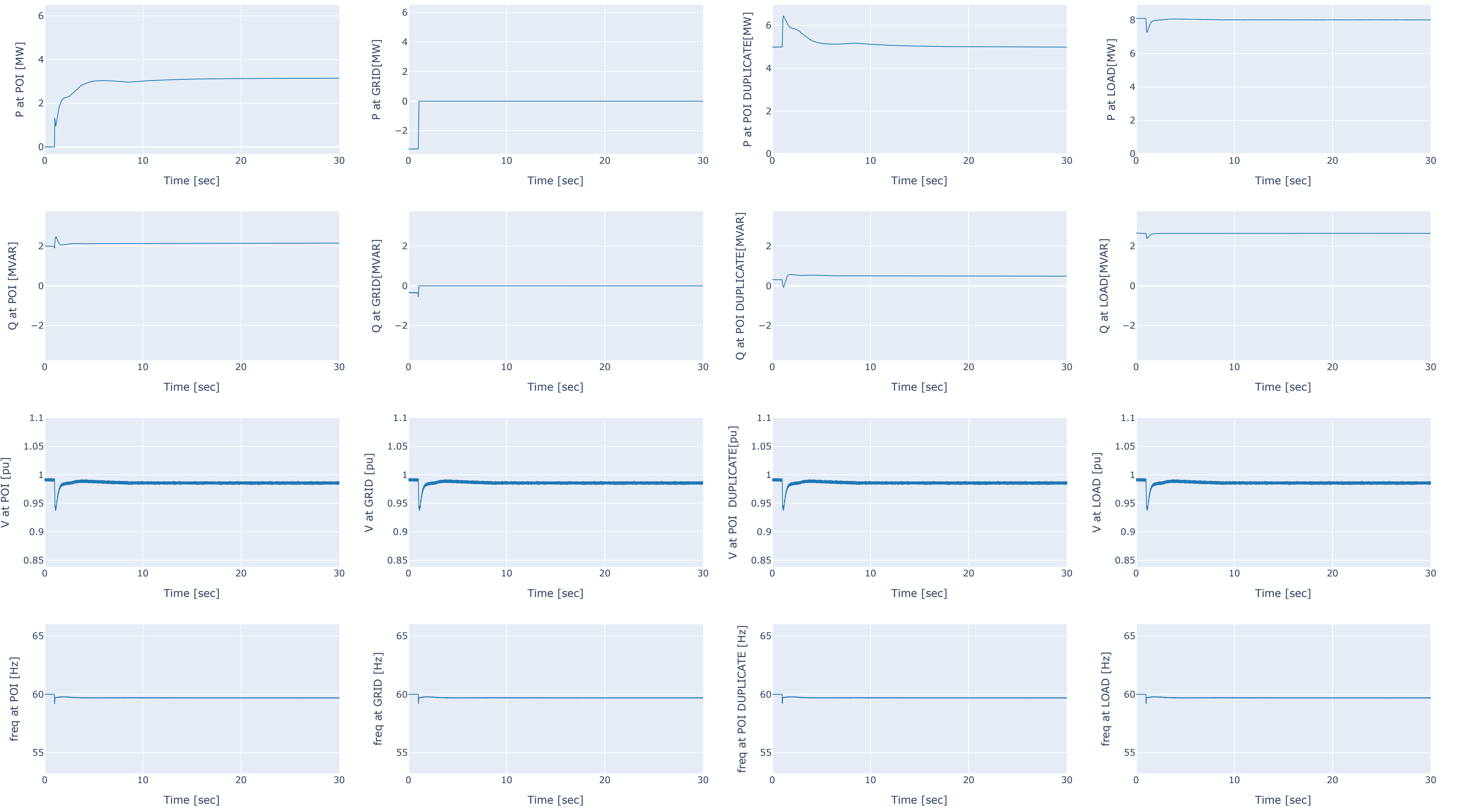
LLSM Bess Charging

PSCAD (Jul1_BB)



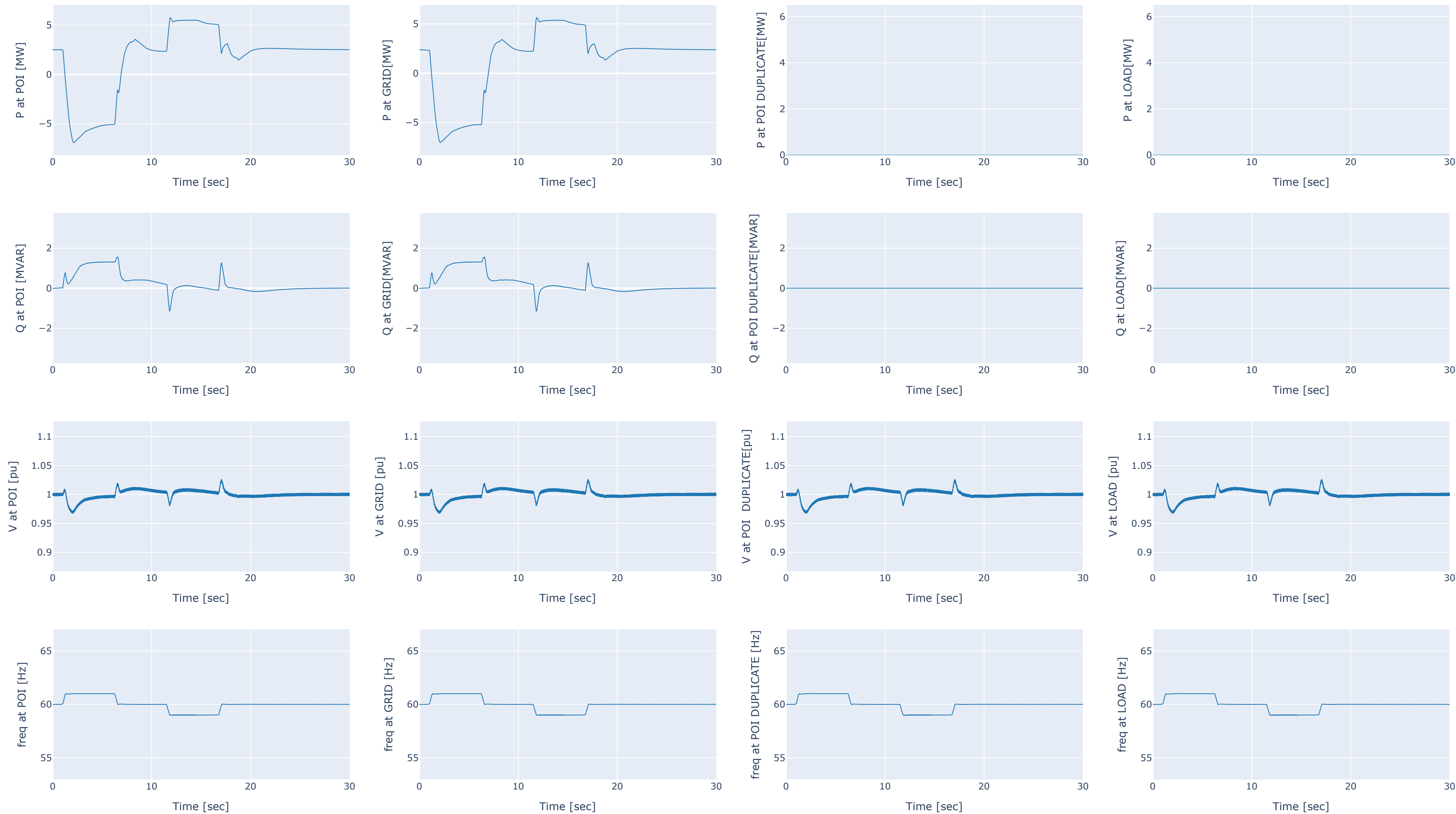
LLSM Limit Test

PSCAD (Jul1_BB)



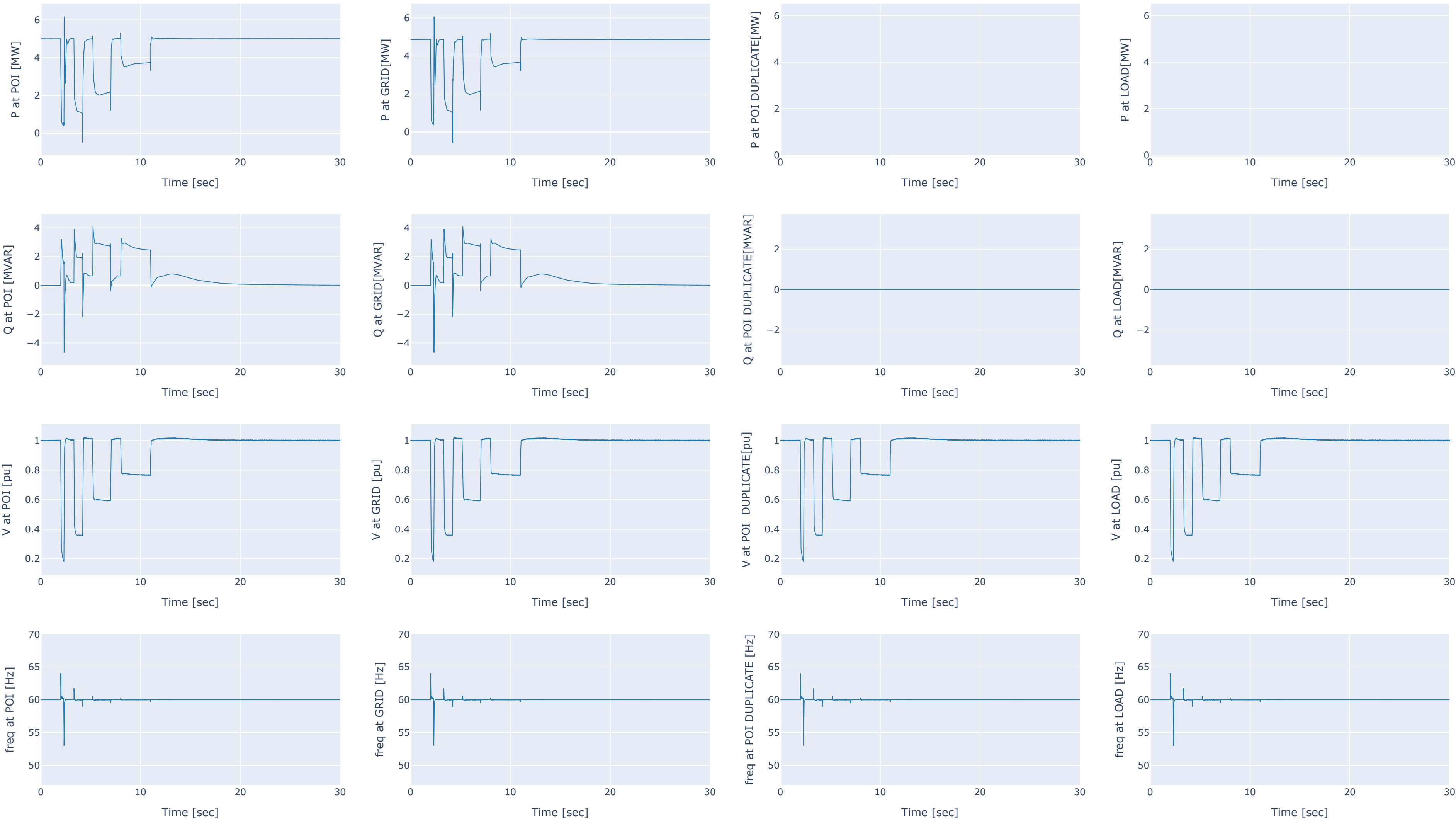
Rate of Change of Frequency Test

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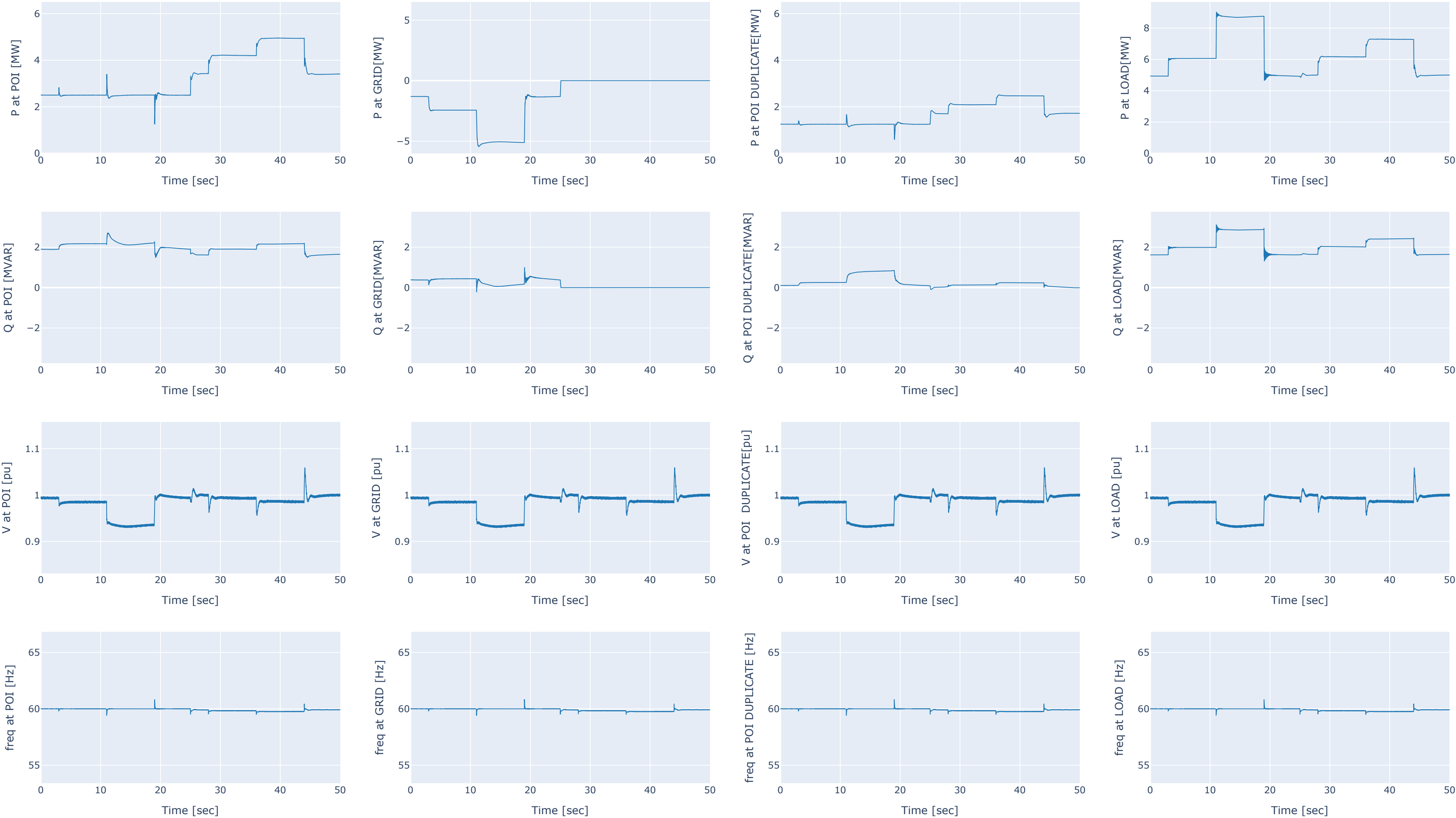
Consecutive Voltage Disturbance Ride Through Test

— PSCAD (Jul1_BB)



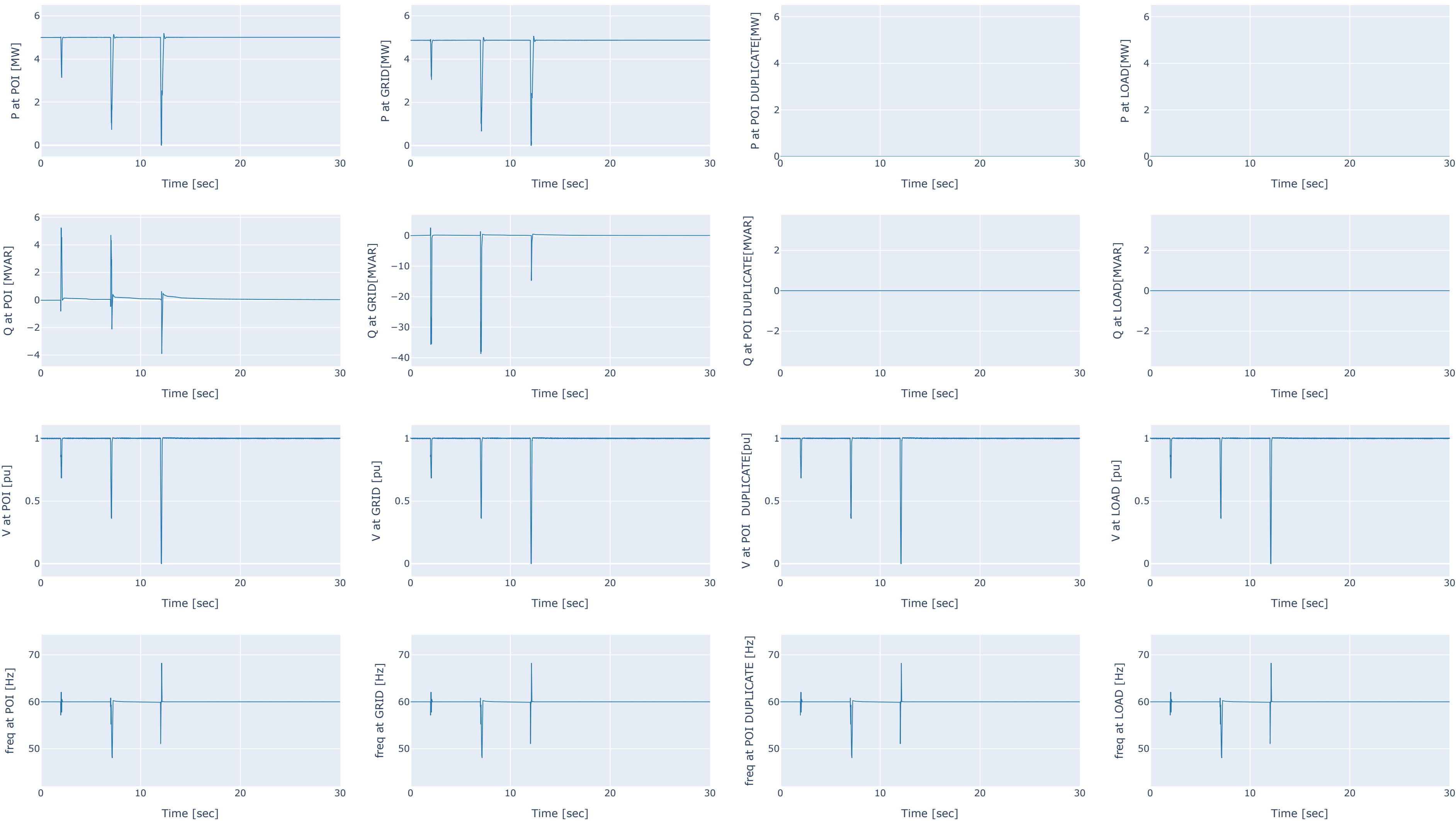
Load Profile 100% to 50% and back to 100% grid connected

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Fault Ride Through

PSCAD (Jul1_BB)



Fault Ride Through with Loss of Synchronous Machine

— PSCAD (Jul1_BB)

